Tensor Formats in Banach Spaces

Minimal Subspaces, Geometry,
and Variational Principles

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Contents

Preface

This monograph presents a unified and self-contained treatment of the mathematical foundations of low-rank tensor formats in Banach spaces. It synthesises and extends the results of four research papers by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy, published between 2012 and 2023.

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Notation and Conventions

See the file `notation.tex` for the complete list of macros and symbols used throughout.

Part I

Algebraic and Topological Foundations

Chapter 1

Introduction and Motivation

1.1 The curse of dimensionality

Many of the central problems in modern mathematics and its applications involve functions, operators, or data defined on high-dimensional product domains. A d-variate function $f : I_1 \times \cdots \times I_d \to \mathbb{R}$ defined on a product of d intervals, for example, lives in a space whose dimension grows exponentially with d: if each factor I_α is discretised with n grid points, the natural approximation space has dimension n^d. For $d = 100$ and $n = 10$ this is 10^{100}, a number that no computer will ever handle directly. This exponential explosion of complexity was famously termed by Bellman the *curse of dimensionality*.

High-dimensional problems of this kind arise throughout science and engineering:

- *Quantum chemistry:* the electronic Schrödinger equation for an N-electron molecule lives in $\mathbb{R}^{3N}$; realistic molecules have N in the hundreds.
- *Uncertainty quantification:* the solution of a PDE with d uncertain parameters is a function on $\mathbb{R}^d \times \Omega$, where Ω is the spatial domain.
- *Stochastic control and finance:* value functions in optimal control problems typically satisfy PDEs in $\mathbb{R}^d$ for large d.
- *Machine learning:* the approximation of unknown functions from data in $\mathbb{R}^d$ is the fundamental task of supervised learning.

The common mathematical structure underlying all these problems is that the object of interest — a function, a quantum state, a probability density — is an element of a *tensor product space* ${}_a \bigotimes_{\alpha \in D} V_\alpha$, where $D = \{1, \ldots, d\}$ and V_α is a function space on the α-th variable. The curse of dimensionality is the statement that the dimension of this space grows exponentially with $\#(D) = d$.

The strategy for circumventing the curse is to restrict attention to *structured low-rank tensors*: elements of $\mathbf{V}_D$ that can be represented using far fewer parameters than n^d. The systematic study of such representations — their algebraic structure, their topological properties in function spaces, and their geometry as differentiable manifolds — is the subject of this monograph.

1.2 Tensor formats and the role of subspaces

From matrices to tensors

The simplest non-trivial case is $d = 2$: here $\mathbf{V}_D = V_1 \otimes_a V_2$, and a tensor $\mathbf{v} \in \mathbf{V}_D$ corresponds to a linear map (a "matrix") from V_1^* to V_2. The *rank* of $\mathbf{v}$ — the minimal r such that $\mathbf{v} = \sum_{\nu=1}^{r} u_\nu \otimes w_\nu$ — determines the complexity of the representation. In finite dimensions this is the rank of the associated matrix; the Eckart–Young theorem gives the best rank-r approximation in the spectral or Frobenius norm.

The higher-order analogue of rank, however, is not straightforward. For $d \geq 3$ there is no single notion of "tensor rank" with all the good properties of matrix rank. Instead, several notions of rank have proven useful, each corresponding to a different way of decomposing the factors of $\mathbf{V}_D$.

Tucker format

The *Tucker format* [Tuc66] represents a tensor $\mathbf{v}$ as

$$\mathbf{v} = \sum_{\substack{1 \leq i_\alpha \leq r_\alpha \\ \alpha \in D}} C^{(D)}_{(i_\alpha)_{\alpha \in D}} \bigotimes_{\alpha \in D} u^{(\alpha)}_{i_\alpha},$$

where $u^{(\alpha)}_{i_\alpha} \in V_\alpha$ and $C^{(D)} \in \mathbb{R}^{r_1 \times \cdots \times r_d}$ is the *core tensor*. The storage cost is $\sum_\alpha r_\alpha n_\alpha + \prod_\alpha r_\alpha$, which is linear in d when the ranks r_α are bounded. The Tucker format is equivalent to the *multilinear SVD* (HOSVD) of De Lathauwer, De Moor, and Vandewalle [DLDMV00].

The correct functional-analytic description of the Tucker format is through *minimal subspaces*: the minimal subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of $\mathbf{v}$ is the smallest subspace such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a V_{[\alpha]}$ (Chapter 4). The Tucker rank of $\mathbf{v}$ is the tuple $(r_\alpha)_{\alpha \in D}$ where $r_\alpha = \dim U_\alpha^{\min}(\mathbf{v})$.

Hierarchical Tucker and Tensor Train formats

The Tucker format still suffers from the curse of dimensionality through its core tensor: $\prod_\alpha r_\alpha$ grows exponentially with d. The *Hierarchical Tucker* (HT) format [HK09] and the *Tensor Train* (TT) format [Ose11] resolve this by organising the dimensions into a *hierarchy*, encoded by a rooted tree T_D. Each format restricts the tensor to satisfy rank constraints at every level of the tree, reducing the total storage to $O(dr^3)$ (HT) or $O(dr^2n)$ (TT), where r is the maximal rank.

Tree-based tensor formats

This monograph studies all these formats within a single unified framework: *tree-based Tucker formats*, introduced in [FHN21]. Given a dimension partition tree T_D over D and a rank tuple $\mathfrak{r}$, the set

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

of tensors with fixed tree-based rank $\mathfrak{r}$ is an intersection of Tucker sets at each level of the tree. The Tucker format is the special case $T_D = T_D^{\text{Tucker}}$ (depth 1); HT and TT correspond to binary trees of depth $\lceil \log_2 d \rceil$ and $d-1$ respectively.

1.3 Three fundamental questions

The results of this monograph are organised around three fundamental mathematical questions, each corresponding to one part of the book.

Question 1 (Chapter 4). *Does every tensor* $\mathbf{v}$ *in an infinite-dimensional Banach tensor space* $\mathbf{V}_{D,\|\cdot\|_D}$ *have a best approximation from the set* $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$*?*

For convex sets in reflexive Banach spaces, best approximations always exist. The set $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ is not convex (it is not even a subspace), so the existence of best approximations is non-trivial. The answer, established in Chapters 4 and 7, is *yes*: under a natural condition on the norm (condition (Inj), which holds for all L^p and Sobolev spaces), $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ is weakly closed in $\mathbf{V}_{D,\|\cdot\|_D}$, and reflexivity gives the best approximation.

Question 2 (Chapters 8 and 9). *Does the set* $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *(or* $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$*) of tensors with fixed rank carry the structure of a differentiable manifold, even in infinite dimensions?*

In finite dimensions, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is known to be a smooth manifold (a quotient of a matrix group). In infinite dimensions, the standard quotient construction fails because the stabiliser group is infinite-dimensional and the quotient topology may not be well-behaved. The answer, established in [FHN19] for Tucker formats and in [FHN23] for tree-based formats, is *yes*: an explicit atlas with $\mathcal{C}^\infty$ transition maps can be constructed using the Grassmann–Banach manifold structure of the minimal subspaces.

Question 3 (Chapter 10). *Can the Dirac–Frenkel variational principle — the projection of the vector field onto the tangent space of the manifold — be extended from Hilbert to Banach tensor spaces?*

The answer is *yes*, provided the tangent space $Z^{(D)}(\mathbf{v})$ is a *split subspace* of $\mathbf{V}_{D,\|\cdot\|_D}$ (i.e., has a bounded projection onto it). The immersion theorems of Chapters 8 and 9 — the deepest results of the monograph — establish exactly this.

1.4 The four source papers

This monograph synthesises and expands four research papers written by the author and his collaborators Wolfgang Hackbusch and Anthony Nouy:

[FH2012] *On Minimal Subspaces in Tensor Representations* [FH12]. This paper introduces minimal subspaces as the fundamental algebraic object in tensor representations, proves their existence and uniqueness in general vector spaces, and establishes the existence of best Tucker approximations in reflexive Banach tensor spaces. It constitutes Part I (Chapters 3–4) of this monograph.

[FHN2019] *On the Dirac–Frenkel Variational Principle on Tensor Banach Spaces* [FHN19]. This paper constructs the $\mathcal{C}^\infty$-Banach manifold structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ via explicit local charts, proves that the manifold is immersed in $\mathbf{V}_{D,\|\cdot\|_D}$ under condition (Inj), and extends the Dirac–Frenkel variational principle to Banach tensor spaces. It forms the basis of Chapters 8 and 10.

[FHN2021] *Tree-based Tensor Formats* [FHN21]. This paper introduces the general framework of dimension partition trees, proves the existence of best tree-based approximations under conditions weaker than [FH12], and characterises tensors of fixed tree-based rank by means of transfer tensors. It underlies Chapters 5–7.

[FHN2023] *Geometry of Tree-based Tensor Formats in Tensor Banach Spaces* [FHN23]. This paper provides the geometric description of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as an intersection of Tucker manifolds, proves that it is a $\mathcal{C}^\infty$-Banach manifold compatible with the Tucker structure at each level of the tree, and extends the Dirac–Frenkel principle to tree-based formats. It forms the basis of Chapter 9.

1.5 What this monograph adds

Beyond collecting and unifying the results of the four source papers, this monograph contributes the following:

1. **Unified notation.** The four papers were written over a decade and use slightly different notational conventions. This monograph adopts a single coherent system (detailed in the Notation chapter), in which the Tucker format is the special case $T_D = T_D^{\text{Tucker}}$ of the tree-based format, minimal subspaces are always denoted $U_\alpha^{\min}(\mathbf{v})$, and the model Banach space of the Tucker manifold is always $\mathbf{E}_D$.

2. **Expanded proofs.** Several results in the papers are stated with proof sketches or references to earlier work. This monograph provides complete proofs of the central theorems, including the full proof of the local chart system (Lemma 8.2.1), the $\mathcal{C}^\infty$ transition maps (Lemma 8.3.1), the tangent space identification (Proposition 8.4.2), and the immersion theorems (Theorems 8.5.3 and 9.6.1).

3. **Self-contained background.** Chapters 2 and 3 provide the Banach space and algebraic tensor product prerequisites in the form and notation used throughout, so that the monograph is accessible without consulting external references for the background.

4. **Historical context and comparison.** Each chapter closes with historical notes that place the results in context and compare them with alternative approaches in the literature.

5. **Open problems.** Chapter 11 collects ten concrete open problems arising naturally from the theory, ranging from the global topology of tensor manifolds to learning theory on tree-based manifolds.

1.6 Overview of the main results

We summarise the principal theorems of the monograph, grouped by theme.

Existence of best approximations (Chapters 4, 7).

- *Theorem 4.5.4* [FH2012]: In a reflexive Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ satisfying (Inj), every $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ has a best Tucker approximation of any prescribed rank.

- *Theorem 7.5.3* [FHN2021]: The same holds for tree-based formats under the weaker local condition (Inj)[T_D].

Manifold structure (Chapters 8, 9).

- *Theorem 8.3.2* [FHN2019]: If the tensor product map is continuous, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathbf{E}_D = \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times r_\alpha}$. It is a fibre bundle over $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ with fibre $\mathbb{R}_*^{\times r_\alpha}$.

- *Theorem 9.4.1* [FHN2023]: Under (Inj) at each level, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$-Banach manifold, with an atlas compatible with the Tucker atlases at each level of T_D.

Immersion theorems (Chapters 8, 9).

- *Theorem 8.5.3* [FHN2019]: Under (Inj), the tangent subspace $Z^{(D)}(\mathbf{v})$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$, and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$.
- *Theorem 9.6.1* [FHN2023]: The same holds for $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, with tangent space $\bigcap_k Z^{(D)}([])\mathcal{P}_k(T_D)]\mathbf{v}$.

Dirac–Frenkel variational principle (Chapter 10).

- *Theorem 10.4.1* [FHN2019]: In a reflexive, strictly convex $\mathbf{V}_{D,\|\cdot\|_D}$ satisfying (Inj), the Dirac–Frenkel ODE on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ has local solutions, and they are characterised by the metric projection onto $Z^{(D)}(\mathbf{v}_r(t))$.
- *Theorem 10.6.1* [FHN2023]: The same holds on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

1.7 Guide to the reader

The monograph is organised into four parts.

Part I: Algebraic and Topological Foundations (Chapters 2–4). Chapter 2 collects the necessary background from Banach space theory and infinite-dimensional differential geometry. Chapter 3 develops the algebraic theory of tensor products. Chapter 4 is the algebraic core of the monograph: it introduces minimal subspaces, proves their fundamental properties, and establishes the existence of best approximations. *Readers familiar with Banach manifolds may skip Chapter 2 and refer back as needed.*

Part II: Tree-Based Tensor Formats (Chapters 5–7). Chapter 5 introduces dimension partition trees and proves the intersection representation $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}()$. Chapter 6 develops the algebraic theory of tree-based formats, culminating in the transfer tensor representation (Theorem 6.4.3). Chapter 7 treats the topological theory: the T_D-continuity condition, the identification theorem, and the existence of best tree-based approximations. *Readers primarily interested in the Tucker format may focus on Chapters 4 and then jump to Part III.*

Part III: Differential Geometry of Tensor Formats (Chapters 8–9). These two chapters contain the geometrically deepest results of the monograph. Chapter 8 constructs the $\mathcal{C}^\infty$ Tucker manifold and proves its immersion in $\mathbf{V}_{D,\|\cdot\|_D}$. Chapter 9 extends these results to tree-based formats via the Laplacian map and the intersection of Tucker atlases. *The key technical result is the immersion theorem (Theorems 8.5.3 and 9.6.1), which relies on Lemmas 8.5.1 and 8.5.2.*

Part IV: Variational Principles and Applications (Chapters 10–11). Chapter 10 applies the geometric results to the Dirac–Frenkel variational principle, establishing the existence and characterisation of solutions to the reduced ODE on tensor manifolds. Chapter 11 collects ten open problems.

The three appendices (Appendices A–D) provide quick references for Banach manifold theory, functional analysis, and the notation.

Prerequisites. The reader is assumed to be familiar with functional analysis at the level of a standard graduate course (Banach spaces, weak convergence, Hilbert spaces) and with differential manifolds at the level of a first course using smooth finite-dimensional manifolds. The infinite-dimensional theory (Banach manifolds, Grassmannians) is developed from scratch in Chapter 2.

Notation. A complete list of symbols is given in the Notation chapter following this Introduction. All macros used in the text are defined in the file `notation.tex`.

1.8 Historical context

The mathematical study of tensor approximation has roots in several independent traditions.

In *multilinear algebra*, Hitchcock [Hit27] introduced the notion of tensor rank in 1927. Tucker [Tuc66] introduced the Tucker format in psychometrics in 1966. The HOSVD was introduced by De Lathauwer, De Moor, and Vandewalle [DLDMV00] in 2000.

In *quantum physics*, the matrix product state (MPS) representation appeared in Vidal's work [Vid03] on quantum simulation and was independently proposed as the TT decomposition by Oseledets and Tyrtyshnikov [Ose11]. The DMRG algorithm of White, which is equivalent to optimisation over MPS/TT manifolds, dates from 1992.

In *numerical analysis*, the HT format was introduced by Hackbusch and Kühn [HK09] with the explicit goal of controlling storage costs for high-dimensional PDEs. The existence of best approximations in infinite-dimensional tensor spaces was first established in the present framework by Falcó and Hackbusch [FH12].

The *differential-geometric* treatment of tensor manifolds began with Koch and Lubich [KL07] in the finite-dimensional Hilbert space setting, motivated by the MCTDH method. The extension to infinite-dimensional Banach spaces in [FHN19, FHN23] required new ideas — notably the explicit chart construction via Grassmann charts and the immersion theorem — that form the core of this monograph.

Chapter 2

Banach Space Prerequisites

This chapter collects the functional-analytic and differential-geometric background required throughout the monograph. The reader familiar with Banach space theory and infinite-dimensional differential geometry may proceed directly to Chapter 3, referring back as needed. All vector spaces are over the field $\mathbb{K}$, which is either $\mathbb{R}$ or $\mathbb{C}$; unless stated otherwise, $\mathbb{K} = \mathbb{R}$.

2.1 Normed spaces and Banach spaces

We begin by fixing notation and recalling the central notions of normed and Banach space theory that will be used throughout.

Basic definitions

Let X be a vector space over $\mathbb{K}$. A *norm* on X is a map $\|\cdot\| : X \to [0, \infty)$ satisfying:

(N1) $\|x\| = 0$ if and only if $x = 0$,

(N2) $\|\lambda x\| = |\lambda| \, \|x\|$ for all $\lambda \in \mathbb{K}$, $x \in X$,

(N3) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$.

The pair $(X, \|\cdot\|)$ is called a *normed space*. A sequence $(x_n)_{n\in\mathbb{N}}$ in X is a *Cauchy sequence* if $\|x_n - x_m\| \to 0$ as $n, m \to \infty$. A normed space in which every Cauchy sequence converges is called a *Banach space*.

Two norms $\|\cdot\|$ and $\|\cdot\|'$ on X are *equivalent* if there exist constants $c, C > 0$ such that $c\|x\| \leq \|x\|' \leq C\|x\|$ for all $x \in X$. Equivalent norms define the same topology, and hence the same Cauchy sequences, so they produce the same Banach space structure.

Example 2.1.1. Let $I \subset \mathbb{R}$ be a measurable set and $1 \leq p < \infty$. The Lebesgue space $L^p(I)$, equipped with the norm $\|f\|_{L^p} = \left(\int_I |f(x)|^p \, dx\right)^{1/p}$, is a Banach space. For

$p = \infty$ one uses the essential supremum norm. The case $p = 2$ gives a Hilbert space with inner product $\langle f, g\rangle = \int_I f(x)\overline{g(x)}\,dx$.

Example 2.1.2. For a bounded open set $\Omega \subset \mathbb{R}^n$, $1 \le p < \infty$, and $k \in \mathbb{N}_0$, the Sobolev space $W^{k,p}(\Omega)$ consists of all functions $f \in L^p(\Omega)$ whose distributional derivatives $D^\alpha f$ of order $|\alpha| \le k$ belong to $L^p(\Omega)$. Equipped with the norm

$$\|f\|_{W^{k,p}} = \Big(\sum_{|\alpha|\le k} \|D^\alpha f\|_{L^p}^p\Big)^{1/p},$$

it is a Banach space. For $p = 2$ one writes $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space. These spaces appear as natural component spaces in high-dimensional tensor problems; see Example 4.5.7 and Section 7.5.

Continuous linear maps

Let X and Y be normed spaces. A linear map $A : X \to Y$ is *continuous* if and only if it is *bounded*, meaning

$$\|A\|_{Y\leftarrow X} := \sup_{x\neq 0} \frac{\|Ax\|_Y}{\|x\|_X} = \sup_{\|x\|_X\le 1} \|Ax\|_Y < \infty.$$

The quantity $\|A\|_{Y\leftarrow X}$ is the *operator norm* of A. The space of all continuous linear maps from X to Y, denoted $\mathcal{L}(X, Y)$, is itself a normed space under the operator norm; it is a Banach space whenever Y is complete. We write $\mathcal{L}(X) := \mathcal{L}(X, X)$. The set of bounded linear automorphisms (bounded invertible maps with bounded inverse) is

$$\mathrm{GL}(X) := \{A \in \mathcal{L}(X) : A \text{ invertible}\}.$$

This is an open subset of $\mathcal{L}(X)$, and hence a Banach manifold modelled on $\mathcal{L}(X)$; see Example 2.4.3.

Dual spaces and reflexivity

The *topological dual* of a normed space $(X, \|\cdot\|)$ is $X^* := \mathcal{L}(X, \mathbb{K})$, equipped with the dual norm

$$\|\varphi\|_{X^*} := \sup_{\|x\|\le 1} |\varphi(x)|. \tag{2.1}$$

The dual X^* is always a Banach space, regardless of whether X is complete. The *bidual* $X^{**} := (X^*)^*$ contains X as a canonically embedded subspace via the isometric embedding $\iota : X \to X^{**}$, $\iota(x)(\varphi) = \varphi(x)$. When ι is surjective, X is called *reflexive*. All L^p spaces with $1 < p < \infty$ are reflexive; L^1 and L^∞ are not.

Reflexivity plays an important role in the existence of best approximations. Recall that a closed set C in a normed space X is *proximinal* if every point of X has at least one nearest point in C. In a reflexive Banach space, every closed convex set is proximinal (see, e.g., [Hac19, p. 61]). This fact will be used extensively in Chapter 4 to prove the existence of best low-rank tensor approximations.

Convexity and smoothness

Two geometric properties of the norm are of particular importance for the Dirac–Frenkel variational principle in Chapter 10.

Definition 2.1.3. A normed space $(X, \|\cdot\|)$ is *strictly convex* (or *rotund*) if $\|x+y\|/2 < 1$ for all $x, y \in X$ with $\|x\| = \|y\| = 1$ and $x \neq y$.

Definition 2.1.4. A normed space $(X, \|\cdot\|)$ is *uniformly convex* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x + y\|/2 \leq 1 - \delta$ whenever $\|x\|, \|y\| \leq 1$ and $\|x - y\| \geq \varepsilon$.

Every uniformly convex Banach space is reflexive and strictly convex (Clarkson's theorem). In particular, L^p spaces with $1 < p < \infty$ are uniformly convex. Strict convexity guarantees uniqueness of metric projections onto closed convex sets, a property needed for the well-posedness of the Dirac–Frenkel equation.

Definition 2.1.5. A Banach space X is *smooth* if the limit $\lim_{t\to 0}\big(\|x + ty\| - \|x\|\big)/t$ exists for all $x, y \in X$ with $\|x\| = 1$.

Smoothness is the dual property to strict convexity in the following sense: X is smooth if and only if X^* is strictly convex, and X is strictly convex if and only if X^* is smooth.

2.2 Complemented subspaces and projections

The central geometric notion underlying all manifold structures in this monograph is that of a complemented (or split) subspace. Unlike the Hilbert space setting, complementability is a non-trivial condition in general Banach spaces.

Projections

A bounded linear map $P \in \mathcal{L}(X)$ is a *projection* if $P^2 = P$. In that case $\operatorname{Im} P$ is a closed subspace, $\operatorname{Ker} P$ is closed, and $X = \operatorname{Im} P \oplus \operatorname{Ker} P$ (as a direct sum of closed subspaces). The complementary projection is $\mathrm{id}_X - P$.

Definition 2.2.1. Let X be a Banach space. A closed subspace $U \subset X$ is a *split subspace* (or *complemented subspace*) of X if there exists a closed subspace $W \subset X$ such that $X = U \oplus W$. The pair (U, W) is then called a *pair of complementary subspaces*, and W is a *(topological) complement* of U in X. We write $P_{U \oplus W}$ for the unique continuous projection from X onto U along W.

The following proposition characterises split subspaces in terms of projections and provides three equivalent conditions that will each be used at different points of the text.

Proposition 2.2.2 ([FHN15, Proposition 4.2])**.** *Let X be a Banach space and $U \subset X$ a closed subspace. The following conditions are equivalent:*

(a) *U is a split subspace of X.*

(b) *There exists $P \in \mathcal{L}(X)$ with $P^2 = P$ and $\operatorname{Im} P = U$.*

(c) *There exists $Q \in \mathcal{L}(X)$ with $Q^2 = Q$ and $\operatorname{Ker} Q = U$.*

In condition (b), the projection P is none other than $P_{U \oplus W}$ for any choice of closed complement W of U; in (c), one has $Q = P_{W \oplus U}$.

Proposition 2.2.3 ([FHN15, Theorem 4.5])**.** *Let X be a Banach space. Every finite-dimensional subspace of X is a split subspace.*

In a Hilbert space, every closed subspace is complemented (the orthogonal complement provides the split). This is no longer true in general Banach spaces: by a classical result of Lindenstrauss and Tzafriri, a Banach space in which every closed subspace is complemented is necessarily isomorphic to a Hilbert space. The failure of complementability in non-Hilbert spaces is precisely what makes the immersion theorems of Chapter 8 non-trivial.

The nilpotent algebra associated to a splitting

The following result, central to the construction of local charts in Chapters 8 and 9, identifies a particularly well-behaved sub-algebra of $\mathcal{L}(X)$ associated to any splitting $X = U \oplus W$.

Proposition 2.2.4. *Let X be a Banach space and $U, W \in \mathbb{G}(X)$ with $X = U \oplus W$. Define*

$$\mathcal{L}_{(U,W)}(X) := \left\{ P_{W \oplus U} \circ S \circ P_{U \oplus W} : S \in \mathcal{L}(X) \right\} \cong \mathcal{L}(U, W).$$

Then:

(i) *The natural map $\mathcal{L}(U, W) \to \mathcal{L}_{(U,W)}(X)$, $L \mapsto P_{W \oplus U} \circ L \circ P_{U \oplus W}$, is an isometric isomorphism.*

(ii) $\mathcal{L}_{(U,W)}(X)$ *is a nilpotent sub-algebra of* $\mathcal{L}(X)$*: for every* $L, L' \in \mathcal{L}_{(U,W)}(X)$ *one has* $L \circ L' = 0$*.*

(iii) *As a consequence, for every* $L \in \mathcal{L}_{(U,W)}(X)$*:*

$$\exp(L) \coloneqq \sum_{n=0}^{\infty} \frac{L^n}{n!} = \mathrm{id}_X + L, \qquad \exp(-L) = \mathrm{id}_X - L = (\mathrm{id}_X + L)^{-1}. \tag{2.2}$$

Proof. For the isometry in (i), note that $\|P_{W \oplus U} \circ L \circ P_{U \oplus W}\|_{X \leftarrow X} = \|L\|_{W \leftarrow U}$ by the definition of the operator norm. For (ii): if $L = P_{W \oplus U} \circ S \circ P_{U \oplus W}$ and $L' = P_{W \oplus U} \circ S' \circ P_{U \oplus W}$, then

$$L \circ L' = P_{W \oplus U} \circ S \circ (P_{U \oplus W} \circ P_{W \oplus U}) \circ S' \circ P_{U \oplus W} = 0,$$

since $P_{U \oplus W} \circ P_{W \oplus U} = 0$. Statement (iii) follows immediately from (ii) and the series definition of the exponential. □

The identity $\exp(L) = \mathrm{id}_X + L$ means that the exponential map on the nilpotent algebra $\mathcal{L}_{(U,W)}(X)$ is a polynomial of degree one, and in particular a global analytic diffeomorphism. This is the key that makes the local charts of the Grassmann–Banach manifold and the Tucker manifold globally well-defined; see Section 2.3.

Common complements and chart compatibility

The following lemma, which underlies the compatibility of overlapping charts in the Grassmann–Banach manifold, characterises when two split subspaces share a common complement.

Lemma 2.2.5 ([DD63, Lemma 2.1])**.** *Let* X *be a Banach space and let* $U, U', W \in \mathbb{G}(X)$ *with* $X = U \oplus W = U' \oplus W$*. The following conditions are equivalent:*

(a) U *and* U' *have* W *as a common complement, i.e.,* $X = U \oplus W = U' \oplus W$*.*

(b) *The restriction* $P_{U \oplus W}\big|_{U'} : U' \to U$ *is a linear isomorphism (with bounded inverse).*

Moreover, in this case $\left(P_{U \oplus W}\big|_{U'}\right)^{-1} = P_{U' \oplus W}\big|_{U}$*.*

2.3 The Grassmann–Banach manifold

The Grassmannian of a Banach space is the natural parameter space for subspaces, and it serves as the base manifold for the Tucker and tree-based tensor manifolds.

Definition 2.3.1. Let X be a Banach space. The *Grassmann manifold* (or *Grassmannian*) of X is

$$\mathbb{G}(X) \coloneqq \{U \subset X : U \text{ is a split subspace of } X\}.$$

For $r \in \mathbb{N}_0$, the *finite Grassmannian* is

$$\mathbb{G}_r(X) \coloneqq \{U \in \mathbb{G}(X) : \dim U = r\}.$$

Both X and $\{0\}$ belong to $\mathbb{G}(X)$. By Proposition 2.2.3, every r-dimensional subspace of X belongs to $\mathbb{G}(X)$, so $\mathbb{G}_r(X)$ is well defined.

Theorem 2.3.2 (Grassmann–Banach manifold, [Dou66])**.** *Let X be a Banach space. The Grassmannian $\mathbb{G}(X)$ is an analytic Banach manifold, with atlas $\left\{\mathbb{G}(W,X),\ \Psi_{U\oplus W}\right\}_{U\in\mathbb{G}(X)}$, where for each splitting $X = U \oplus W$:*

(i) *$\mathbb{G}(W,X) \coloneqq \{V \in \mathbb{G}(X) : X = V \oplus W\}$ is the chart domain at U.*

(ii) *The chart map $\Psi_{U\oplus W} : \mathbb{G}(W,X) \to \mathcal{L}(U,W)$ is defined by*

$$\Psi_{U\oplus W}(V) \coloneqq P_{W\oplus U}\big|_V \circ \left(P_{U\oplus W}\big|_V\right)^{-1}, \tag{2.3}$$

with inverse $\Psi^{-1}_{U\oplus W}(L) = G(L) \coloneqq \{(\mathrm{id}_X + L)(u) : u \in U\}$.

The transition maps are analytic. Each chart domain $\mathbb{G}(W,X)$ is a Banach manifold modelled on $\mathcal{L}(U,W)$.

The chart map $\Psi_{U\oplus W}$ sends a subspace $V \in \mathbb{G}(W,X)$ to the linear map $L : U \to W$ whose graph $G(L)$ equals V. The identity $\Psi^{-1}_{U\oplus W}(L) = \exp(L)(U)$, combined with Proposition 2.2.4, shows that the inverse chart is a degree-one polynomial map (hence analytic). Analyticity of the transition maps follows by a direct computation using Lemma 2.2.5.

Remark 2.3.3*.* The full Grassmannian $\mathbb{G}(X)$ is a Banach manifold but is *not* modelled on a single Banach space: if U and U' are finite-dimensional with $\dim U \neq \dim U'$, then $\mathcal{L}(U,W)$ and $\mathcal{L}(U',W')$ are non-isomorphic. This is not a deficiency but a structural fact: $\mathbb{G}(X)$ decomposes into connected components $\mathbb{G}(X) = \bigsqcup_{r\in\mathbb{N}_0} \mathbb{G}_r(X)$, each of which *is* modelled on a single Banach space.

Proposition 2.3.4. *For each $r \in \mathbb{N}_0$, the finite Grassmannian $\mathbb{G}_r(X)$ is a connected component of $\mathbb{G}(X)$, and it is an analytic Banach manifold modelled on $\mathcal{L}(U,W)$ for any $U \in \mathbb{G}_r(X)$ and complement W (the isomorphism class of $\mathcal{L}(U,W)$ does not depend on the choice of U or W).*

Remark 2.3.5 (Grassmannian of a normed space)*.* The construction extends to normed (non-complete) spaces. Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Define $\mathbb{G}_r(X)$

as the set of r-dimensional subspaces of X that are also split in $\bar{X}$. By Lemma 2.5.6, every chart domain $\mathbb{G}(W, \bar{X})$ is contained in $\mathbb{G}_r(X)$, so the atlas of $\mathbb{G}_r(\bar{X})$ restricts to an analytic atlas on $\mathbb{G}_r(X)$. This variant is needed in Section 8.2, where the component spaces V_α are normed but not necessarily complete.

The tangent space to $\mathbb{G}(X)$ at U is identified as follows.

Proposition 2.3.6. *For $U \in \mathbb{G}(X)$ with complement W, the tangent space $T_U\mathbb{G}(X) \cong \mathcal{L}(U, W)$.*

This identification is consistent with the finite-dimensional case: in $\mathbb{R}^n$, the tangent space to the Grassmannian $Gr(r, n)$ at a subspace U is the space of linear maps from U to its orthogonal complement, recovering the classical description.

2.4 Banach manifolds

We recall the definition of a Banach manifold in its general form, since the full Grassmannian $\mathbb{G}(X)$ and the full tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ are manifolds not modelled on a fixed Banach space.

Definition 2.4.1. Let $\mathbb{M}$ be a set. An *atlas of class* $\mathcal{C}^p$ (for $p \geq 0$ or $p = \infty$ or $p = \omega$, the analytic case) on $\mathbb{M}$ is a family of *charts* $\{(M_i, u_i)\}_{i \in A}$ satisfying:

(AT1) $\{M_i\}_{i \in A}$ covers $\mathbb{M}$.

(AT2) For each $i \in A$, the chart $u_i : M_i \to U_i$ is a bijection onto an open set U_i in some Banach space X_i, and for any $i, j \in A$, the set $u_i(M_i \cap M_j)$ is open in X_i.

(AT3) Whenever $M_i \cap M_j \neq \emptyset$, the *transition map* $u_j \circ u_i^{-1} : u_i(M_i \cap M_j) \to u_j(M_i \cap M_j)$ is a diffeomorphism of class $\mathcal{C}^p$.

Two atlases are *compatible* if their union is again an atlas. A $\mathcal{C}^p$*-Banach manifold* is a set $\mathbb{M}$ together with a maximal (complete) atlas of class $\mathcal{C}^p$. If all X_i are isomorphic to a single Banach space X, we say $\mathbb{M}$ is *modelled on* X.

Example 2.4.2. Every Banach space X is a $\mathcal{C}^\infty$-Banach manifold modelled on itself, with the single chart (X, id_X).

Example 2.4.3. If X is a Banach space, then $\mathrm{GL}(X)$ is an open subset of $\mathcal{L}(X)$, and hence an analytic Banach manifold modelled on $\mathcal{L}(X)$. The maps $A \mapsto A^{-1}$ and $(A, B) \mapsto AB$ are analytic (see [Upm85, 2.7, 2.8]).

Definition 2.4.4. Let $\mathbb{M}$ be a $\mathcal{C}^p$-Banach manifold with $p \geq 1$ and $m \in \mathbb{M}$. A *tangent vector* at m is an equivalence class of triples (U_i, u_i, v) where (U_i, u_i) is a chart at m and

$v \in X_i$, under the equivalence $(U_i, u_i, v) \sim (U_j, u_j, w)$ if $D(u_j \circ u_i^{-1})(u_i(m))v = w$. The *tangent space* $T_m\mathbb{M}$ is the set of all tangent vectors at m, which inherits the structure of a Banach space from any chart.

Definition 2.4.5. A $\mathcal{C}^1$-map $f : \mathbb{M} \to \mathbb{N}$ between Banach manifolds is an *immersion* at m if $Df(m) : T_m\mathbb{M} \to T_{f(m)}\mathbb{N}$ is injective with closed split image. It is an *immersion* if it is an immersion at every point. An *immersed submanifold* of $\mathbb{N}$ is the image of an injective immersion.

The closedness and split condition on the image of $Df(m)$ is automatic in finite dimensions and in Hilbert spaces (where every closed subspace is split), but is a genuine constraint in Banach spaces. This is the key difficulty resolved in Section 8.5.

2.5 Multilinear maps and Fréchet differentiability

Since the tensor product map is multilinear, the following general results will be used repeatedly.

Definition 2.5.1. A map $M : X_1 \times \cdots \times X_d \to Y$ between normed spaces is *multilinear* if it is linear in each argument separately. It is *bounded* (equivalently, *continuous*) if

$$\|M\| \coloneqq \sup_{\substack{x_j \in X_j \\ \|x_j\|_j \leq 1}} \|M(x_1, \dots, x_d)\|_Y < \infty.$$

The following proposition shows that continuity implies a much stronger regularity.

Proposition 2.5.2 ([HN12, Proposition 79])**.** *Let $X_1, \dots, X_d$ and Y be normed spaces and let $M : X_1 \times \cdots \times X_d \to Y$ be a continuous multilinear map. Then M is $\mathcal{C}^\infty$-Fréchet differentiable, and $D^k M = 0$ for all $k > d$.*

The vanishing of derivatives beyond order d reflects the polynomial nature of multilinear maps. In the complex case, a stronger conclusion holds.

Proposition 2.5.3 ([HN12, Theorem 160])**.** *Let X, Y be complex Banach spaces and $U \subset X$ open. A map $f : U \to Y$ is analytic if and only if it is $\mathcal{C}^1$-Fréchet differentiable.*

Corollary 2.5.4. *Every continuous multilinear map between complex Banach spaces is analytic.*

In real Banach spaces, $\mathcal{C}^\infty$ and analytic are distinct notions; the tensor product map is $\mathcal{C}^\infty$ but need not be analytic over $\mathbb{R}$. For the real case, $\mathcal{C}^\infty$-regularity is sufficient for the manifold constructions in Chapters 8 and 9.

Remark 2.5.5. Proposition 2.5.2 is the reason why the Tucker and tree-based tensor manifolds are $\mathcal{C}^\infty$-Banach manifolds (and analytic over $\mathbb{C}$): their chart transition maps are compositions of continuous multilinear maps with bounded linear projections, all of which are $\mathcal{C}^\infty$-Fréchet differentiable.

The Grassmannian revisited: Lemma on normed spaces

For later use in Section 8.2, we record the following lemma about Grassmannians of normed (non-complete) spaces.

Lemma 2.5.6. *Let $(X, \|\cdot\|)$ be a normed space with completion $\bar{X}$. Let $U \in \mathbb{G}_r(\bar{X})$ with $U \subset X$, and suppose $\bar{X} = U \oplus W$ for some $W \in \mathbb{G}(\bar{X})$. Then every $V \in \mathbb{G}(W, \bar{X})$ satisfies $V \subset X$.*

Proof. Observe that $X = U \oplus (W \cap X)$, where $W \cap X$ is dense in W. Suppose for contradiction that there exists $V \in \mathbb{G}(W, \bar{X})$ with $V \not\subset X$. Since V is finite-dimensional and $\bar{X} = V \oplus W$, the subspace $V \cap X$ is a proper subspace of V. On the other hand, $X = (V \cap X) \oplus (W \cap X)$ implies $\bar{X} = (V \cap X) \oplus W$, so $V \cap X \in \mathbb{G}(W, \bar{X})$. But $\dim(V \cap X) < \dim V = r$, contradicting the fact that every element of $\mathbb{G}(W, \bar{X})$ has dimension r. □

2.6 Historical notes

The theory of Banach spaces was developed primarily in the 1920s and 1930s, with Stefan Banach's monograph [Ban32] providing the first systematic treatment. The duality theory and reflexivity go back to Banach and Alaoglu; the theorem that uniform convexity implies reflexivity is due to Clarkson (1936) and Milman (1938).

Complemented subspaces. The study of complemented subspaces has its roots in the work of Banach himself, who showed that c_0 has uncomplemented subspaces. The decisive result — that a Banach space in which every closed subspace is complemented must be isomorphic to a Hilbert space — was proved by Lindenstrauss and Tzafriri in 1971. This explains why complementability is a genuine and non-trivial constraint in the Banach tensor setting of this monograph.

The characterisation of split subspaces via bounded projections (Proposition 2.2.2) and the finite-dimensionality criterion (Proposition 2.2.3) are classical; the formulation used here follows [FHN15]. The nilpotent algebra associated to a splitting (Proposition 2.2.4) and its use to identify the chart inverse as a degree-one polynomial map is a key observation of [FHN19].

The lemma on common complements (Lemma 2.2.5) is a version of results going back to Dixmier and Douady [DD63], who studied continuous fields of Hilbert spaces.

The formulation for general Banach spaces and its application to the transition maps of the Tucker manifold appears in [FHN19].

Grassmann–Banach manifold. The infinite-dimensional Grassmannian was constructed as an analytic Banach manifold by Douady [Dou66] in the context of moduli spaces of compact analytic subspaces. The atlas described in Theorem 2.3.2 goes back to this work; the specific formulation in terms of the chart map (2.3) and the identification of the tangent space with $\mathcal{L}(U, W)$ follows the presentation of [Upm85]. The finite-dimensional Grassmannian $Gr(r, n)$ is a smooth compact manifold of dimension $r(n - r)$, a fact well known since Grassmann's original work in the 1840s; the infinite-dimensional version requires the split condition as a substitute for compactness arguments.

Multilinear maps and Fréchet differentiability. The fact that continuous multilinear maps are $\mathcal{C}^\infty$ (Proposition 2.5.2) is a standard result of infinite-dimensional calculus; see [HN12] for a careful treatment. The equivalence between $\mathcal{C}^1$ and analyticity in complex Banach spaces (Proposition 2.5.3) is the Gâteaux–Cauchy theorem; see [HN12]. These two propositions are the main tools that allow one to conclude that the Tucker and tree-based tensor manifolds are $\mathcal{C}^\infty$ (real case) or analytic (complex case) without further calculation: their atlas transition maps are compositions of continuous multilinear maps and bounded linear projections, all of which satisfy the hypotheses of Propositions 2.5.2 and 2.5.3.

Chapter 3

Algebraic Tensor Spaces

This chapter develops the purely algebraic foundations of tensor spaces. We introduce the algebraic tensor product via its universal property, establish the key notions of elementary tensor, matricisation, and multilinear rank, and define the Tucker format and the set $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ of tensors of fixed Tucker rank. No topology is assumed in this chapter; that enters in Chapter 7.

Throughout, $D = \{1, \ldots, d\}$ with $d \geq 2$ is the *dimension index set*, and V_α $(\alpha \in D)$ are vector spaces over $\mathbb{K}$. Concerning the foundational construction of tensor products we follow Greub [Gre81].

3.1 The algebraic tensor product

Universal property and construction

Definition 3.1.1. The *algebraic tensor product* of the spaces V_α, $\alpha \in D$, is a pair $(\mathbf{V}_D, \bigotimes)$ consisting of a vector space $\mathbf{V}_D$ and a multilinear map

$$\bigotimes : \prod_{\alpha \in D} V_\alpha \longrightarrow \mathbf{V}_D, \qquad (v_\alpha)_{\alpha \in D} \mapsto \bigotimes_{\alpha \in D} v_\alpha, \tag{3.1}$$

satisfying the following *universal property*: for every vector space Z and every multilinear map $M : \prod_{\alpha \in D} V_\alpha \to Z$, there exists a unique linear map $\widehat{M} : \mathbf{V}_D \to Z$ such that $M = \widehat{M} \circ \bigotimes$.

We write $\mathbf{V}_D = {}_a \bigotimes_{\alpha \in D} V_\alpha$ and refer to elements of the form $\bigotimes_{\alpha \in D} v_\alpha$ (with $v_\alpha \in V_\alpha$) as *elementary tensors*. Every element of $\mathbf{V}_D$ is a *finite* linear combination of elementary tensors.

The algebraic tensor product exists and is unique up to isomorphism; see [Gre81] for the standard construction. The prefix 'a' in ${}_a \bigotimes$ emphasises the algebraic (non-topological) nature. The underlying field is $\mathbb{K} = \mathbb{R}$ throughout, but all results hold equally for $\mathbb{K} = \mathbb{C}$.

Remark 3.1.2. The set of elementary tensors is *not* a basis of $\mathbf{V}_D$; it generates $\mathbf{V}_D$ but with many linear dependencies. For example, $\lambda v \otimes w = v \otimes \lambda w$ for all $\lambda \in \mathbb{K}$. If $\{e_i\}$ and $\{f_j\}$ are bases of V_1 and V_2 respectively, then $\{e_i \otimes f_j\}$ is a basis of $V_1 \otimes_a V_2$.

Bilinear identification and the α-unfolding

For each $\alpha \in D$, the algebraic tensor space can be rewritten as a two-factor product by grouping all indices except α:

$$\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}, \qquad \mathbf{V}_\alpha^{[\alpha]} := {}_a\bigotimes_{\beta \in D\setminus\{\alpha\}} V_\beta. \tag{3.2}$$

Concretely, there is a canonical linear isomorphism $\Phi_\alpha : \mathbf{V}_D \xrightarrow{\sim} V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and throughout this monograph we identify $\mathbf{v} \in \mathbf{V}_D$ with $\Phi_\alpha(\mathbf{v}) \in V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ without further comment. An elementary tensor splits as $\bigotimes_{\beta \in D} v_\beta \leftrightarrow v_\alpha \otimes \bigotimes_{\beta \neq \alpha} v_\beta$.

Tensor products of linear maps

Given linear maps $A_\alpha : V_\alpha \to W_\alpha$ for $\alpha \in D$, their tensor product is the unique linear map

$$\bigotimes_{\alpha \in D} A_\alpha : \mathbf{V}_D \longrightarrow \mathbf{W}_D := {}_a\bigotimes_{\alpha \in D} W_\alpha, \tag{3.3}$$

determined on elementary tensors by $\left(\bigotimes_{\alpha \in D} A_\alpha\right)\left(\bigotimes_{\alpha \in D} v_\alpha\right) = \bigotimes_{\alpha \in D} A_\alpha v_\alpha$. This is the unique linear extension guaranteed by the universal property. We denote the algebraic dual of V by $V' = L(V, \mathbb{K})$ and the space of all linear maps from V to W by $L(V, W)$.

Partitions and iterated tensor products

Definition 3.1.3. Let $\mathcal{P}$ be a partition of D (i.e., a collection of non-empty, pairwise disjoint subsets of D whose union is D). For each $\alpha \in \mathcal{P}$, set $\mathbf{V}_\alpha := {}_a\bigotimes_{j \in \alpha} V_j$ (with the convention $\mathbf{V}_{\{j\}} = V_j$). Then the *iterated tensor product* associated to $\mathcal{P}$ is

$${}_a\bigotimes_{\alpha \in \mathcal{P}} \mathbf{V}_\alpha \cong \mathbf{V}_D, \tag{3.4}$$

where the isomorphism is canonical and consistent with the identifications of elementary tensors.

This associativity-of-tensor-products fact is fundamental: it allows us to regroup the factors of $\mathbf{V}_D$ according to any hierarchical structure, which is precisely what the dimension partition trees of Chapter 5 will exploit.

Example 3.1.4. For $D = \{1, 2, 3, 4\}$ and the partition $\mathcal{P} = \{\{1, 2\}, \{3, 4\}\}$:

$$(V_1 \otimes_a V_2) \otimes_a (V_3 \otimes_a V_4) \cong V_1 \otimes_a V_2 \otimes_a V_3 \otimes_a V_4.$$

The partition $\mathcal{P}' = \{\{1\}, \{2\}, \{3\}, \{4\}\}$ (trivial) recovers the original four-fold product. The non-trivial partition $\mathcal{P}$ is the structure underlying the Tucker format (depth-2 tree), while more refined partitions correspond to hierarchical formats; see Chapter 5.

3.2 Matricisation and multilinear rank

Matricisation in finite dimensions

When the component spaces are finite-dimensional, the tensor $\mathbf{v} \in \mathbf{V}_D$ can be represented as a multi-dimensional array, and the identification (3.2) induces a matrix.

Definition 3.2.1. Let $D = \{1, \dots, d\}$, $\dim V_\alpha = n_\alpha < \infty$, and fix bases of each V_α. Identify $\mathbf{v} \in \mathbf{V}_D$ with its coordinate array $C \in \mathbb{R}^{n_1 \times \cdots \times n_d}$. For each $\alpha \in D$, the *α-matricisation* (or *α-unfolding*) of $\mathbf{v}$ is the matrix

$$\mathcal{M}_\alpha(\mathbf{v}) \in \mathbb{R}^{n_\alpha \times \prod_{\beta \neq \alpha} n_\beta}$$

obtained by reordering the coordinate array C so that the α-index labels the rows and all other indices, in lexicographic order, label the columns. The rank of this matrix, $\operatorname{rank} \mathcal{M}_\alpha(\mathbf{v}) \in \{0, 1, \dots, n_\alpha\}$, is the *$\alpha$-rank* of $\mathbf{v}$.

The notion of α-rank was introduced by Hitchcock [Hit27, p. 170] in 1927 under the name "rank on the jth index". In the infinite-dimensional case, the correct replacement of $\operatorname{rank} \mathcal{M}_\alpha(\mathbf{v})$ is the dimension of the minimal subspace $U_\alpha^{\min}(\mathbf{v})$, which we define in Section 3.3 and develop fully in Chapter 4.

Full-rank core tensors

The following definition isolates the tensors whose coordinate arrays have maximal rank in every direction, a condition that will characterise elements of the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

Definition 3.2.2. For a rank tuple $\mathfrak{r} = (r_\alpha)_{\alpha \in D} \in \mathbb{N}^d$, define

$$\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha} := \Big\{ C \in \mathbb{R}^{\times_{\alpha \in D} r_\alpha} : \operatorname{rank} \mathcal{M}_\alpha(C) = r_\alpha \ \text{ for all } \alpha \in D \Big\}. \tag{3.5}$$

Elements of $\mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ are called *full-rank core tensors* of shape $\mathfrak{r}$.

Remark 3.2.3. Since $\operatorname{rank}\mathcal{M}_\alpha(C) = r_\alpha$ if and only if $\mathcal{M}_\alpha(C)\mathcal{M}_\alpha(C)^T \in \mathrm{GL}(\mathbb{R}^{r_\alpha})$, and the determinant is a continuous function, $\mathbb{R}_*^{\times_{\alpha\in D} r_\alpha}$ is an open subset of $\mathbb{R}^{\times_{\alpha\in D} r_\alpha}$, hence a smooth finite-dimensional manifold in its own right. In the scalar case $r_\alpha = 1$ for all α, we recover $\mathbb{R}_* = \mathbb{R} \setminus \{0\} = \mathrm{GL}(\mathbb{R})$.

3.3 The Tucker format

Minimal subspaces: first appearance

We introduce minimal subspaces here in their algebraic form; the full theory (existence, uniqueness, lattice properties, best approximation) is developed in Chapter 4.

Definition 3.3.1. Let $\mathbf{v} \in \mathbf{V}_D = {}_a\bigotimes_{\alpha\in D} V_\alpha$ and $\alpha \in D$. Under the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, the *minimal subspace* of $\mathbf{v}$ in direction α is the unique subspace $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ of smallest dimension such that $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The *$\alpha$-rank* of $\mathbf{v}$ is $r_\alpha(\mathbf{v}) \coloneqq \dim U_\alpha^{\min}(\mathbf{v})$.

The key properties — existence, uniqueness, and the symmetry $\dim U_\alpha^{\min}(\mathbf{v}) = \dim U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ — are established in Chapter 4. Here we proceed taking these properties for granted.

The Tucker format and the set $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$

Definition 3.3.2. Let $\mathfrak{r} = (r_\alpha)_{\alpha\in D} \in \mathbb{N}_0^d$ and let $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ for each $\alpha \in D$. A tensor $\mathbf{v} \in \mathbf{V}_D$ is said to be in *Tucker format* with subspace tuple $(U_\alpha)_{\alpha\in D}$ if

$$\mathbf{v} \in {}_a\bigotimes_{\alpha\in D} U_\alpha.$$

The *Tucker rank* of $\mathbf{v}$ is the tuple $\mathfrak{r}(\mathbf{v}) = (r_\alpha(\mathbf{v}))_{\alpha\in D}$, and the set of tensors with fixed Tucker rank $\mathfrak{r}$ is

$$\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \coloneqq \Big\{\mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in D\Big\}. \tag{3.6}$$

The Tucker format thus represents $\mathbf{v}$ as an element of a subspace $U_1 \otimes \cdots \otimes U_d$ spanned by r_α-dimensional subspaces $U_\alpha \subset V_\alpha$. Unlike the subspaces, the representation is not unique; the set $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ collects all tensors for which the *minimal* such subspaces have dimension exactly r_α.

Admissible Tucker ranks

Definition 3.3.3. A tuple $\mathfrak{r} = (r_\alpha)_{\alpha\in D} \in \mathbb{N}_0^d$ is an *admissible Tucker rank* for $\mathbf{V}_D$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $r_\alpha(\mathbf{v}) = r_\alpha$ for all $\alpha \in D$. The set of all admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) := \Big\{(r_\alpha(\mathbf{v}))_{\alpha\in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D\Big\}. \tag{3.7}$$

The zero tuple $\mathbf{0} = (0, \dots, 0)$ is always admissible (corresponding to $\mathbf{v} = 0$). The tuple $\mathbf{1} = (1, \dots, 1)$ is admissible if and only if $\dim V_\alpha \geq 1$ for all α. Since the r_α are bounded above by $\dim V_\alpha$ (in finite dimensions), $\mathcal{AD}(\mathbf{V}_D)$ is a finite set when all V_α are finite-dimensional.

The following decomposition is one of the foundational structural facts of the whole theory.

Proposition 3.3.4. *The algebraic tensor space decomposes as a disjoint union:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r}\in\mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D). \tag{3.8}$$

Proof. Since every $\mathbf{v} \in \mathbf{V}_D$ has a well-defined Tucker rank tuple (Chapter 4), the sets $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ cover $\mathbf{V}_D$; they are disjoint by definition of the rank. □

The decomposition (3.8) is the algebraic precursor of the manifold decomposition established in Chapter 8: each component $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ will be shown to carry the structure of an infinite-dimensional $\mathcal{C}^\infty$-Banach manifold.

Explicit representation of Tucker tensors

The following lemma, which combines the definition of $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ with the properties of minimal subspaces, gives three equivalent characterisations of Tucker tensors. It is the algebraic heart of the Tucker manifold construction.

Lemma 3.3.5. *Let $\mathbf{V}_D = {}_a\bigotimes_{\alpha\in D} V_\alpha$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following conditions are equivalent.*

(a) $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$.

(b) (Tucker representation) *For each $\alpha \in D$ there exists a basis $\{u^{(\alpha)}_{i_\alpha} : 1 \leq i_\alpha \leq r_\alpha\}$ of $U^{\min}_\alpha(\mathbf{v})$ and a unique full-rank core tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha\in D} r_\alpha}$, once the bases are fixed, such that*

$$\mathbf{v} = \sum_{\substack{1\leq i_\alpha\leq r_\alpha\\ \alpha\in D}} C^{(D)}_{(i_\alpha)_{\alpha\in D}} \bigotimes_{\alpha\in D} u^{(\alpha)}_{i_\alpha}. \tag{3.9}$$

(c) (Bilinear form) *For each $\alpha \in D$ there exist linearly independent vectors $\{u^{(\alpha)}_{i_\alpha}\}_{i_\alpha=1}^{r_\alpha}$ in V_α and linearly independent vectors $\{\mathbf{U}^{(\alpha)}_{i_\alpha}\}_{i_\alpha=1}^{r_\alpha}$ in $\mathbf{V}_\alpha^{[\alpha]}$ such that*

$$\mathbf{v} = \sum_{i_\alpha=1}^{r_\alpha} u^{(\alpha)}_{i_\alpha} \otimes \mathbf{U}^{(\alpha)}_{i_\alpha}. \tag{3.10}$$

Moreover, when (3.9) holds, the vectors $\mathbf{U}^{(\alpha)}_{i_\alpha}$ are given by

$$\mathbf{U}^{(\alpha)}_{i_\alpha} = \sum_{\substack{1\le i_\beta\le r_\beta\\ \beta\in D\setminus\{\alpha\}}} C^{(D)}_{i_\alpha,(i_\beta)_{\beta\neq\alpha}} \bigotimes_{\beta\neq\alpha} u^{(\beta)}_{i_\beta}. \tag{3.11}$$

Proof. **(a) ⇔ (c).** If (a) holds, then by Proposition 4.2.2 (Chapter 4), $\mathbf{v}\in U^{\min}_\alpha(\mathbf{v})\otimes_a U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v})$ with $\dim U^{\min}_\alpha(\mathbf{v}) = \dim U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v}) = r_\alpha$, which gives (c). Conversely, (c) implies $\dim U^{\min}_\alpha(\mathbf{v}) = r_\alpha$ for all α, hence (a).

(b) ⇒ (c). Immediate.

(c) ⇒ (b). From (c), one has $U^{\min}_\alpha(\mathbf{v}) = \operatorname{span}\{u^{(\alpha)}_{i_\alpha}\}$ for each α, and since $\mathbf{v}\in {}_a\bigotimes_{\alpha\in D} U^{\min}_\alpha(\mathbf{v})$, there exists $C^{(D)}\in\mathbb{R}^{\times_{\alpha\in D} r_\alpha}$ satisfying (3.9). From (3.11) and the linear independence of the $\mathbf{U}^{(\alpha)}_{i_\alpha}$, one deduces $\operatorname{rank}\mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for all α, so $C^{(D)}\in\mathbb{R}^{\times_{\alpha\in D} r_\alpha}_*$. □

Remark 3.3.6. From Lemma 3.3.5 and Remark 3.2.3, once bases of the minimal subspaces $U^{\min}_\alpha(\mathbf{v})$ are fixed ($\alpha\in D$), there is a natural identification

$$\mathfrak{M}_{\mathfrak{r}}\Big)\Big({}_a \bigotimes_{\alpha\in D} U^{\min}_\alpha(\mathbf{v})\Big) \xrightarrow{\sim} \mathbb{R}^{\times_{\alpha\in D} r_\alpha}_*, \qquad \mathbf{u}\mapsto E^{(D)}, \tag{3.12}$$

where $E^{(D)}$ is the coordinate array of $\mathbf{u}$ in the chosen bases. This identification is a diffeomorphism (both directions are polynomial), and it is the local model for the Tucker manifold chart; see Section 8.2.

The Tucker format as a flat case of tree-based formats

The Tucker format corresponds to the *shallowest* possible hierarchical structure: the dimension partition tree T^{Tucker}_D of depth 1, whose root is D and whose children are the singletons $\{1\},\ldots,\{d\}$:

$$T^{\text{Tucker}}_D = \Big\{D,\{1\},\{2\},\ldots,\{d\}\Big\}, \qquad S(D) = \Big\{\{1\},\ldots,\{d\}\Big\} = \mathcal{L}(T_D)[T^{\text{Tucker}}_D].$$

In Chapter 6 we will define tree-based formats for arbitrary dimension partition trees T_D and show that $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) = \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T^{\text{Tucker}}_D)$.

3.4 The tensor product map and its differential

When the component spaces are normed, the tensor product map inherits both topological and differential properties that underpin the manifold constructions in Part III.

Definition 3.4.1. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$. The *tensor product map*

$$\bigotimes : \prod_{\alpha\in D} (V_\alpha, \|\cdot\|_\alpha) \longrightarrow (\mathbf{V}_D, \|\cdot\|_D), \qquad (v_\alpha)_{\alpha\in D} \mapsto \bigotimes_{\alpha\in D} v_\alpha, \tag{3.13}$$

(where the product is equipped with the max norm $\|(v_\alpha)\|_\times = \max_\alpha \|v_\alpha\|_\alpha$) is said to be *continuous* if $\|\bigotimes_\alpha v_\alpha\|_D \le C \prod_\alpha \|v_\alpha\|_\alpha$ for some constant $C > 0$.

Continuity of the tensor product map is equivalent to $\|\bigotimes\| < \infty$; it holds whenever $\|\cdot\|_D$ is a crossnorm (Definition 4.5.1) on $\mathbf{V}_D$.

Proposition 3.4.2. *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ such that the tensor product map (3.13) is continuous. Then:*

(i) *The map (3.13) is $\mathcal{C}^\infty$-Fréchet differentiable, with differential*

$$D\bigotimes(v_1, \dots, v_d)(w_1, \dots, w_d) = \sum_{\alpha\in D} v_1 \otimes \cdots \otimes v_{\alpha-1} \otimes w_\alpha \otimes v_{\alpha+1} \otimes \cdots \otimes v_d. \tag{3.14}$$

(ii) *The tensor product of bounded linear maps,*

$$\bigotimes : \prod_{\alpha\in D} \mathcal{L}(U_\alpha, V_\alpha) \longrightarrow \mathcal{L}\Big({}_a\bigotimes_{\alpha\in D} U_\alpha,\, \mathbf{V}_D\Big), \quad (A_\alpha) \mapsto \bigotimes_{\alpha\in D} A_\alpha,$$

is also continuous and $\mathcal{C}^\infty$-Fréchet differentiable for any finite-dimensional subspaces $U_\alpha \subset V_\alpha$.

Proof. Part (i) follows directly from Proposition 2.5.2, since $\bigotimes$ is a continuous multilinear map. The formula for the differential follows by the Leibniz rule for multilinear maps. For (ii), one checks boundedness of the map using the continuity of (3.13):

$$\|\bigotimes_\alpha A_\alpha\|_{\mathbf{V}_D \leftarrow \bigotimes U_\alpha} \le C \prod_\alpha \|A_\alpha\|_{V_\alpha \leftarrow U_\alpha},$$

where C is the constant in Definition 3.4.1, and the finite-dimensionality of U_α ensures $L(U_\alpha, V_\alpha) = \mathcal{L}(U_\alpha, V_\alpha)$. Fréchet differentiability then follows again from Proposition 2.5.2. $\square$

Remark 3.4.3. Proposition 3.4.2 is the reason why the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is $\mathcal{C}^\infty$ (not merely continuous): all coordinate change maps between charts are built from compositions of projections and the tensor product map, all of which are $\mathcal{C}^\infty$-Fréchet differentiable. See Section 8.3 for the precise argument.

Banach tensor spaces

When a norm is fixed on $\mathbf{V}_D$, its completion is the natural ambient space.

Definition 3.4.4. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D = {}_a\bigotimes_{\alpha\in D} V_\alpha$. The *Banach tensor space* associated to this data is the completion

$$\mathbf{V}_{D,\|\cdot\|_D} := {}_{\|\cdot\|_D}\bigotimes_{\alpha\in D} V_\alpha := \overline{\mathbf{V}_D}^{\|\cdot\|_D}.$$

If $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space, it is called a *Hilbert tensor space*.

Example 3.4.5. For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 \le p < \infty$, the Lebesgue spaces $L^p(I_\alpha)$ satisfy

$$L^p\Big(\prod_{\alpha\in D} I_\alpha\Big) = {}_{\|\cdot\|_{0,p}}\bigotimes_{\alpha\in D} L^p(I_\alpha),$$

where $\|\cdot\|_{0,p}$ denotes the standard L^p norm on the product domain. For $p = 2$ this is a Hilbert tensor space.

Example 3.4.6. For Sobolev spaces $H^{N,p}(I_\alpha)$ with $1 < p < \infty$ and $\mathbf{I} = \prod_\alpha I_\alpha$:

$$H^{N,p}(\mathbf{I}) = {}_{\|\cdot\|_{N,p}}\bigotimes_{\alpha\in D} H^{N,p}(I_\alpha).$$

This is reflexive and separable. For $p = 2$ it is a Hilbert tensor space. These spaces arise naturally in the numerical treatment of high-dimensional PDEs; see Section 4.5 and Section 7.5.

Remark 3.4.7. The algebraic tensor space $\mathbf{V}_D$ is dense in $\mathbf{V}_{D,\|\cdot\|_D}$ by construction. The Tucker manifold $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ lives inside the algebraic space, but its closure (as a subset of $\mathbf{V}_{D,\|\cdot\|_D}$) and its topology as a manifold are determined by the norm $\|\cdot\|_D$ and, in particular, by whether $\|\cdot\|_D$ is a crossnorm. This interplay between algebraic and topological structure is a recurring theme throughout the monograph.

3.5 Historical notes

Algebraic tensor products. The tensor product of vector spaces was introduced in its modern form by Whitney in 1938 and placed within the framework of multilinear algebra by Bourbaki in the 1940s. The presentation here follows the approach of Greub [Gre81], which is well suited to the infinite-dimensional setting: the algebraic tensor product is defined by the universal property, without any choice of basis or norm.

The projective and injective tensor norms were introduced by Grothendieck in his 1953 Résumé, which also established the fundamental inequality $\|\cdot\|_\varepsilon \le \|\cdot\|_D \le \|\cdot\|_\pi$ for any crossnorm $\|\cdot\|_D$, and identified the dual of $V_1 \otimes_\pi V_2$ as the space of bounded bilinear forms on $V_1 \times V_2$. This work, largely unrecognised at the time, later became the foundation of Banach space theory through the notion of tensor norms and operator ideals. A systematic treatment appears in [Hac19], Chapters 3–4.

Crossnorms and tensor product norms. The crossnorm condition (Cross) and the injective norm condition (Inj) are formulated in [FH12] as the minimal hypotheses under which the algebraic structure of minimal subspaces extends to the topological setting. The condition (Inj) (the norm dominates the injective norm) is the natural analogue of the "reasonable crossnorm" condition in Grothendieck's theory and holds for all the standard function-space tensor products (L^p, Sobolev) used in numerical applications; see Remark 4.5.2.

Tucker format and tensor rank. The Tucker decomposition [Tuc66] was introduced as a psychometric tool for three-mode factor analysis. It generalises the singular value decomposition (SVD) to higher-order tensors: just as every matrix $A \in \mathbb{R}^{m\times n}$ can be written as $A = U\Sigma V^\top$ with $U \in \mathbb{R}^{m\times r}$, $V \in \mathbb{R}^{n\times r}$ orthogonal and Σ diagonal, a d-th order Tucker tensor is represented by a core array and a set of factor matrices in each mode.

The notion of tensor rank for $d \geq 3$ is far more subtle than for matrices: the maximum rank of a real $n \times n \times n$ tensor is not known in general, and computing the rank is NP-hard (Håstad, 1990). The minimal-subspace framework of Chapter 4 avoids these difficulties by working with the subspace rank $(r_\alpha)_{\alpha\in D}$ rather than the canonical (CP) rank; the former has a clean algebraic characterisation and leads to a well-posed approximation problem.

Tensor product map and the manifold structure. The differential of the tensor product map (Section 3.4) is the key ingredient in the construction of the tangent space to the Tucker manifold. The formula $D\tau(\mathbf{v})[\delta\mathbf{v}] = \sum_j \tau^{(j)}(\delta v_j)$, where $\tau^{(j)}$ inserts δv_j in the j-th slot, is the multilinear analogue of the product rule and appears in [FHN19] as the starting point for the identification $T_\mathbf{v}\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \cong Z^{(D)}(\mathbf{v})$.

Chapter 4

Minimal Subspaces in Tensor Representations

In Chapter 3 we introduced the Tucker format and defined the minimal subspace $U_\alpha^{\min}(\mathbf{v})$ informally, taking its existence and properties for granted. This chapter provides the complete theory: a discussion of reflexivity and why it is the right hypothesis for best-approximation problems (Section 4.1), existence and uniqueness of minimal subspaces (Section 4.2), the lattice structure and inclusion under refinement of partitions (Section 4.3), admissible ranks and the disjoint decomposition of $\mathbf{V}_D$ (Section 4.4), and the existence of best approximations in Banach tensor spaces (Section 4.5). The results in this chapter are taken from [FH12], expanded with complete proofs and additional examples.

4.1 Reflexivity and best approximations

The main existence result of this chapter (Theorem 4.5.4) requires the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ to be *reflexive*. This section explains why reflexivity is the right hypothesis, states the key characterisation theorem of James, and verifies reflexivity for the tensor spaces that arise in applications.

Why reflexivity?

The standard strategy for proving existence of best approximations in a non-convex set $A \subset X$ is:

(i) take a minimising sequence $(\mathbf{v}_n) \subset A$, $\|\mathbf{u} - \mathbf{v}_n\| \to \inf_{\mathbf{v}\in A} \|\mathbf{u} - \mathbf{v}\|$;

(ii) extract a convergent subsequence;

(iii) show the limit belongs to A.

In step (ii), norm convergence cannot be guaranteed (the set $A = \mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)$ is not compact for infinite-dimensional V_α). The substitute is *weak* convergence: in a reflexive Banach space, every bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian theorem). The norm is then weakly lower semicontinuous, so step (ii)–(iii) go through provided A is weakly closed. Theorem 4.5.3 establishes exactly this for $A = \mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)$ under condition (Inj).

Reflexivity is not merely a technical convenience: it is also *necessary* for certain best-approximation properties, as expressed by the following classical result.

Theorem 4.1.1 (James, [Jam57])**.** *A Banach space X is reflexive if and only if every bounded linear functional $f \in X^*$ attains its norm on the closed unit ball $B_X = \{x \in X : \|x\| \le 1\}$, i.e., there exists $x_f \in B_X$ such that $f(x_f) = \|f\|_{X^*}$.*

Proof. The "only if" direction follows from the Krein–Milman theorem: in a reflexive space B_X is weakly compact, and every continuous linear functional on a weakly compact convex set attains its supremum. For the "if" direction — that norm-attainment of all functionals forces reflexivity — we refer to the original argument of [Jam57], which uses a subtle diagonal construction. A streamlined proof can be found in [Meg98]. □

Remark 4.1.2 (James's theorem and tensor approximation)*.* James's theorem has a direct bearing on tensor approximation: if $\mathbf{V}_{D,\|\cdot\|_D}$ is not reflexive, there exists a linear functional $F \in (\mathbf{V}_{D,\|\cdot\|_D})^*$ that does not attain its norm. By duality, one can construct a tensor $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ whose best approximation from a rank-one slice $\{v \otimes \mathbf{w} : v \in V_\alpha,\ \mathbf{w} \in \mathbf{V}_\alpha^{[\alpha]}\}$ does not exist — the infimum is not achieved. For example, in $L^1([0,1]) \otimes L^1([0,1])$ (which is not reflexive), best rank-r approximations may fail to exist even for simple target functions. This is why Theorem 4.5.4 requires reflexivity.

Characterisations and stability of reflexivity

Proposition 4.1.3 (Stability of reflexivity)**.** *The class of reflexive Banach spaces is stable under:*

(i) *Closed subspaces: every closed subspace of a reflexive space is reflexive.*

(ii) *Quotients: every quotient of a reflexive space by a closed subspace is reflexive.*

(iii) *Finite direct sums: $X \oplus Y$ is reflexive if and only if both X and Y are reflexive.*

(iv) *Isomorphisms: reflexivity is preserved under bounded linear isomorphisms.*

Proof. Statements (i)–(iv) are classical; see [Meg98, Theorem 1.11.17]. Property (i) will be used in Chapter 10: the tangent spaces $Z^{(D)}(\mathbf{v})$, being closed subspaces of the reflexive space $\mathbf{V}_{D,\|\cdot\|_D}$, are themselves reflexive, ensuring that the metric projection onto $Z^{(D)}(\mathbf{v})$ exists. □

Reflexivity of tensor product spaces

The key question for applications is whether the ambient tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive when the component spaces V_α are.

Proposition 4.1.4 (Reflexivity of tensor products). *Let $(V_\alpha, \|\cdot\|_\alpha)$ be Banach spaces for $\alpha \in D$ and let $\|\cdot\|_D$ be a reasonable crossnorm on $\mathbf{V}_D$.*

(i) *If all V_α are Hilbert spaces and $\|\cdot\|_D$ is the Hilbert tensor norm, then $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space and hence reflexive.*

(ii) *If all $V_\alpha = L^p(I_\alpha)$ with $1 < p < \infty$ and $\|\cdot\|_D = \|\cdot\|_{L^p}$, then $\mathbf{V}_{D,\|\cdot\|_D} \cong L^p(\prod_\alpha I_\alpha)$ is reflexive.*

(iii) *If all V_α are reflexive and $\|\cdot\|_D$ is the* projective norm *$\|\cdot\|_\pi$, then $\mathbf{V}_{D,\|\cdot\|_D}$ is in general* not *reflexive, even for $d = 2$ and $V_1 = V_2 = \ell^2$.*

Proof. Part (i) follows because the Hilbert tensor norm makes $\mathbf{V}_{D,\|\cdot\|_D}$ a Hilbert space (completion of the algebraic tensor product under the inner product $\langle \bigotimes u_\alpha, \bigotimes v_\alpha \rangle = \prod_\alpha \langle u_\alpha, v_\alpha \rangle_\alpha$). Part (ii) follows from the Fubini isomorphism $L^p(I_1) \otimes \cdots \otimes L^p(I_d) \cong L^p(\prod I_\alpha)$ and the reflexivity of L^p for $1 < p < \infty$. For part (iii): $\ell^2 \otimes_\pi \ell^2$ contains an isomorphic copy of ℓ^1 (via the trace-class operators), which is not reflexive; see [Rya02, Example 2.19]. $\square$

Remark 4.1.5 (Which norm to choose?). Proposition 4.1.4 shows that the choice of norm $\|\cdot\|_D$ on $\mathbf{V}_D$ matters for reflexivity. For the purposes of this monograph, we work with norms satisfying (Inj) that also make $\mathbf{V}_{D,\|\cdot\|_D}$ reflexive. The typical examples — L^p spaces with $1 < p < \infty$, Sobolev spaces $W^{k,p}$ with $1 < p < \infty$, and Hilbert spaces — all fall into this category. The projective norm is excluded by Proposition 4.1.4(iii), and so are the endpoint cases L^1 and L^∞.

Dual characterisation of the injective norm

The injective norm condition (Inj) has a natural dual interpretation that will be used in the proof of Theorem 4.5.3.

Proposition 4.1.6 (Dual characterisation of $\|\cdot\|_\varepsilon$). *For any $\mathbf{v} \in \mathbf{V}_D$:*

$$\|\mathbf{v}\|_\varepsilon = \sup\Big\{ \Big| (\textstyle\bigotimes_{\alpha \in D} \varphi_\alpha)(\mathbf{v}) \Big| : \varphi_\alpha \in V_\alpha^*,\ \|\varphi_\alpha\|_{V_\alpha^*} \le 1 \Big\}. \tag{4.1}$$

In particular, if $\|\cdot\|_D$ satisfies (Inj), then for any $\varphi_\alpha \in V_\alpha^*$:

$$\|\varphi_1 \otimes \cdots \otimes \varphi_d\|_{(\mathbf{V}_{D,\|\cdot\|_D})^*} \le \prod_{\alpha \in D} \|\varphi_\alpha\|_{V_\alpha^*}. \tag{4.2}$$

Proof. Equation (4.1) is the definition of the injective norm. For (4.2): since $|(\bigotimes_\alpha \varphi_\alpha)(\mathbf{v})| \leq \|\mathbf{v}\|_\varepsilon \cdot \prod_\alpha \|\varphi_\alpha\|$ (from the definition) and $\|\mathbf{v}\|_\varepsilon \leq \|\mathbf{v}\|_D$ (from (Inj)), we get $|(\bigotimes_\alpha \varphi_\alpha)(\mathbf{v})| \leq \prod_\alpha \|\varphi_\alpha\| \cdot \|\mathbf{v}\|_D$, which is exactly (4.2).

That $\|\mathbf{v}\|_\varepsilon = 0$ implies $\mathbf{v} = 0$ follows from the Hahn–Banach theorem: for any $\mathbf{v} \neq 0$, by Corollary B.7.2 there exists a functional that separates $\mathbf{v}$ from 0, and this functional can be chosen as a tensor product of functionals (using the tensor product structure of $\mathbf{V}_D$). □

Remark 4.1.7 (Role of (Inj) in the proof of weak closedness). Proposition 4.1.6 is the precise reason why (Inj) appears as a hypothesis in Theorem 4.5.3: it ensures that the contraction operators $T_\varphi = \overline{\varphi \otimes \mathrm{id}_{\mathbf{V}_\alpha^{[\alpha]}}}$ are bounded on $\mathbf{V}_{D,\|\cdot\|_D}$, which is needed for the weak continuity argument in Step 1 of the proof. Without (Inj), the contraction operators need not extend to bounded operators on $\mathbf{V}_{D,\|\cdot\|_D}$, and the weak convergence argument breaks down.

4.2 Existence and uniqueness of minimal subspaces

The case of two factors $\mathbf{V}_D = V_1 \otimes_a V_2$ is classical and serves as the inductive base. The key observation is a lattice property of tensor subspaces.

Lemma 4.2.1 (Intersection lemma)**.** *Let X_i, Y_i be subspaces of V_i for $i = 1, 2$. Then*

$$(X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2) = (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2). \tag{4.3}$$

Proof. The inclusion $(X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2) \subseteq (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ is immediate. For the reverse inclusion, let $\mathbf{v} \in (X_1 \otimes_a X_2) \cap (Y_1 \otimes_a Y_2)$ with representations

$$\mathbf{v} = \sum_{\nu=1}^{n_x} x_\nu^{(1)} \otimes x_\nu^{(2)} = \sum_{\nu=1}^{n_y} y_\nu^{(1)} \otimes y_\nu^{(2)}, \quad x_\nu^{(i)} \in X_i,\ y_\nu^{(i)} \in Y_i.$$

By adding terms with zero coefficient, assume $n_x = n_y = n$ and that $\{x_\nu^{(1)}\}$ and $\{y_\nu^{(1)}\}$ are each linearly independent (passing to a minimal representation). The two representations yield the same matrix $M_1(\mathbf{v})$ under any identification with a bilinear form, so $\mathrm{span}\{x_\nu^{(1)}\} = \mathrm{span}\{y_\nu^{(1)}\} \subset X_1 \cap Y_1$. A symmetric argument for the second factor gives $\mathrm{span}\{x_\nu^{(2)}\} \subset X_2 \cap Y_2$, so $\mathbf{v} \in (X_1 \cap Y_1) \otimes_a (X_2 \cap Y_2)$. □

Proposition 4.2.2 (Minimal subspace: case $d = 2$)**.** *Let $\mathbf{v} \in V_1 \otimes_a V_2$. There exist unique subspaces $U_i \subset V_i$ of finite dimension such that:*

(i) $\mathbf{v} \in U_1 \otimes_a U_2$.

(ii) *If $\mathbf{v} \in X_1 \otimes_a X_2$ for any subspaces X_i, then $U_i \subset X_i$.*

Moreover, $\dim U_1 = \dim U_2$.

Proof. **Existence.** Fix any representation $\mathbf{v} = \sum_\nu v_\nu^{(1)} \otimes v_\nu^{(2)}$ and set $U_i = \operatorname{span}\{v_\nu^{(i)}\}$. By passing to a representation with linearly independent $v_\nu^{(1)}$, one can also take $v_\nu^{(2)}$ linearly independent (otherwise row reduce). Then $\dim U_1 = \dim U_2$ (both equal the rank of the bilinear form).

Minimality. If $\mathbf{v} \in X_1 \otimes_a X_2$, then by the intersection lemma, $\mathbf{v} \in (U_1 \cap X_1) \otimes_a (U_2 \cap X_2)$, which forces $U_i \subset X_i$ by minimality of U_i.

Uniqueness. If U_i and U_i' both satisfy (i)–(ii), minimality of U_i applied to U_i' gives $U_i \subset U_i'$, and vice versa. □

The general case: existence and uniqueness

Theorem 4.2.3 (Existence and uniqueness of minimal subspaces)**.** *Let* $D = \{1, \ldots, d\}$, $\mathbf{V}_D = {}_a\bigotimes_{\alpha \in D} V_\alpha$, *and* $\mathbf{v} \in \mathbf{V}_D$. *For each* $\alpha \in D$, *under the identification* $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, *there exists a unique subspace* $U_\alpha^{\min}(\mathbf{v}) \subset V_\alpha$ *of finite dimension* $r_\alpha < \infty$ *such that:*

(a) $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a \mathbf{V}_\alpha^{[\alpha]}$.

(b) *If* $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ *for any subspace* $U_\alpha \subset V_\alpha$, *then* $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$.

Furthermore, there exists a unique subspace $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha^{[\alpha]}$ *of dimension* r_α *such that* $\mathbf{v} \in U_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$.

Proof. Apply Proposition 4.2.2 to the two-factor product $V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$. The subspace $U_\alpha^{\min}(\mathbf{v})$ is the minimal subspace in the V_α-factor. Uniqueness follows from the same argument as in $d = 2$. □

Definition 4.2.4. The subspace $U_\alpha^{\min}(\mathbf{v})$ of Theorem 4.2.3 is called the *minimal subspace* of $\mathbf{v}$ in direction α. The integer $r_\alpha(\mathbf{v}) := \dim U_\alpha^{\min}(\mathbf{v})$ is the α-*rank* of $\mathbf{v}$.

Characterisation via contractions

The following characterisation, which is central to both the proof of Chapter 8 and the tree-based theory of Chapter 6, expresses $U_\alpha^{\min}(\mathbf{v})$ as the range of a family of contractions.

Theorem 4.2.5 (Contraction characterisation)**.** *Let* $\mathbf{v} \in \mathbf{V}_D$ *and* $\alpha \in D$. *Then*

$$U_\alpha^{\min}(\mathbf{v}) = \operatorname{span}\Big\{(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (\mathbf{V}_\alpha^{[\alpha]})^*\Big\}, \tag{4.4}$$

where $(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in V_\alpha$ *is the contraction of* $\mathbf{v}$ *with the functional* $\varphi^{[\alpha]}$ *in all directions except* α.

Proof. Write $\mathbf{v} = \sum_\nu u_\nu \otimes \mathbf{U}_\nu$ with $u_\nu \in V_\alpha$ and $\mathbf{U}_\nu \in \mathbf{V}_\alpha^{[\alpha]}$. For $\varphi \in (\mathbf{V}_\alpha^{[\alpha]})^*$: $(\mathrm{id}_{V_\alpha} \otimes \varphi)(\mathbf{v}) = \sum_\nu \varphi(\mathbf{U}_\nu) u_\nu \in \mathrm{span}\{u_\nu\} = U_\alpha^{\min}(\mathbf{v})$. Conversely, by taking $\varphi = \varphi_\nu$ dual to a basis of $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$, one recovers a basis of $U_\alpha^{\min}(\mathbf{v})$, so equality holds. □

Remark 4.2.6. In Theorem 4.2.5, one can replace $(\mathbf{V}_\alpha^{[\alpha]})^*$ by $(U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}))^*$ without changing the span, since $\mathbf{v} \in V_\alpha \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$. This gives the tighter formula

$$U_\alpha^{\min}(\mathbf{v}) = \mathrm{span}\Big\{(\mathrm{id}_{V_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}))^*\Big\}, \tag{4.5}$$

which is often more useful in practice.

4.3 Lattice properties and inclusion under partitions

Monotonicity

Proposition 4.3.1 (Monotonicity of minimal subspaces)**.** *Let* $\mathbf{v} \in \mathbf{V}_D$ *and* $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ *for each* $\alpha \in D$*. Then:*

$$\mathbf{v} \in {}_a\bigotimes_{\alpha\in D} U_\alpha \quad \Longrightarrow \quad U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha \quad \forall\, \alpha \in D.$$

Proof. Under $\mathbf{v} \in {}_a\bigotimes_{\alpha\in D} U_\alpha$, the identification $\mathbf{V}_D \cong V_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ gives $\mathbf{v} \in U_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$, so the minimality of $U_\alpha^{\min}(\mathbf{v})$ (Theorem 4.2.3(b)) yields $U_\alpha^{\min}(\mathbf{v}) \subset U_\alpha$. □

Inclusion under partition refinement

The following proposition is the key algebraic fact underlying the tree-based format: it shows that minimal subspaces at a coarser level of a tree are contained in tensor products of minimal subspaces at finer levels.

Proposition 4.3.2 (Inclusion under refinement)**.** *Let* $\mathbf{v} \in \mathbf{V}_D$ *and let* $\alpha \subset D$ *with* $\#(\alpha) \geq 2$*. Let* $\mathcal{P}_\alpha$ *be a non-trivial partition of* α*. Then*

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v}). \tag{4.6}$$

Proof. From the canonical identification $\mathbf{V}_D \cong \mathbf{V}_\alpha \otimes_a \mathbf{V}_\alpha^{[\alpha]}$ and Theorem 4.2.5,

$$U_\alpha^{\min}(\mathbf{v}) = \mathrm{span}\Big\{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) : \varphi^{[\alpha]} \in (U_{\alpha^c}^{\min}(\mathbf{v}))^*\Big\}.$$

For each such element $w_\alpha = (\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})(\mathbf{v}) \in \mathbf{V}_\alpha$, we apply Proposition 4.3.1 to the tensor space $\mathbf{V}_\alpha \cong {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} \mathbf{V}_\beta$. Since $w_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and $\mathbf{V}_\alpha = {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} \mathbf{V}_\beta$, one has $w_\alpha \in {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} U_\beta^{\min}(w_\alpha)$. Moreover, from the minimality of $U_\beta^{\min}(\mathbf{v})$ and the relation between contractions at different levels, $U_\beta^{\min}(w_\alpha) \subset U_\beta^{\min}(\mathbf{v})$ for each $\beta \in \mathcal{P}_\alpha$. Hence $w_\alpha \in {}_a\bigotimes_{\beta\in\mathcal{P}_\alpha} U_\beta^{\min}(\mathbf{v})$, and taking the span over all such w_α gives (4.6). □

Remark 4.3.3. Proposition 4.3.2 applied iteratively at every internal vertex $\alpha \in T_D$ of a dimension partition tree (Chapter 5) gives the fundamental compatibility of minimal subspaces with the tree structure:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \quad \forall\, \alpha \in T_D \setminus \mathcal{L}(T_D). \tag{4.7}$$

This is the algebraic foundation of the tree-based format in Chapter 6.

New characterisation of minimal subspaces

The following result, proved in [FHN21] and used in the construction of tree-based tensor formats, gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for $\alpha \supset \beta$.

Proposition 4.3.4. *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \subset D$ with $\#(\alpha) \geq 2$, and $\mathcal{P}_\alpha$ a non-trivial partition of α. Suppose that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in \mathcal{P}_\alpha$ are normed spaces. Then for each $\beta \in \mathcal{P}_\alpha$:*

$$U_\beta^{\min}(\mathbf{v}) = \operatorname{span}\Big\{(\mathrm{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)})(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}),\ \varphi^{(\alpha\setminus\beta)} \in ({}_a\bigotimes_{\gamma\in\mathcal{P}_\alpha\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma])^*\Big\}. \tag{4.8}$$

Proof. Applying Theorem 4.2.5 twice — first for the full index set D and then for α — one obtains two nested contraction characterisations. The formula (4.8) follows by composing these and using that $\mathbf{v} \in \mathbf{V}_\alpha \otimes_a U_{\alpha^c}^{\min}(\mathbf{v})$ allows the replacement of the ambient dual by that of $U_{\alpha^c}^{\min}(\mathbf{v})$. □

4.4 Admissible ranks and the disjoint decomposition

Admissible Tucker ranks

Recall from Chapter 3 (Definition 3.3.3) that the set of admissible Tucker ranks is

$$\mathcal{AD}(\mathbf{V}_D) = \Big\{(r_\alpha(\mathbf{v}))_{\alpha\in D} \in \mathbb{N}_0^d : \mathbf{v} \in \mathbf{V}_D\Big\}.$$

The following proposition characterises which tuples are admissible.

Proposition 4.4.1. *A tuple $\mathfrak{r} = (r_\alpha)_{\alpha\in D} \in \mathbb{N}_0^d$ belongs to $\mathcal{AD}(\mathbf{V}_D)$ if and only if there exist subspaces $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$ and a tensor $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}({}_a\bigotimes_{\alpha\in D} U_\alpha)$, i.e., a tensor whose α-unfolding has rank exactly r_α for all $\alpha \in D$.*

Proposition 4.4.2. *For $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$:*

(i) $r_\alpha \le \dim V_\alpha$ *for all* $\alpha \in D$.

(ii) $r_\alpha = r_{\alpha^c} := \dim U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v})$ *for all* $\alpha \in D$ *(symmetry of marginal ranks).*

(iii) $r_\alpha \le \prod_{\beta\neq\alpha} r_\beta$ *for all* $\alpha \in D$.

Proof. Statements (i) and (iii) follow from the representation formula (3.9) and rank properties of multi-index arrays. Statement (ii) follows from Proposition 4.2.2 applied to $V_\alpha \otimes_a \mathbf{V}^{[\alpha]}_\alpha$. □

The disjoint decomposition

Theorem 4.2.3 immediately gives the following structural result.

Theorem 4.4.3. *The algebraic tensor space* $\mathbf{V}_D$ *admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r}\in\mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D). \tag{4.9}$$

Each $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *is non-empty for* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$*. The zero tensor* $\mathbf{0}$ *belongs to* $\mathfrak{M}_{\mathbf{0}}(\mathbf{V}_D)$ *and each elementary tensor* $\bigotimes_\alpha v_\alpha$ *(with all* $v_\alpha \neq 0$*) belongs to* $\mathfrak{M}_{\mathbf{1}}(\mathbf{V}_D)$*.*

Remark 4.4.4 (Topology of the decomposition). While $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is algebraically well-defined, its topological structure is non-trivial. The key result — that each $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Banach manifold immersed in the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ — is established in Chapter 8. The immersion requires the condition (Inj) on the norm $\|\cdot\|_D$ (Definition 4.5.1).

Stratification by rank

Definition 4.4.5. For $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, the set of tensors with Tucker rank *bounded by* $\mathfrak{r}$ is

$$\mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D) := \bigsqcup_{\substack{\mathfrak{s}\in\mathcal{AD}(\mathbf{V}_D)\\ \mathfrak{s}\le\mathfrak{r}}} \mathfrak{M}_{\mathfrak{s}}(\mathbf{V}_D) = \Big\{\mathbf{v} \in \mathbf{V}_D : r_\alpha(\mathbf{v}) \le r_\alpha \ \forall\, \alpha \in D\Big\}, \tag{4.10}$$

where $\mathfrak{s} \le \mathfrak{r}$ means $s_\alpha \le r_\alpha$ for all $\alpha \in D$.

The set $\mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)$ is not a linear subspace (it is not closed under addition) but it is closed under scalar multiplication. The key property for applications — that best approximations from $\mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)$ exist in Banach tensor spaces — is proved in Section 4.5.

4.5 Best approximations in Banach tensor spaces

In numerical applications, one needs to approximate a given tensor $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$ by a tensor of bounded rank. The existence of such a best approximation is non-trivial: the set $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ is not convex, so general theorems about proximinal sets do not apply. The proof uses the weak closedness of $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$.

Crossnorms and the injective norm condition

Definition 4.5.1. Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces and $\|\cdot\|_D$ a norm on $\mathbf{V}_D$. We say $\|\cdot\|_D$ is a *crossnorm* if

$$\Big\| \bigotimes_{\alpha \in D} v_\alpha \Big\|_D = \prod_{\alpha \in D} \|v_\alpha\|_\alpha \quad \forall\, v_\alpha \in V_\alpha. \tag{4.11}$$

The *injective norm* $\|\cdot\|_\varepsilon$ on $\mathbf{V}_D$ is defined by

$$\|\mathbf{v}\|_\varepsilon := \sup\Big\{ |(\bigotimes_{\alpha \in D} \varphi_\alpha)(\mathbf{v})| : \varphi_\alpha \in V_\alpha^*,\ \|\varphi_\alpha\|_{V_\alpha^*} \leq 1 \Big\}.$$

We say $\|\cdot\|_D$ satisfies condition (Inj) if $\|\mathbf{v}\|_D \geq \|\mathbf{v}\|_\varepsilon$ for all $\mathbf{v} \in \mathbf{V}_D$.

A crossnorm satisfies (Inj) if and only if it is a *reasonable crossnorm* in the sense that its dual norm is also a crossnorm. The injective norm is the smallest reasonable crossnorm (see [Hac19], Chapter 4).

Remark 4.5.2. The L^p norm on $L^p(\prod_\alpha I_\alpha)$ is a reasonable crossnorm for $1 \leq p \leq \infty$ (see [Hac19], Example 4.72). The Sobolev norm $\|\cdot\|_{(0,1),p}$ on $H^{1,p}(I_1) \otimes_a H^{1,p}(I_2)$ is a crossnorm (Example 4.42 in [Hac19]), hence satisfies (Inj).

Weak closedness of bounded-rank sets

Theorem 4.5.3 (Weak closedness, [FH12, Theorem 5.1])**.** *Let* $(V_\alpha, \|\cdot\|_\alpha)$ *be normed spaces and let* $\|\cdot\|_D$ *be a norm on* $\mathbf{V}_D$ *satisfying condition* (Inj)*. Then the set* $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ *is weakly closed in* $\mathbf{V}_{D,\|\cdot\|_D}$*.*

Proof. Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ converge weakly to $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$. We must show $r_\alpha(\mathbf{v}) \leq r_\alpha$ for all $\alpha \in D$.

Step 1: Weak continuity of contraction operators.

Fix $\alpha \in D$. For each $\varphi \in V_\alpha^*$, the *contraction operator* $T_\varphi := \overline{\varphi \otimes \mathrm{id}_{\mathbf{V}_\alpha^{[\alpha]}}} : \mathbf{V}_{D,\|\cdot\|_D} \to \mathbf{V}_\alpha^{[\alpha]}$ is bounded: for any $\psi \in (\mathbf{V}_\alpha^{[\alpha]})^*$,

$$|\langle T_\varphi(\mathbf{v}), \psi \rangle| = |(\varphi \otimes \psi)(\mathbf{v})| \leq \|\varphi\|_{V_\alpha^*} \|\psi\|_{(\mathbf{V}_\alpha^{[\alpha]})^*} \|\mathbf{v}\|_D,$$

where the last inequality uses Proposition 4.1.6 (4.2). Hence T_φ is bounded with $\|T_\varphi\| \leq \|\varphi\|_{V_\alpha^*}$, so weak convergence $\mathbf{v}_n \rightharpoonup \mathbf{v}$ implies $T_\varphi(\mathbf{v}_n) \rightharpoonup T_\varphi(\mathbf{v})$ weakly in $\mathbf{V}_\alpha^{[\alpha]}$ for all $\varphi \in V_\alpha^*$.

Step 2: Rank is weakly lower semicontinuous.

By the contraction characterisation (Theorem 4.2.5), $U_\alpha^{\min}(\mathbf{v}_n) = \overline{\operatorname{span}\{T_\varphi(\mathbf{v}_n) : \varphi \in V_\alpha^*\}}$ and $\dim U_\alpha^{\min}(\mathbf{v}_n) \le r_\alpha$. Suppose for contradiction that $\dim U_\alpha^{\min}(\mathbf{v}) > r_\alpha$. Then there exist $r_\alpha+1$ linearly independent vectors $w_k = T_{\varphi_k}(\mathbf{v})$, $k = 1, \dots, r_\alpha+1$, in $U_\alpha^{\min}(\mathbf{v})$. Step 1 gives $T_{\varphi_k}(\mathbf{v}_n) \rightharpoonup w_k$ weakly. Since $T_{\varphi_k}(\mathbf{v}_n) \in U_\alpha^{\min}(\mathbf{v}_n)$ and $\dim U_\alpha^{\min}(\mathbf{v}_n) \le r_\alpha$, the vectors $T_{\varphi_1}(\mathbf{v}_n), \dots, T_{\varphi_{r_\alpha+1}}(\mathbf{v}_n)$ are linearly dependent for each n. Linear dependence is preserved under weak limits (it is a closed algebraic condition), so $w_1, \dots, w_{r_\alpha+1}$ are linearly dependent — contradiction. Therefore $r_\alpha(\mathbf{v}) \le r_\alpha$.

Since the argument holds for every $\alpha \in D$, we conclude $\mathbf{v} \in \mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$. □

Existence of best approximations

Theorem 4.5.4 (Best approximation, [FH12, Theorem 5.3])**.** *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces, let $\|\cdot\|_D$ be a norm on $\mathbf{V}_D$ satisfying* (Inj)*, and assume that $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$, there exists $\mathbf{v}_* \in \mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})} \|\mathbf{u} - \mathbf{v}\|_D. \tag{4.12}$$

Proof. Since $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive, any bounded sequence has a weakly convergent subsequence (Eberlein–Šmulian theorem). Let $(\mathbf{v}_n) \subset \mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ be a minimising sequence for the infimum in (4.12). The sequence $(\mathbf{v}_n)$ is bounded (the infimum is attained or approached by a bounded set), so by reflexivity there is a weakly convergent subsequence $\mathbf{v}_{n_k} \rightharpoonup \mathbf{v}_*$. By Theorem 4.5.3, $\mathbf{v}_* \in \mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$. Since the norm is weakly lower semicontinuous, $\|\mathbf{u} - \mathbf{v}_*\|_D \le \liminf_k \|\mathbf{u} - \mathbf{v}_{n_k}\|_D = \inf$, so $\mathbf{v}_*$ is a best approximation. □

Corollary 4.5.5. *For $I_\alpha \subset \mathbb{R}$ ($\alpha \in D$) and $1 < p < \infty$, the Banach tensor space $L^p(\prod_\alpha I_\alpha)$ is reflexive and its norm satisfies* (Inj) *(Example 4.5.2). Hence every $f \in L^p(\prod_\alpha I_\alpha)$ has a best Tucker approximation of any prescribed rank.*

Corollary 4.5.6. *For $1 < p < \infty$, $N \ge 1$, and bounded intervals I_α:*

(i) *$H^{N,p}(\prod_\alpha I_\alpha)$ with the standard Sobolev norm is reflexive and satisfies* (Inj).

(ii) *The space $H^{1,p}(I_1) \otimes_{\|\cdot\|_{(0,1),p}} H^{1,p}(I_2)$ with the partial-derivative crossnorm satisfies* (Inj).

In both cases, best Tucker approximations of any rank exist.

Example 4.5.7. Consider $d = 2$, $V_1 = H^1(0,1)$, $V_2 = H^1(0,1)$, and the ambient space $V_1 \otimes_{H^1} V_2$ with the norm $\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|\partial_{x_2} f\|_{L^2}^2$. This Banach tensor space is reflexive and satisfies (Inj), so every $f \in V_1 \otimes_{H^1} V_2$ has a best approximation of rank $\le r$ for any $r \ge 1$. This setting arises in parametric PDEs where x_1 is the spatial variable and x_2 is the stochastic parameter.

Extension to intersections

In many applications (including the tree-based format), the ambient space is an intersection of Banach tensor spaces, each with its own norm.

Theorem 4.5.8 (Best approximation in intersections, [FH12, Theorem 5.6]). *Let $\mathbf{V}^{(1)}_{D,\|\cdot\|_D}, \ldots, \mathbf{V}^{(k)}_{D,\|\cdot\|_D}$ be reflexive Banach tensor spaces over the same algebraic tensor space $\mathbf{V}_D$, each with a norm satisfying* (Inj)*. Equip the intersection $Z = \mathbf{V}^{(1)}_{D,\|\cdot\|_D} \cap \cdots \cap \mathbf{V}^{(k)}_{D,\|\cdot\|_D}$ with the norm $\|\mathbf{v}\|_Z = \max_i \|\mathbf{v}\|_i$. If Z is reflexive (which holds, e.g., when $k = 2$ and both norms are reflexive), then every $\mathbf{u} \in Z$ has a best approximation in $\mathfrak{M}_{\leq \mathfrak{r}}(Z)$.*

4.6 Historical notes and comparison with the literature

The notion of minimal subspace was introduced in [FH12]. It generalises to infinite dimensions the classical concept of matrix rank: for $d = 2$ and finite-dimensional V_α, the α-rank $r_\alpha(\mathbf{v}) = \operatorname{rank}\mathcal{M}_\alpha(\mathbf{v})$ coincides with the classical matrix rank of the α-unfolding, introduced by Hitchcock [Hit27] in 1927. The higher-order version for finite-dimensional spaces was studied by Tucker [Tuc66], De Lathauwer et al. [DLDMV00], and Kolda–Bader [KB09].

The infinite-dimensional theory of [FH12] is motivated by the numerical treatment of high-dimensional PDEs, where $V_\alpha = L^2(\mathbb{R}^3)$ or Sobolev spaces, and the ambient tensor space is infinite-dimensional. The existence of best approximations (Theorem 4.5.4) removes a fundamental obstacle in the convergence analysis of alternating least-squares and related iterative methods.

The weak closedness argument of Theorem 4.5.3 is modelled on Uschmajew's result for the case of the spectral norm on finite-dimensional spaces [UV13a]. The innovation of [FH12] is the extension to general reflexive Banach spaces under the crossnorm condition (Inj), which is the natural condition for tensor product norms.

The connection between minimal subspaces and the geometry of Tucker manifolds is developed in Chapter 8, where the rank map $\varrho : \mathbf{V}_D \to \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ defined by $\varrho(\mathbf{v}) = (U^{\min}_\alpha(\mathbf{v}))_{\alpha \in D}$ plays a central role in the construction of local charts.

Part II

Tree-Based Tensor Formats

Chapter 5

Dimension Partition Trees

The Tucker format developed in Chapters 3 and 4 corresponds to the shallowest possible hierarchical structure: all d factor spaces are placed at the same level below the root. Richer formats arise by organising the dimensions into a *hierarchy*, encoded by a rooted tree. This chapter introduces dimension partition trees, establishes their basic properties, and presents the principal examples (Tucker, Hierarchical Tucker, Tensor Train). The material prepares the ground for the algebraic (Chapter 6) and topological (Chapter 7) development of tree-based tensor formats.

Throughout, $D = \{1, \dots, d\}$ with $d \geq 2$.

5.1 Definition and basic properties

Definition 5.1.1 (Dimension partition tree)**.** A finite rooted tree T_D is called a *dimension partition tree of D* if:

(a) Every vertex $\alpha \in T_D$ is a non-empty subset of D.

(b) The root of T_D is D itself.

(c) Every non-leaf vertex $\alpha \in T_D$ (i.e., every α with $\#(\alpha) \geq 2$) has at least two children, and the set of children of α, denoted $S(\alpha)$, forms a *non-trivial partition* of α (i.e., a partition into at least two non-empty subsets whose union is α).

(d) Every vertex α with $\#(\alpha) = 1$ has no children; such vertices are called *leaves*.

The set of all leaves of T_D is denoted $\mathcal{L}(T_D)$. Condition (c) forces every non-leaf to have at least two children, so the tree is never a path of singletons. The following observation is immediate from the definition.

Proposition 5.1.2. *For any dimension partition tree T_D, the set of leaves coincides with the set of singletons of D:*

$$\mathcal{L}(T_D) = \Big\{ \{j\} : j \in D \Big\}.$$

In particular, $\mathcal{L}(T_D)$ is the trivial partition of D into singletons.

Proof. By (d), leaves are exactly the vertices with $\#(\alpha) = 1$, i.e., the singletons $\{j\}$ for $j \in D$. Since the root is D and each non-leaf splits into a partition of its label set, every $j \in D$ is eventually reached as a leaf by iterating the partition. □

Levels, depth, and level-partitions

Definition 5.1.3. The *level* of a vertex $\alpha \in T_D$ is defined recursively by

$$\mathrm{level}(D) = 0, \qquad \mathrm{level}(\beta) = \mathrm{level}(\alpha) + 1 \quad \text{for } \beta \in S(\alpha).$$

The *depth* of the tree is

$$\mathrm{depth}(T_D) := \max_{\alpha \in T_D} \mathrm{level}(\alpha). \tag{5.1}$$

Since leaves are the only vertices with $\#(\alpha) = 1$, and every path from root to leaf passes through a strictly decreasing chain of subset sizes, the depth satisfies $1 \leq \mathrm{depth}(T_D) \leq d - 1$.

Definition 5.1.4. For $1 \leq k \leq \mathrm{depth}(T_D)$, the *$k$-th level partition* of D induced by T_D is

$$\mathcal{P}_k(T_D) := \Big\{\alpha \in T_D : \mathrm{level}(\alpha) = k\Big\}. \tag{5.2}$$

By convention, $\mathcal{P}_0(T_D) = \{D\}$.

Proposition 5.1.5. *For each $1 \leq k \leq \mathrm{depth}(T_D)$, the collection $\mathcal{P}_k(T_D)$ is a partition of D. Moreover:*

(i) *Every element of $\mathcal{P}_k(T_D)$ is a subset of some element of $\mathcal{P}_{k-1}(T_D)$ (refinement property).*

(ii) *$\mathcal{P}_{\mathrm{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$, the trivial partition of D.*

Proof. The elements of $\mathcal{P}_k(T_D)$ are precisely the vertex labels at level k. Since each such vertex α is the child of a unique parent $\beta \in \mathcal{P}_{k-1}(T_D)$ satisfying $\alpha \subset \beta$, and since $S(\beta)$ partitions β, the elements of $\mathcal{P}_k(T_D)$ at level k partition D by induction on k. Parts (i) and (ii) follow directly. □

The sequence of partitions $\mathcal{P}_0(T_D) \succ \mathcal{P}_1(T_D) \succ \cdots \succ \mathcal{P}_{\mathrm{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$ (where $\mathcal{P} \succ \mathcal{Q}$ means $\mathcal{Q}$ refines $\mathcal{P}$) captures the hierarchical structure of the tree. This is the central object connecting dimension partition trees to tensor spaces: for each level k, the identification $\mathbf{V}_D \cong \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ (Equation (3.4)) allows one to view $\mathbf{v} \in \mathbf{V}_D$ as a Tucker tensor at level k.

5.2 Special trees and standard formats

The following three families of dimension partition trees correspond to the three most widely used tensor formats.

The Tucker tree

Example 5.2.1 (Tucker tree)**.** The *Tucker tree* T_D^{Tucker} over D has depth 1 and satisfies

$$T_D^{\text{Tucker}} = \Big\{D,\, \{1\},\, \{2\},\, \dots,\, \{d\}\Big\}, \qquad S(D) = \mathcal{L}(T_D)[T_D^{\text{Tucker}}] = \Big\{\{j\} : j \in D\Big\}.$$

The only non-trivial partition step is $D \to \{\{1\}, \dots, \{d\}\}$.

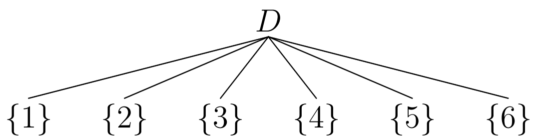

Figure 5.1: Tucker tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 1, one level partition $\mathcal{P}_1 = \mathcal{L}(T_D)$.

The Tucker tree is the unique tree of depth 1 over D. The corresponding tensor format is exactly the Tucker format of Chapter 3: $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

Binary trees and the Hierarchical Tucker format

Definition 5.2.2. A dimension partition tree T_D is *binary* if every non-leaf vertex has exactly two children: $\#(S(\alpha)) = 2$ for all $\alpha \in T_D$ with $\#(\alpha) \geq 2$.

Binary trees are the natural hierarchical structure for the *Hierarchical Tucker* (HT) format introduced by Hackbusch and Kühn [HK09].

Example 5.2.3 (Hierarchical Tucker tree)**.** For $D = \{1, 2, 3, 4\}$, the balanced binary HT tree is

$$T_D^{\text{HT}} = \Big\{D,\, \{1,2\},\, \{3,4\},\, \{1\},\, \{2\},\, \{3\},\, \{4\}\Big\},$$

with $S(D) = \{\{1, 2\}, \{3, 4\}\}$, $S(\{1, 2\}) = \{\{1\}, \{2\}\}$, $S(\{3, 4\}) = \{\{3\}, \{4\}\}$. Here $\text{depth}(T_D^{\text{HT}}) = 2$, with level partitions $\mathcal{P}_1 = \{\{1, 2\}, \{3, 4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D) = \{\{1\}, \{2\}, \{3\}, \{4\}\}$.

Example 5.2.4 (Non-binary tree)**.** For $D = \{1, 2, 3, 4, 5, 6\}$:

$$T_D = \Big\{D,\, \{1,2,3\},\, \{4,5\},\, \{6\},\, \{1\},\, \{2,3\},\, \{4\},\, \{5\},\, \{2\},\, \{3\}\Big\},$$

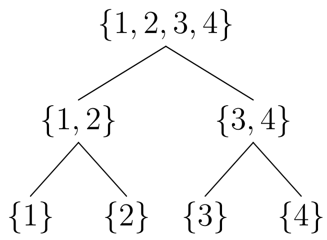

Figure 5.2: Balanced binary tree over $D = \{1, 2, 3, 4\}$: depth 2, corresponding to the HT format.

where $S(D) = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$ (three children), $S(\{1, 2, 3\}) = \{\{1\}, \{2, 3\}\}$, $S(\{2, 3\}) = \{\{2\}, \{3\}\}$, $S(\{4, 5\}) = \{\{4\}, \{5\}\}$. Here $\mathrm{depth}(T_D) = 3$, with $\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5\}, \{6\}\}$, $\mathcal{P}_2 = \{\{1\}, \{2, 3\}, \{4\}, \{5\}, \{6\}\}$, $\mathcal{P}_3 = \mathcal{L}(T_D)$.

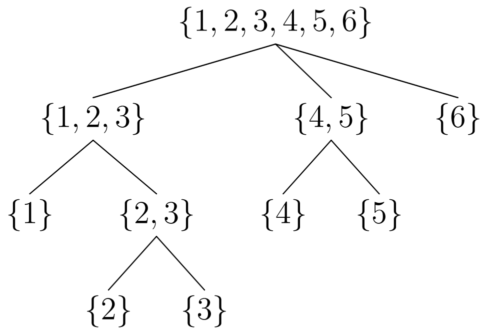

Figure 5.3: Non-binary dimension partition tree over $D = \{1, 2, 3, 4, 5, 6\}$: depth 3.

The Tensor Train tree

Example 5.2.5 (Tensor Train tree)**.** For $D = \{1, \ldots, d\}$, $d \geq 2$, the *Tensor Train* (TT) tree [Ose11] is the linear (chain) binary tree

$$T_D^{\mathrm{TT}} = \{D\} \cup \Big\{\{j\} : 1 \leq j \leq d\Big\} \cup \Big\{\{1, \ldots, j\} : 2 \leq j \leq d - 1\Big\},$$

with $S(\{1, \ldots, j\}) = \{\{1, \ldots, j - 1\}, \{j\}\}$ for $j \geq 2$. Here $\mathrm{depth}(T_D^{\mathrm{TT}}) = d - 1$. The level partitions are:

$$\mathcal{P}_k(T_D^{\mathrm{TT}}) = \Big\{\{1, \ldots, d - k\}, \{d - k + 1\}, \ldots, \{d\}\Big\}, \quad 1 \leq k \leq d - 1.$$

Remark 5.2.6. The TT format is a special case of the HT format: T_D^{TT} is a binary tree with maximal depth $d - 1$ and a linear (path) structure, whereas HT trees are typically balanced with depth $\lceil \log_2 d \rceil$. This is reflected in their computational cost: TT has storage $O(dr^2 n)$ while balanced HT has $O(dr^3)$, with r the maximal rank and $n = \max_j \dim V_j$ [HK09, Ose11].

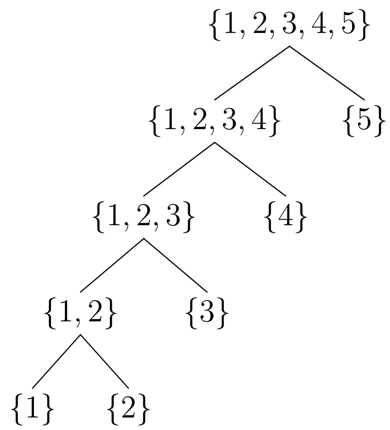

Figure 5.4: Tensor Train tree over $D = \{1, 2, 3, 4, 5\}$: depth 4, linear binary structure.

5.3 Graph-theoretic properties

We record here the combinatorial facts about dimension partition trees that will be used in Chapters 6 and 9.

Proposition 5.3.1. *Let T_D be a dimension partition tree over $D = \{1, \dots, d\}$. Then:*

(i) $\#(\mathcal{L}(T_D)) = d$. *The number of internal (non-leaf) vertices satisfies* $\#(T_D \setminus \mathcal{L}(T_D)) \leq d - 1$.

(ii) *If T_D is binary, then $\#(T_D) = 2d - 1$, with exactly $d - 1$ internal vertices.*

(iii) $\mathrm{depth}(T_D) \leq d - 1$, *with equality if and only if $T_D = T_D^{\mathrm{TT}}$ (up to re-labelling).*

Proof. Part (i): every leaf is a singleton $\{j\}$ for a unique $j \in D$, so $\#(\mathcal{L}(T_D)) = d$. The bound on internal vertices follows since each internal vertex contributes at least two children which are eventually singletons. Part (ii): in a binary tree with d leaves, the number of internal nodes is exactly $d - 1$, giving $2d - 1$ total vertices. Part (iii): the depth is maximised when each step reduces the largest set by exactly one element, giving the TT structure. □

Proposition 5.3.2 (Recursive subtree structure)**.** *Let T_D be a dimension partition tree over D and $\alpha \in T_D$ with $\#(\alpha) \geq 2$. Then the restriction*

$$T_D|_\alpha \coloneqq \{\beta \in T_D : \beta \subseteq \alpha\}$$

is itself a dimension partition tree over α (with root α).

This recursive structure is fundamental to the inductive proofs in Chapter 6: properties of tree-based tensor formats are proved by induction on $\mathrm{depth}(T_D)$, reducing at each step to the subtrees $T_D|_\beta$ for $\beta \in S(D)$.

5.4 The intersection representation

The central structural theorem of the tree-based theory expresses the set of tree-based tensors with fixed rank as an intersection of Tucker manifolds — one at each level of the tree. This is the key result connecting the tree structure to the geometry of Chapters 9.

For each level partition $\mathcal{P}_k(T_D)$, recall from Section 3.3 that the identification $\mathbf{V}_D \cong \bigotimes_{\alpha\in\mathcal{P}_k(T_D)} \mathbf{V}_\alpha$ allows one to define the Tucker set

$$\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) := \Big\{ \mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in \mathcal{P}_k(T_D) \Big\}, \tag{5.3}$$

where $\mathfrak{r}_k = (r_\alpha)_{\alpha\in\mathcal{P}_k(T_D)}$.

Theorem 5.4.1 (Intersection representation, [FHN23, Theorem 3.12]). *Let T_D be a dimension partition tree over D and $\mathfrak{r} = (r_\alpha)_{\alpha\in T_D} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then*

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\operatorname{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)), \tag{5.4}$$

where $\mathfrak{r}_k = (r_\alpha)_{\alpha\in\mathcal{P}_k(T_D)}$.

Proof. We prove both inclusions.

($\supseteq$) Forward inclusion: $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \subseteq \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$.

By definition, $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ means $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for every $\alpha \in T_D \setminus \{D\}$. Since each partition $\mathcal{P}_k(T_D)$ consists of vertices of T_D (at level k, plus any leaves at earlier levels), the condition $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ holds in particular for each $\alpha \in \mathcal{P}_k(T_D)$. Hence $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ for every k, and $\mathbf{v} \in \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$.

($\subseteq$) Reverse inclusion: $\bigcap_k \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) \subseteq \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

Let $\mathbf{v} \in \bigcap_k \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$. We must show that $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for every $\alpha \in T_D \setminus \{D\}$.

We proceed by upward induction on the level of α, starting from the leaves (deepest level) and going toward the root.

Base case (leaves): Every leaf $\alpha \in \mathcal{L}(T_D)$ belongs to the level-depth(T_D) partition $\mathcal{P}_{\operatorname{depth}(T_D)}(T_D) = \mathcal{L}(T_D)$. Since $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}_{\operatorname{depth}(T_D)}}(\mathbf{V}_D, \mathcal{L}(T_D))$, we have $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for all $\alpha \in \mathcal{L}(T_D)$.

Inductive step: Let $\alpha \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ at level $k = \operatorname{level}(\alpha)$, and suppose $\dim U_\beta^{\min}(\mathbf{v}) = r_\beta$ for all $\beta \in S(\alpha)$ (inductive hypothesis at level $k+1$).

Since $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ and $\alpha \in \mathcal{P}_k(T_D)$, we have $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$. (The membership in the k-th Tucker set directly gives the rank condition at level k.)

This establishes $\dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha$ for every internal vertex α at level k, and by induction from the leaves up, for every $\alpha \in T_D \setminus \{D\}$.

Remark on the inductive structure. The key point is that the rank conditions at different levels are not independent: Proposition 4.3.2 (and its tree version (4.7)) gives

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \quad \forall\, \alpha \in T_D \setminus \mathcal{L}(T_D),$$

so the dimension satisfies $r_\alpha = \dim U_\alpha^{\min}(\mathbf{v}) \le \prod_{\beta \in S(\alpha)} r_\beta$. The admissibility condition $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ precisely requires this inequality to hold, so the intersection is non-empty and the inductive argument is consistent. □

Remark 5.4.2. For $T_D = T_D^{\text{Tucker}}$ (depth 1), the intersection (5.4) reduces to a single factor: $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{L}(T_D)) = \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$, recovering the Tucker format.

Example 5.4.3. For $D = \{1,2,3,4\}$ and the balanced HT tree T_D^{HT} of Example 5.2.3 with $\mathfrak{r} = (r_D, r_{\{1,2\}}, r_{\{3,4\}}, r_{\{1\}}, r_{\{2\}}, r_{\{3\}}, r_{\{4\}})$:

$$\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D^{\text{HT}}) = \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1) \cap \mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{P}_2),$$

where $\mathcal{P}_1 = \{\{1,2\},\{3,4\}\}$ and $\mathcal{P}_2 = \mathcal{L}(T_D)$. Concretely:

$$\begin{aligned} \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1) &= \Big\{\mathbf{v} \in \mathbf{V}_D : \dim U_{\{1,2\}}^{\min}(\mathbf{v}) = r_{\{1,2\}},\ \dim U_{\{3,4\}}^{\min}(\mathbf{v}) = r_{\{3,4\}}\Big\}, \\ \mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{P}_2) &= \Big\{\mathbf{v} \in \mathbf{V}_D : \dim U_{\{j\}}^{\min}(\mathbf{v}) = r_{\{j\}},\ j = 1,2,3,4\Big\}. \end{aligned}$$

The intersection captures tensors that are simultaneously Tucker at the level of pairs $(\{1,2\},\{3,4\})$ and at the level of singletons.

Remark 5.4.4 (Degenerate case). Not all intersections define distinct sets. If for some internal vertex $\alpha \in T_D$ the rank r_α equals $\prod_{\beta \in S(\alpha)} r_\beta$, the rank constraint at α is automatically satisfied whenever the constraints at the children are. In such cases one of the Tucker factors in (5.4) is redundant. This situation is analysed in detail in [FHN23, Theorem 3.12].

5.5 Historical notes

Dimension partition trees formalise the hierarchical structure of tensor representations that had appeared in various forms in the literature. The HT format was introduced by Hackbusch and Kühn in [HK09] using a binary tree structure, with the specific aim of reducing the exponential storage of full Tucker tensors to $O(dr^3)$. The TT format, independently introduced by Vidal [Vid03] in quantum information (under the name *matrix product states*) and by Oseledets [Ose11] in numerical mathematics (as *TT decomposition*), corresponds to the linear chain binary tree.

The general notion of a dimension partition tree, unifying these formats under a common combinatorial framework, was introduced in [FHN21]. The intersection representation (Theorem 5.4.1) was established in [FHN23] and is the key to the geometric theory of Chapter 9: it reduces the analysis of the tree-based manifold $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ to the intersection of Tucker manifolds, each of which has the structure established in Chapter 8.

The connection between dimension partition trees and quantum physics is deeper than a mere formal analogy. In quantum information, a tensor network is a collection of tensors connected by contraction along shared indices, and the tree structure encodes the entanglement structure of a quantum state. The area law for entanglement entropy in one-dimensional quantum systems [Vid03] is the physical motivation for the TT format; balanced binary trees encode area laws in higher-dimensional systems. This connection is discussed further in Chapter 11.

Chapter 6

Algebraic Tree-Based Tensor Formats

This chapter develops the algebraic theory of tree-based tensor formats. Given a dimension partition tree T_D over D and a tensor space $\mathbf{V}_D = {}_a\bigotimes_{j\in D} V_j$, we associate to each vertex $\alpha \in T_D$ a subspace $\mathbf{V}_\alpha$ and a minimal subspace $U_\alpha^{\min}(\mathbf{v})$, define the tree-based rank and the sets $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$, and establish the central characterisation theorem (Theorem 6.4.3) expressing every tensor of fixed tree-based rank in terms of a hierarchical system of *transfer tensors*. The chapter closes with a discussion of the special cases Tucker, HT, and TT.

Throughout, T_D is a dimension partition tree over $D = \{1, \ldots, d\}$ with $d \geq 2$, and V_j $(j \in D)$ are vector spaces over $\mathbb{K}$.

6.1 Tree-based tensor spaces

Associated spaces at each vertex

For each $\alpha \in T_D$ we associate a vector space $\mathbf{V}_\alpha$ defined recursively from the leaves upward:

- For leaves $\{j\} \in \mathcal{L}(T_D)$: $\mathbf{V}_{\{j\}} = V_j$.
- For internal vertices $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_\alpha = {}_a\bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$.

Since $S(\alpha)$ partitions α, the canonical identification $\mathbf{V}_\alpha = {}_a\bigotimes_{j\in\alpha} V_j$ holds at every vertex by the associativity of Equation (3.4). In particular, $\mathbf{V}_D = \mathbf{V}_D$.

Definition 6.1.1. The *tensor space in tree-based format* associated to $(\mathbf{V}_D, T_D)$ is the pair $(\mathbf{V}_D, T_D)$ together with the collection of spaces $\{\mathbf{V}_\alpha\}_{\alpha\in T_D\setminus\{D\}}$.

Minimal subspaces compatible with the tree

Recall from Proposition 4.3.2 that for any $\alpha \in T_D \setminus \mathcal{L}(T_D)$ and any $\mathbf{v} \in \mathbf{V}_D$:

$$U_\alpha^{\min}(\mathbf{v}) \subset {}_a \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}) \subset \mathbf{V}_\alpha. \tag{6.1}$$

This compatibility is the algebraic backbone of the tree-based format: the minimal subspace at each internal vertex is contained in the tensor product of the minimal subspaces of its children.

6.2 Tree-based rank and admissible tuples

Definition 6.2.1 (Tree-based rank)**.** For $\mathbf{v} \in \mathbf{V}_D$, its *tree-based rank with respect to* T_D is the tuple

$$\operatorname{rank}_{T_D}(\mathbf{v}) := \left(\dim U_\alpha^{\min}(\mathbf{v})\right)_{\alpha \in T_D} \in \mathbb{N}_0^{\#(T_D)}. \tag{6.2}$$

Definition 6.2.2 (Admissible tuples)**.** A tuple $\mathfrak{r} = (r_\alpha)_{\alpha \in T_D} \in \mathbb{N}^{\#(T_D)}$ is *admissible for* $(\mathbf{V}_D, T_D)$ if there exists $\mathbf{v} \in \mathbf{V}_D$ with $\operatorname{rank}_{T_D}(\mathbf{v}) = \mathfrak{r}$. The set of all admissible tuples is

$$\mathcal{AD}(\mathbf{V}_D, T_D) := \left\{\operatorname{rank}_{T_D}(\mathbf{v}) : \mathbf{v} \in \mathbf{V}_D\right\}. \tag{6.3}$$

Remark 6.2.3. Admissible tuples satisfy the compatibility constraints inherited from Proposition 4.4.2 and Equation (6.1). Specifically, for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$r_\alpha \leq \prod_{\beta \in S(\alpha)} r_\beta. \tag{6.4}$$

Equality in (6.4) is the generic case; strict inequality indicates that the tensor has additional structure at vertex α.

Sets of tensors with fixed and bounded tree-based rank

Definition 6.2.4. Let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Define:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) := \left\{\mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) = r_\alpha \text{ for all } \alpha \in T_D\right\}, \tag{6.5}$$

$$\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) := \left\{\mathbf{v} \in \mathbf{V}_D : \dim U_\alpha^{\min}(\mathbf{v}) \leq r_\alpha \text{ for all } \alpha \in T_D\right\}. \tag{6.6}$$

The relation between these two sets is:

$$\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigsqcup_{\substack{\mathfrak{s} \in \mathcal{AD}(\mathbf{V}_D, T_D) \\ \mathfrak{s} \leq \mathfrak{r}}} \mathcal{FT}_{\mathfrak{s}}(\mathbf{V}_D, T_D). \tag{6.7}$$

Remark 6.2.5 (Tucker as special case). For $T_D = T_D^{\text{Tucker}}$ (depth 1, $S(D) = \mathcal{L}(T_D)$), the tree-based rank reduces to the Tucker rank: $\operatorname{rank}_{T_D^{\text{Tucker}}}(\mathbf{v}) = (r_D, r_{\{1\}}, \ldots, r_{\{d\}})$. Since $r_D = 1$ for all non-zero $\mathbf{v}$ (the root has a single rank), the effective tuple is $(r_{\{1\}}, \ldots, r_{\{d\}})$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D^{\text{Tucker}}) = \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

6.3 Minimal subspaces compatible with a tree: new characterisation

The following proposition, proved in [FHN21], gives a characterisation of $U_\beta^{\min}(\mathbf{v})$ for an internal vertex β in terms of contractions of $U_\alpha^{\min}(\mathbf{v})$ for its parent $\alpha = \operatorname{parent}(\beta)$. This is the tree-based analogue of the contraction characterisation Theorem 4.2.5.

Proposition 6.3.1 (Tree-based characterisation of minimal subspaces, [FHN21, Proposition 2.3])**.** *Let $\mathbf{v} \in \mathbf{V}_D$, $\alpha \in T_D \setminus \mathcal{L}(T_D)$ with $\#(S(\alpha)) \geq 2$, and assume that $\mathbf{V}_\alpha$ and $\mathbf{V}_\beta$ for $\beta \in S(\alpha)$ are normed spaces. Then for each $\beta \in S(\alpha)$:*

$$U_\beta^{\min}(\mathbf{v}) = \operatorname{span}\left\{ \left(\mathrm{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)}\right)(\mathbf{v}_\alpha) : \mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v}),\ \varphi^{(\alpha\setminus\beta)} \in \left({}_a \bigotimes_{\gamma\in S(\alpha)\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma] \right)^* \right\}. \tag{6.8}$$

Proof. This follows directly from Proposition 4.3.4 applied at level k of the tree, with α playing the role of the block and $S(\alpha)$ playing the role of the partition $\mathcal{P}_\alpha$. □

6.4 The transfer tensor representation

The main result of this section is the characterisation of every $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ by a system of *transfer tensors* $\{C^{(\alpha)}\}_{\alpha\in T_D}$, one at each vertex. This is the algebraic heart of the tree-based format and the starting point for all computational algorithms (ALS, DMRG, etc.).

Transfer tensors and the representation formula

Definition 6.4.1. Let $\alpha \in T_D$ with $\#(S(\alpha)) \geq 2$ and let $\mathfrak{r}_\alpha = (r_\beta)_{\beta\in S(\alpha)}$. A *transfer tensor at vertex α* is an array

$$C^{(\alpha)} \in \mathbb{R}^{r_\alpha \times \prod_{\beta\in S(\alpha)} r_\beta},$$

where we view it as a matrix with r_α rows and $\prod_{\beta\in S(\alpha)} r_\beta$ columns (indexed by the multi-index $(i_\beta)_{\beta\in S(\alpha)}$). The transfer tensor $C^{(\alpha)}$ is said to be *full-rank* if $\operatorname{rank}\mathcal{M}_\alpha(C^{(\alpha)}) = r_\alpha$.

For the root vertex D, the transfer tensor $C^{(D)}$ plays a special role:

Definition 6.4.2. The *root transfer tensor* is

$$C^{(D)} \in \mathbb{R}^{\times_{\alpha\in S(D)} r_\alpha},$$

a multi-dimensional array of shape $\prod_{\alpha\in S(D)} r_\alpha$ (no "row" index since the root has no parent). The full-rank condition at the root is $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha\in S(D)} r_\alpha}$, the open set of Definition 3.2.2.

The main characterisation theorem

Theorem 6.4.3 (Transfer tensor characterisation, [FHN21, Theorem 2.6])**.** *Let T_D be a dimension partition tree over D with* $\operatorname{depth}(T_D) = L$*, and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathfrak{r} \neq \mathbf{0}$. For $\mathbf{v} \in \mathbf{V}_D$, the following statements are equivalent.*

(a) $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

(b) *Given bases $\{u^{(k)}_{i_k} : 1 \le i_k \le r_k\}$ of $U^{\min}_k(\mathbf{v})$ for each $k \in \mathcal{L}(T_D)$, there exist:*

- *A unique root transfer tensor $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha\in S(D)} r_\alpha}$ such that*

$$\mathbf{v} = \sum_{\substack{1\le i_\alpha\le r_\alpha\\ \alpha\in S(D)}} C^{(D)}_{(i_\alpha)_{\alpha\in S(D)}} \bigotimes_{\alpha\in S(D)} \mathbf{u}^{(\alpha)}_{i_\alpha}, \tag{6.9}$$

- *For each internal vertex $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, a unique transfer tensor $C^{(\mu)} \in \mathbb{R}^{r_\mu \times \prod_{\beta\in S(\mu)} r_\beta}$ with $\operatorname{rank}\mathcal{M}_\mu(C^{(\mu)}) = r_\mu$, and a basis $\{\mathbf{u}^{(\mu)}_{i_\mu} : 1 \le i_\mu \le r_\mu\}$ of $U^{\min}_\mu(\mathbf{v})$ given by:*

$$\mathbf{u}^{(\mu)}_{i_\mu} = \sum_{\substack{1\le i_\beta\le r_\beta\\ \beta\in S(\mu)}} C^{(\mu)}_{i_\mu,(i_\beta)_{\beta\in S(\mu)}} \bigotimes_{\beta\in S(\mu)} \mathbf{u}^{(\beta)}_{i_\beta}, \quad 1 \le i_\mu \le r_\mu. \tag{6.10}$$

Proof. **(b) ⇒ (a).** Clear: the representation gives $\dim U^{\min}_\alpha(\mathbf{v}) \le r_\alpha$ at each level, and the full-rank conditions force equality.

(a) ⇒ (b). We construct the transfer tensors level by level from the leaves upward.

Step 1 (Root). Since $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v} \in {}_a\bigotimes_{\alpha\in S(D)} U^{\min}_\alpha(\mathbf{v})$ (by the compatibility condition (6.1)), there exists a unique $C^{(D)} \in \mathbb{R}^{\times_{\alpha\in S(D)} r_\alpha}$ such that (6.9) holds (choosing any bases of the $U^{\min}_\alpha(\mathbf{v})$). The full-rank condition $C^{(D)} \in \mathbb{R}_*^{\times_{\alpha\in S(D)} r_\alpha}$ follows from $\dim U^{\min}_\alpha(\mathbf{v}) = r_\alpha$ for $\alpha \in S(D)$: the rank $\operatorname{rank}\mathcal{M}_\alpha(C^{(D)}) = r_\alpha$ for each α because otherwise one could find a subspace of $U^{\min}_\alpha(\mathbf{v})$ of smaller dimension still containing $\mathbf{v}$, contradicting minimality.

Step 2 (Internal vertices). For each $\mu \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ with $S(\mu) \neq \emptyset$, the inclusion $U^{\min}_\mu(\mathbf{v}) \subset {}_a\bigotimes_{\beta\in S(\mu)} U^{\min}_\beta(\mathbf{v})$ (Proposition 4.3.2) gives the existence of $C^{(\mu)}$

satisfying (6.10). The full-rank condition rank$\mathcal{M}_\mu(C^{(\mu)}) = r_\mu$ follows from the linear independence of $\{\mathbf{u}^{(\mu)}_{i_\mu}\}$ (as a basis of $U^{\min}_\mu(\mathbf{v})$). □

Remark 6.4.4 (Uniqueness up to gauge). The transfer tensors $\{C^{(\alpha)}\}$ are uniquely determined once the bases $\{u^{(k)}_{i_k}\}_{k\in\mathcal{L}(T_D)}$ are fixed. Changing the bases introduces a *gauge transformation*: if $\{(\tilde{u}^{(k)}_{i_k})\}$ is another set of bases related by invertible matrices $G_k \in \mathrm{GL}(\mathbb{R}^{r_k})$ (via $\tilde{u}^{(k)}_{i_k} = \sum_j (G_k)_{ij} u^{(k)}_j$), then the transfer tensors transform by contracting with the corresponding G_k. The freedom to choose orthonormal bases (when the V_j are Hilbert spaces) is exploited in the HOSVD and related algorithms.

Recursive computation

Equation (6.10) has a natural recursive interpretation: the basis vectors of $U^{\min}_\mu(\mathbf{v})$ are built bottom-up from the leaf bases, level by level, using the transfer tensors as "gluing" coefficients. The full representation can be written compactly as:

$$\mathbf{v} = \sum_{(i_j)_{j\in D}} \Big[\text{network contraction of } \{C^{(\alpha)}\}_{\alpha\in T_D}\Big]_{(i_j)_{j\in D}} \bigotimes_{j\in D} u^{(\{j\})}_{i_j}, \tag{6.11}$$

where the "network contraction" is the tensor network obtained by connecting the transfer tensors along the edges of T_D.

6.5 Special cases

We verify Theorem 6.4.3 in the three standard formats, recovering known representations.

Tucker format

For $T_D = T^{\text{Tucker}}_D$ (depth 1), $S(D) = \mathcal{L}(T_D) = \{\{j\} : j \in D\}$, and there are no internal vertices other than the root. The only transfer tensor is the root tensor $C^{(D)}$, and Theorem 6.4.3 reduces to:

$$\mathbf{v} = \sum_{\substack{1\le i_j\le r_j\\ j\in D}} C^{(D)}_{(i_j)_{j\in D}} \bigotimes_{j\in D} u^{(\{j\})}_{i_j}, \quad C^{(D)} \in \mathbb{R}^{\times_{j\in D} r_j}_{*}.$$

This is exactly the Tucker representation (3.9) of Lemma 3.3.5, with $C^{(D)}$ playing the role of the core tensor.

Hierarchical Tucker format ($d = 4$, balanced binary)

For T^{HT}_D of Example 5.2.3 with $D = \{1,2,3,4\}$, $S(D) = \{\{1,2\},\{3,4\}\}$, and the two internal vertices $\{1,2\}$ and $\{3,4\}$ each have $S(\{i,j\}) = \{\{i\},\{j\}\}$. Theorem 6.4.3 gives:

- Leaf bases: $\{u^{(\{j\})}_{i_j}\}_{i_j=1}^{r_j}$ for $j = 1, 2, 3, 4$.
- Transfer tensors at level 2: $C^{(\{1,2\})} \in \mathbb{R}^{r_{\{1,2\}} \times r_1 r_2}$ and $C^{(\{3,4\})} \in \mathbb{R}^{r_{\{3,4\}} \times r_3 r_4}$, defining bases of $U^{\min}_{\{1,2\}}(\mathbf{v})$ and $U^{\min}_{\{3,4\}}(\mathbf{v})$:

$$\mathbf{u}^{(\{1,2\})}_{i_{12}} = \sum_{i_1=1}^{r_1} \sum_{i_2=1}^{r_2} C^{(\{1,2\})}_{i_{12}, i_1 i_2}\, u^{(\{1\})}_{i_1} \otimes u^{(\{2\})}_{i_2}.$$

- Root transfer tensor: $C^{(D)} \in \mathbb{R}_*^{r_{\{1,2\}} \times r_{\{3,4\}}}$, giving:

$$\mathbf{v} = \sum_{i_{12}=1}^{r_{\{1,2\}}} \sum_{i_{34}=1}^{r_{\{3,4\}}} C^{(D)}_{i_{12}, i_{34}}\, \mathbf{u}^{(\{1,2\})}_{i_{12}} \otimes \mathbf{u}^{(\{3,4\})}_{i_{34}}.$$

The total storage for this representation is $r_1 r_2 r_{\{1,2\}} + r_3 r_4 r_{\{3,4\}} + r_{\{1,2\}} r_{\{3,4\}} + (r_1 + r_2 + r_3 + r_4)n$, where $n = \max_j \dim V_j$, compared to n^4 for a full tensor.

Tensor Train format

For T_D^{TT} over $D = \{1, \ldots, d\}$, with $S(\{1, \ldots, k\}) = \{\{1, \ldots, k-1\}, \{k\}\}$ for $k \geq 2$, Theorem 6.4.3 gives the *TT decomposition*:

$$\mathbf{v} = \sum_{\substack{1 \leq i_j \leq r_j \\ j \in D}} G^{(1)}_{i_1} G^{(2)}_{i_1 i_2} \cdots G^{(d-1)}_{i_{d-2} i_{d-1}} G^{(d)}_{i_{d-1} i_d} \bigotimes_{j \in D} u^{(\{j\})}_{i_j}, \tag{6.12}$$

where $G^{(k)} \in \mathbb{R}^{r_{k-1} \times r_k}$ are the transfer matrices at level k (with $r_0 = r_d = 1$). This is the standard TT format of [Ose11], expressed in terms of minimal subspace bases.

6.6 Disjoint decomposition and rank stratification

Theorem 6.6.1. *The algebraic tensor space $\mathbf{V}_D$ admits the disjoint decomposition*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)} \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D). \tag{6.13}$$

Every non-zero $\mathbf{v} \in \mathbf{V}_D$ belongs to exactly one stratum $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ for $\mathfrak{r} = \mathrm{rank}_{T_D}(\mathbf{v})$.

Proof. Follows directly from the uniqueness of minimal subspaces (Theorem 4.2.3) and the definition of tree-based rank. □

Proposition 6.6.2 (Relation to Tucker decomposition)**.** *For any T_D and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$, the set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is contained in the Tucker set of rank $\mathfrak{r}_{\mathrm{leaf}}$:*

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \subset \mathfrak{M}_{\mathfrak{r}_{\mathrm{leaf}}}(\mathbf{V}_D), \quad \mathfrak{r}_{\mathrm{leaf}} = (r_{\{j\}})_{j \in D}.$$

Moreover, from Theorem 5.4.1:

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \mathfrak{M}_{\mathfrak{r}_{\text{leaf}}}(\mathbf{V}_D) \cap \mathfrak{M}_{\mathfrak{r}_1}(\mathbf{V}_D, \mathcal{P}_1(T_D)) \cap \cdots \cap \mathfrak{M}_{\mathfrak{r}_{\text{depth}()-1}}(\mathbf{V}_D, \mathcal{P}_{\text{depth}()-1}(T_D)).$$

The tree-based format imposes additional *rank constraints beyond the Tucker format at each intermediate level of the tree.*

6.7 Historical notes and comparison

The characterisation of tree-based tensors by transfer tensors (Theorem 6.4.3) generalises the Tucker core representation (for depth 1 trees) and the HT representation of Hackbusch–Kühn [HK09] (for binary trees). The TT case (6.12) was established by Oseledets [Ose11]; the connection to the minimal subspace framework was made explicit in [FH12].

The general tree-based format, unifying all these cases, was introduced in [FHN21]. The key innovation there is the *new characterisation of minimal subspaces* (Proposition 6.3.1), which expresses $U_\beta^{\min}(\mathbf{v})$ via contractions of $U_\alpha^{\min}(\mathbf{v})$ for the parent vertex α. This extends the results of [FH12] (where only the bilinear splitting $V_\alpha \otimes V_{\alpha^c}$ was used) to the full tree structure.

The gauge freedom (Remark 6.4.4) is the algebraic manifestation of the non-uniqueness of tensor decompositions. In the TT case, canonical gauges (left-orthogonal, right-orthogonal, and mixed-canonical) are used in the DMRG algorithm [HRS12]. For general trees, the analogous gauge conditions lead to the HOSVD (Higher-Order SVD) of De Lathauwer et al. [DLDMV00].

Chapter 7

Topological Tree-Based Tensor Spaces

The algebraic theory of Chapter 6 operates entirely within the algebraic tensor space $\mathbf{V}_D$. For applications — numerical solution of PDEs, best approximation, dynamical systems — one needs a *topological* framework: a norm on $\mathbf{V}_D$ (or on its completion) and conditions guaranteeing that the tensor product maps at each level of the tree are continuous. This chapter introduces topological tree-based tensor spaces (Section 7.1), the T_D-continuity condition (Section 7.2), the key identification theorem (Section 7.3), and the existence of best approximations in the tree-based setting (Section 7.5). The main results are drawn from [FHN21], which significantly weakens the norm conditions required in [FH12].

7.1 Topological tree-based tensor spaces

Definition 7.1.1 (Topological tree-based tensor space)**.** Let T_D be a dimension partition tree over D and let $(V_j, \|\cdot\|_j)$ be a normed space for each $j \in D$, with completion $V_{j,\|\cdot\|_j}$ (a Banach space). A *topological tree-based representation* of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D} = {}_{\|\cdot\|_D} \bigotimes_{j\in D} V_j$ is a collection $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha\in T_D\setminus\{D\}}$ such that:

(i) For each leaf $\{j\} \in \mathcal{L}(T_D)$: $V_{\{j\},\|\cdot\|_j} = V_{j,\|\cdot\|_j}$.

(ii) For each internal vertex $\alpha \in T_D \setminus \mathcal{L}(T_D)$: $\mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ is a Banach tensor space

$$\mathbf{V}_{\alpha,\|\cdot\|_\alpha} = {}_{\|\cdot\|_\alpha} \bigotimes_{\beta\in S(\alpha)} V_\beta, \tag{7.1}$$

i.e., the completion of the algebraic space $\mathbf{V}_\alpha = {}_a \bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$ under the norm $\|\cdot\|_\alpha$.

The key question is when the norm $\|\cdot\|_\alpha$ on the algebraic tensor space ${}_a \bigotimes_{\beta\in S(\alpha)} \mathbf{V}_\beta$

is *compatible* with completing each factor: that is, when does

$$ {}_{\|\cdot\|_\alpha} \bigotimes_{\beta \in S(\alpha)} V_{\beta,\|\cdot\|_\beta} = {}_{\|\cdot\|_\alpha} \bigotimes_{\beta \in S(\alpha)} V_\beta = {}_{\|\cdot\|_\alpha} \bigotimes_{j \in \alpha} V_j \tag{7.2} $$

hold? The answer is provided by the T_D-continuity condition.

7.2 The T_D-continuity condition

Definition 7.2.1 (T_D-continuous tensor product)**.** Let $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha \in T_D \setminus \{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D,\|\cdot\|_D}$. The tensor product map $\otimes$ is T_D*-continuous* if for each internal vertex $\alpha \in T_D \setminus \mathcal{L}(T_D)$, the map

$$ \bigotimes : \prod_{\beta \in S(\alpha)} (V_{\beta,\|\cdot\|_\beta}, \|\cdot\|_\beta) \longrightarrow (\mathbf{V}_{\alpha,\|\cdot\|_\alpha}, \|\cdot\|_\alpha) \tag{7.3} $$

is continuous (equivalently, bounded as a multilinear map).

T_D-continuity is equivalent to requiring that each norm $\|\cdot\|_\alpha$ is a crossnorm on the product $\prod_{\beta \in S(\alpha)} V_{\beta,\|\cdot\|_\beta}$ (or at least bounds it from above up to a constant).

Lemma 7.2.2 ([Hac19, Lemma 4.34])**.** *Let $(V_j, \|\cdot\|_j)$ be normed spaces and $\|\cdot\|$ a norm on ${}_a \bigotimes_{j=1}^d V_j$ such that the tensor product map $\otimes : \prod_j (V_{j,\|\cdot\|_j}, \|\cdot\|_j) \to ({}_a \bigotimes_j V_{j,\|\cdot\|_j}, \|\cdot\|)$ is continuous. Then*

$$ {}_{\|\cdot\|} \bigotimes_{j=1}^d V_{j,\|\cdot\|_j} = {}_{\|\cdot\|} \bigotimes_{j=1}^d V_j. $$

7.3 The identification theorem

The following theorem is the topological counterpart of the algebraic associativity: it shows that, under T_D-continuity, completing at each vertex of the tree gives the same result as completing all factors simultaneously.

Theorem 7.3.1 (Identification theorem, [FHN21, Theorem 3.2])**.** *Let $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}_{\alpha \in T_D \setminus \{D\}}$ be a topological tree-based representation of $\mathbf{V}_{D,\|\cdot\|_D}$ such that the tensor product map $\otimes$ is T_D-continuous. Then for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:*

$$ \mathbf{V}_{\alpha,\|\cdot\|_\alpha} = {}_{\|\cdot\|_\alpha} \bigotimes_{\beta \in S(\alpha)} V_{\beta,\|\cdot\|_\beta} = {}_{\|\cdot\|_\alpha} \bigotimes_{\beta \in S(\alpha)} V_\beta = {}_{\|\cdot\|_\alpha} \bigotimes_{j \in \alpha} V_j. \tag{7.4} $$

Proof. Apply Lemma 7.2.2 at each internal vertex α, using the T_D-continuity of the tensor product map at α. The third equality in (7.4) follows from the algebraic identity $\mathbf{V}_\alpha = {}_a \bigotimes_{j \in \alpha} V_j$ (associativity of tensor products, Proposition 3.1.3). $\square$

Example 7.3.2 (Anisotropic Sobolev space)**.** Consider $D = \{1, 2, 3\}$, $V_1 = L^p(I_1)$, $V_2 = H^{N,p}(I_2)$, $V_3 = H^{N,p}(I_3)$, and the tree

$$T_D : D \to \Big\{\{1\}, \{2, 3\}\Big\} \to \Big\{\{1\}, \{2\}, \{3\}\Big\},$$

with norms $\|\cdot\|_{23}$ on $V_2 \otimes_a V_3$ and $\|\cdot\|_{123}$ on V_D. If both tensor product maps are continuous (i.e., $\bigotimes$ is T_D-continuous), then

$$_{\|\cdot\|_{123}}\bigotimes_{j=1}^{3} V_j = L^p(I_1) \otimes_{\|\cdot\|_{123}} \Big(H^{N,p}(I_2) \otimes_{\|\cdot\|_{23}} H^{N,p}(I_3)\Big),$$

which is the anisotropic Sobolev space of Figure 7.1.

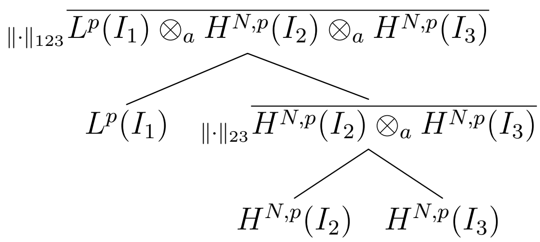

Figure 7.1: Topological tree-based tensor space for an anisotropic Sobolev product. The intermediate space at vertex $\{2, 3\}$ is the Banach completion of $H^{N,p}(I_2) \otimes_a H^{N,p}(I_3)$ under $\|\cdot\|_{23}$.

7.4 The tree-based injective norm condition

For the existence of best approximations, a stronger condition than T_D-continuity is required. The following condition is the tree-based analogue of condition (Inj) from Section 4.5.

Definition 7.4.1 (T_D-injective norm condition)**.** A topological tree-based representation $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ satisfies the T_D*-injective norm condition* (Inj)[T_D] if for each $\alpha \in T_D \setminus \mathcal{L}(T_D)$:

$$\|\cdot\|_\alpha \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}, \tag{7.5}$$

where $\|\cdot\|_{\varepsilon(S(\alpha))}$ is the injective norm on $_a\bigotimes_{\beta \in S(\alpha)} V_{\beta,\|\cdot\|_\beta}$.

Proposition 7.4.2 ([FHN21, Proposition 3.3])**.** *Condition* (Inj)*[T_D] implies T_D-continuity.*

Proof. The injective norm is a reasonable crossnorm (Proposition 4.4.2), so $\|\cdot\|_\alpha \gtrsim \|\cdot\|_{\varepsilon(S(\alpha))}$ implies that $\bigotimes : \prod_{\beta \in S(\alpha)} V_{\beta,\|\cdot\|_\beta} \to \mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ is continuous. □

Remark 7.4.3. The condition (Inj)[T_D] holds for all common tensor product norms arising from L^p and Sobolev spaces. In particular, the L^p norm on $L^p(\prod_j I_j)$ satisfies

(Inj)$[T_D]$ for any binary tree over D (this follows from the fact that L^p norms are reasonable crossnorms; see [Hac19], Example 4.72).

Minimal subspaces in the completion

Under condition (Inj)$[T_D]$, the definition of minimal subspaces can be extended from $\mathbf{V}_D$ to the Banach completion $\mathbf{V}_{D,\|\cdot\|_D}$. This extension is needed for tensors in $\mathbf{V}_{D,\|\cdot\|_D} \setminus \mathbf{V}_D$, which arise naturally as weak limits of sequences in $\mathbf{V}_D$.

Lemma 7.4.4 (Extension of minimal subspaces, [FH12, Lemma 3.8]). *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[*T_D*]. For* $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$ *and* $\alpha \in S(D)$*, define*

$$U_\alpha^{\min}(\mathbf{v}) := \overline{\operatorname{span}\Big\{\overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})}(\mathbf{v}) : \varphi^{[\alpha]} \in {}_a \bigotimes_{\beta \in S(D)\setminus\{\alpha\}} \mathbf{V}_\beta^*\Big\}}^{\|\cdot\|_\alpha}, \tag{7.6}$$

where $\overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_{\alpha,\|\cdot\|_\alpha})$ *is the continuous extension of the contraction map. Then* $U_\alpha^{\min}(\mathbf{v})$ *is a well-defined closed subspace of* $\mathbf{V}_{\alpha,\|\cdot\|_\alpha}$ *with* $\dim U_\alpha^{\min}(\mathbf{v}) < \infty$*.*

The lemma allows one to define the tree-based rank $\operatorname{rank}_{T_D}(\mathbf{v})$ for any $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$, not just for $\mathbf{v} \in \mathbf{V}_D$. This is essential for the best approximation theorem.

7.5 Weak closedness and best approximations

Theorem 7.5.1 (Weak lower semicontinuity of rank, [FHN21, Theorem 3.4]). *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy* (Inj)*[*T_D*], and let* $(\mathbf{v}_n) \subset \mathbf{V}_{D,\|\cdot\|_D}$ *be a sequence converging weakly to* $\mathbf{v} \in \mathbf{V}_{D,\|\cdot\|_D}$*. Then for each* $\alpha \in T_D \setminus \{D\}$*:*

$$\dim U_\alpha^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n). \tag{7.7}$$

Proof. The proof proceeds in two stages: first for vertices $\alpha \in S(D)$ (children of the root), then by downward induction over the tree.

Stage 1: Vertices in $S(D)$.

Fix $\alpha \in S(D)$. By the extension formula (7.6), for any $\varphi^{[\alpha]} \in {}_a \bigotimes_{\beta \in S(D)\setminus\{\alpha\}} \mathbf{V}_\beta^*$, the operator

$$T_{\varphi^{[\alpha]}} := \overline{(\mathrm{id}_{\mathbf{V}_\alpha} \otimes \varphi^{[\alpha]})} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_{\alpha,\|\cdot\|_\alpha})$$

is bounded, with $\|T_{\varphi^{[\alpha]}}\| \le \|\varphi^{[\alpha]}\|_*$ by condition (Inj)$[T_D]$ (which implies the contraction maps are controlled by the dual functional norms; see the proof of Lemma 7.4.4). Hence weak convergence $\mathbf{v}_n \rightharpoonup \mathbf{v}$ in $\mathbf{V}_{D,\|\cdot\|_D}$ implies

$$T_{\varphi^{[\alpha]}}(\mathbf{v}_n) \rightharpoonup T_{\varphi^{[\alpha]}}(\mathbf{v}) \quad \text{weakly in } \mathbf{V}_{\alpha,\|\cdot\|_\alpha}, \qquad \forall\, \varphi^{[\alpha]}. \tag{7.8}$$

Now suppose for contradiction that $\dim U_\alpha^{\min}(\mathbf{v}) > \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n)$. Then there exists a subsequence (still denoted $(\mathbf{v}_n)$) and an integer $m > 0$ such that $\dim U_\alpha^{\min}(\mathbf{v}_n) \le m$ for all n, while $\dim U_\alpha^{\min}(\mathbf{v}) \ge m+1$.

Choose $m+1$ functionals $\varphi_1^{[\alpha]}, \dots, \varphi_{m+1}^{[\alpha]}$ such that the vectors $w_k := T_{\varphi_k^{[\alpha]}}(\mathbf{v}) \in U_\alpha^{\min}(\mathbf{v})$ are linearly independent (possible since $\dim U_\alpha^{\min}(\mathbf{v}) \ge m+1$). By (7.8), $T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) \rightharpoonup w_k$ for each k.

Since $T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) \in U_\alpha^{\min}(\mathbf{v}_n)$ and $\dim U_\alpha^{\min}(\mathbf{v}_n) \le m$, the $m+1$ vectors $T_{\varphi_1^{[\alpha]}}(\mathbf{v}_n), \dots, T_{\varphi_{m+1}^{[\alpha]}}(\mathbf{v}_n)$ are linearly dependent for every n. That is, there exist scalars $\lambda_1^n, \dots, \lambda_{m+1}^n$, not all zero, with $\sum_k \lambda_k^n T_{\varphi_k^{[\alpha]}}(\mathbf{v}_n) = 0$. Normalising so that $\max_k |\lambda_k^n| = 1$ and passing to a subsequence, we may assume $\lambda_k^n \to \lambda_k$ for each k, with $\max_k |\lambda_k| = 1$. The weak limit gives $\sum_k \lambda_k w_k = 0$, contradicting the linear independence of $w_1, \dots, w_{m+1}$. Therefore

$$\dim U_\alpha^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\alpha^{\min}(\mathbf{v}_n) \quad \forall\, \alpha \in S(D). \tag{7.9}$$

Stage 2: Downward induction over T_D.

We now extend (7.9) to every $\alpha \in T_D \setminus \{D\}$ by induction on the level $\operatorname{level}(\alpha)$ of the vertex, proceeding from the root downward.

Base case: $\alpha \in S(D)$. This is (7.9).

Inductive step: Let $\alpha \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$ and assume that (7.7) holds for α (established at the previous level). Fix $\beta \in S(\alpha)$.

By the tree-based characterisation of minimal subspaces (Proposition 6.3.1), for any $\mathbf{v}_\alpha \in U_\alpha^{\min}(\mathbf{v})$ and any functional $\varphi^{(\alpha\setminus\beta)} \in ({}_a\bigotimes_{\gamma\in S(\alpha)\setminus\{\beta\}} \mathbf{V}_\alpha[\gamma])^*$, the contraction

$$S_{\varphi^{(\alpha\setminus\beta)}}(\mathbf{v}_\alpha) := (\mathrm{id}_{\mathbf{V}_\beta} \otimes \varphi^{(\alpha\setminus\beta)})(\mathbf{v}_\alpha) \in U_\beta^{\min}(\mathbf{v}).$$

Under condition (Inj)$[T_D]$, the composition

$$R_\phi := S_{\varphi^{(\alpha\setminus\beta)}} \circ T_{\varphi^{[\alpha]}} \in \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}, \mathbf{V}_\beta)$$

is bounded, and $\mathbf{v}_n \rightharpoonup \mathbf{v}$ implies $R_\phi(\mathbf{v}_n) \rightharpoonup R_\phi(\mathbf{v})$ weakly in $\mathbf{V}_\beta$.

Since $R_\phi(\mathbf{v}_n) \in U_\beta^{\min}(\mathbf{v}_n)$ and $\dim U_\beta^{\min}(\mathbf{v}_n) \le \dim U_\alpha^{\min}(\mathbf{v}_n)$, the same contradiction argument as in Stage 1 applies: if $\dim U_\beta^{\min}(\mathbf{v}) > \liminf_n \dim U_\beta^{\min}(\mathbf{v}_n)$, we find linearly independent weak-limit vectors in $U_\beta^{\min}(\mathbf{v})$ that are linear combinations of weakly convergent sequences confined to a subspace of smaller dimension, yielding a contradiction.

Hence $\dim U_\beta^{\min}(\mathbf{v}) \le \liminf_{n\to\infty} \dim U_\beta^{\min}(\mathbf{v}_n)$ for all $\beta \in S(\alpha)$.

Applying this argument level by level from the root D down to the leaves gives (7.7) for all $\alpha \in T_D \setminus \{D\}$. □

Theorem 7.5.2 (Weak closedness, [FHN21, Theorem 3.5])**.** *Let* $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ *satisfy*

(Inj)*[T_D] and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$. Then $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ is weakly closed in $\mathbf{V}_{D,\|\cdot\|_D}$.*

Proof. Immediate from Theorem 7.5.1: if $\mathbf{v}_n \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ and $\mathbf{v}_n \rightharpoonup \mathbf{v}$, then $\dim U_\alpha^{\min}(\mathbf{v}) \leq \liminf \dim U_\alpha^{\min}(\mathbf{v}_n) \leq r_\alpha$ for all $\alpha \in T_D$, so $\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$. □

Theorem 7.5.3 (Best approximation, [FHN21, Theorem 3.6])**.** *Let $\{\mathbf{V}_{\alpha,\|\cdot\|_\alpha}\}$ satisfy* (Inj)*[T_D] and assume $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive. Then for every $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$, there exists $\mathbf{v}_* \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$ such that*

$$\|\mathbf{u} - \mathbf{v}_*\|_D = \inf_{\mathbf{v} \in \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)} \|\mathbf{u} - \mathbf{v}\|_D. \tag{7.10}$$

Proof. Identical to the proof of Theorem 4.5.4: take a minimising sequence $(\mathbf{v}_n) \subset \mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$, extract a weakly convergent subsequence (by reflexivity), and apply Theorem 7.5.2 to conclude the limit is in $\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$. Lower semicontinuity of the norm then gives the minimum. □

Remark 7.5.4 (Improvement over prior results)*.* Theorem 7.5.3 improves the earlier result of Theorem 4.5.4 in two ways:

1. It applies to *any* tree T_D, not just the Tucker tree T_D^{Tucker}.

2. The norm condition (Inj)[T_D] is *weaker* than the condition of [FH12]: it requires the injective norm bound only *locally* at each vertex, whereas [FH12] required a global injective norm bound on the full tensor product.

In particular, Theorem 7.5.3 applies to the anisotropic Sobolev space of Example 7.3.2, which is *not* covered by Theorem 4.5.4.

Corollary 7.5.5. *For $1 < p < \infty$, $L^p(\prod_{j \in D} I_j)$ with any dimension partition tree T_D satisfies the hypotheses of Theorem 7.5.3. Hence every $f \in L^p(\prod_j I_j)$ has a best tree-based approximation of any prescribed tree-based rank.*

7.6 Historical notes

The first best approximation results for tensor formats in infinite- dimensional spaces were proved in [FH12] for the Tucker format, under the assumption that $\|\cdot\|_D \gtrsim \|\cdot\|_\varepsilon$ globally. The extension to tree-based formats under the weaker condition (Inj)[T_D] was achieved in [FHN21]. The key improvement is the use of the *local* injective norm bound at each vertex of the tree, combined with the new characterisation of minimal subspaces (Proposition 6.3.1) to propagate weak lower semicontinuity up the tree.

The anisotropic Sobolev space of Example 7.3.2 motivates this improvement: in parametric PDEs where the spatial variable $x_1 \in I_1$ and the parameter $x_2 \in I_2$ have different regularity, the natural norm on $L^p(I_1) \otimes H^{N,p}(I_2)$ is not a global injective norm, but it does satisfy (Inj)[T_D] for the two-level tree. The existence of best approximations in such spaces is crucial for the convergence analysis of greedy algorithms for parametric PDEs (cf. [Nou17]).

Part III

Differential Geometry of Tensor Formats

Chapter 8

The Tucker Manifold

This chapter establishes the central geometric result of the monograph: the set $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ of tensors with fixed Tucker rank carries the structure of a $\mathcal{C}^\infty$-Banach manifold, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$. Section 8.1 introduces the rank map and motivates the manifold structure. Section 8.2 constructs the local charts explicitly. Section 8.3 proves the $\mathcal{C}^\infty$ atlas (Theorem 8.3.2). Section 8.6 develops the fibre bundle structure in detail: local trivialisations, transition functions, and the structure group $G_{\mathfrak{r}} = \prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$. Section 8.4 identifies the tangent space. Section 8.5 proves the immersion theorem (Theorem 8.5.3). Throughout, $D = \{1, \ldots, d\}$, $d \geq 2$, and $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces.

8.1 The rank map and the decomposition of $\mathbf{V}_D$

Recall from Theorem 4.4.3 that

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D).$$

The map

$$\varrho : \mathbf{V}_D \longrightarrow \bigcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \qquad \mathbf{v} \mapsto \left(U_\alpha^{\min}(\mathbf{v})\right)_{\alpha \in D}, \tag{8.1}$$

sends each tensor to its tuple of minimal subspaces. For fixed $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, the restriction

$$\varrho_{\mathfrak{r}} := \varrho\Big|_{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \longrightarrow \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha) \tag{8.2}$$

is surjective: for any tuple $(U_\alpha)_{\alpha \in D}$ with $U_\alpha \in \mathbb{G}_{r_\alpha}(V_\alpha)$, any full-rank tensor in ${}_a\bigotimes_\alpha U_\alpha$ belongs to $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and maps to (U_α).

The product of Grassmannians $\prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha)$ is a Banach manifold (Theorem 2.3.2 applied factor by factor), and the rank map $\varrho_{\mathfrak{r}}$ is its natural base for the bundle structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

8.2 Local charts

Fix $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$ and $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. Let $V_{\alpha,\|\cdot\|_\alpha}$ denote the completion of V_α and choose for each $\alpha \in D$ a closed complement $W_\alpha^{\min}(\mathbf{v})$ of $U_\alpha^{\min}(\mathbf{v})$ in $V_{\alpha,\|\cdot\|_\alpha}$:

$$V_{\alpha,\|\cdot\|_\alpha} = U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v}).$$

Such a complement exists by Proposition 2.2.3, since $U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional.

Chart domain

Define the *chart domain at* $\mathbf{v}$ by

$$\mathcal{U}(\mathbf{v}) := \Big\{\mathbf{w} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) : U_\alpha^{\min}(\mathbf{w}) \in \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha),\ \forall \alpha \in D\Big\}, \tag{8.3}$$

where $\mathbb{G}(W, X) = \{U \in \mathbb{G}(X) : X = U \oplus W\}$ is the chart domain of the Grassmannian at $U_\alpha^{\min}(\mathbf{v})$ (see Section 2.3).

Chart map

Fix bases $\{u_{i_\alpha}^{(\alpha)}\}_{i_\alpha=1}^{r_\alpha}$ of $U_\alpha^{\min}(\mathbf{v})$ for each $\alpha \in D$. By Lemma 3.3.5, there is a unique $C^{(D)}(\mathbf{v}) \in \mathbb{R}_*^{\times r_\alpha}$ such that $\mathbf{v}$ has the Tucker representation (3.9). For each $\mathbf{w} \in \mathcal{U}(\mathbf{v})$, define the *chart map*

$$\xi_{\mathbf{v}} : \mathcal{U}(\mathbf{v}) \longrightarrow \prod_{\alpha \in D} \mathcal{L}\Big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\Big) \times \mathbb{R}_*^{\times r_\alpha}, \tag{8.4}$$

by sending $\mathbf{w} = (\bigotimes_\alpha \exp(L_\alpha))(\mathbf{u}(E^{(D)}))$ to the pair $((L_\alpha)_{\alpha\in D},\, E^{(D)})$, where $(L_\alpha, E^{(D)})$ are uniquely determined by Lemma 8.2.1 below.

Lemma 8.2.1 (Local representation, [FHN19, Lemma 3.7])**.** *Under the conditions above, for* $\mathbf{w} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *the following are equivalent:*

(a) $\mathbf{w} \in \mathcal{U}(\mathbf{v})$.

(b) *There exists a unique pair* $((L_\alpha)_{\alpha\in D},\, E^{(D)}) \in \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_\alpha}$ *such that*

$$\mathbf{w} = \Big(\bigotimes_{\alpha \in D} \exp(L_\alpha)\Big)(\mathbf{u}(E^{(D)})), \tag{8.5}$$

where $\mathbf{u}(E^{(D)}) \in \mathfrak{M}_{\mathfrak{r}}(()_a \otimes_\alpha U_\alpha^{\min}(\mathbf{v}))$ *is the tensor with core* $E^{(D)}$ *in the fixed bases.*

Here $\exp(L_\alpha) = \mathrm{id}_{V_\alpha} + L_\alpha$ by the nilpotency of $L_\alpha \in \mathcal{L}_{(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))}(V_\alpha)$ (Proposition 2.2.4), so the representation is polynomial and $\mathcal{C}^\infty$.

Remark 8.2.2. Formula (8.5) has a clean geometric interpretation: $\mathbf{w}$ is obtained from the "reference" tensor $\mathbf{u}(E^{(D)})$ (which lives in the fixed subspaces $U_\alpha^{\min}(\mathbf{v})$) by applying the "rotation" $\bigotimes_\alpha \exp(L_\alpha)$ which moves each subspace $U_\alpha^{\min}(\mathbf{v})$ to $U_\alpha^{\min}(\mathbf{w}) = G(L_\alpha) = \operatorname{span}\{(\mathrm{id} + L_\alpha)(u_{i_\alpha}^{(\alpha)})\}$.

8.3 The $\mathcal{C}^\infty$-Banach manifold structure

Lemma 8.3.1 (Smooth transitions, [FHN19, Lemma 3.8])**.** *Under the continuity assumption on the tensor product map* (3.13)*, for* $\mathbf{v}, \mathbf{v}' \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ *with* $\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}') \neq \emptyset$*, the transition map*

$$\xi_{\mathbf{v}'} \circ \xi_{\mathbf{v}}^{-1} : \xi_{\mathbf{v}}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')) \longrightarrow \xi_{\mathbf{v}'}(\mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}'))$$

is $\mathcal{C}^\infty$*-Fréchet differentiable (analytic if the* V_α *are complex Banach spaces).*

Sketch. For $\mathbf{w} \in \mathcal{U}(\mathbf{v}) \cap \mathcal{U}(\mathbf{v}')$, the transition map is

$$(L_\alpha, E^{(D)}) \mapsto \Big((\Psi_{\mathbf{v}'}^{(\alpha)} \circ (\Psi_{\mathbf{v}}^{(\alpha)})^{-1})(L_\alpha),\ f(\mathfrak{L}, \mathbf{u}(E^{(D)})) \Big),$$

where $\Psi_{\mathbf{v}}^{(\alpha)}$ is the Grassmann chart map at $U_\alpha^{\min}(\mathbf{v})$ (Theorem 2.3.2), and $f(\mathfrak{L}, \mathbf{u}(E^{(D)})) = (\bigotimes_\alpha \exp(L_\alpha - L'_\alpha))(\mathbf{u}(E^{(D)}))$. The first component is analytic (transition map of the Grassmannian). The second is $\mathcal{C}^\infty$ by Proposition 3.4.2 (tensor product of $\mathcal{C}^\infty$ maps) and the fact that the evaluation map is multilinear and continuous. See [FHN19] for the complete argument. □

Theorem 8.3.2 ($\mathcal{C}^\infty$ Tucker manifold, [FHN19, Theorem 3.17])**.** *Let* $(V_\alpha, \|\cdot\|_\alpha)$ *be normed spaces for* $\alpha \in D$ *and let* $\|\cdot\|_D$ *be a norm on* $\mathbf{V}_D$ *such that the tensor product map* (3.13) *is continuous. Then:*

(i) *The collection* $\{(\mathcal{U}(\mathbf{v}), \xi_{\mathbf{v}})\}_{\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)}$ *is a* $\mathcal{C}^\infty$*-atlas on* $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$*, which is thus a* $\mathcal{C}^\infty$*-Banach manifold modelled on the Banach space*

$$\mathbf{E}_D \coloneqq \prod_{\alpha \in D} \mathcal{L}\Big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\Big) \times \mathbb{R}^{\times_{\alpha \in D} r_\alpha}. \tag{8.6}$$

(ii) *The triple* $(\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D), \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \varrho_\mathfrak{r})$ *is a* $\mathcal{C}^\infty$*-fibre bundle with typical fibre* $\mathbb{R}_*^{\times r_\alpha}$*.*

If the V_α *are complex Banach spaces, the manifold is analytic.*

Proof. The atlas conditions AT1–AT3 are verified as follows. AT1: $\{\mathcal{U}(\mathbf{v})\}$ covers $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ since every $\mathbf{v}$ belongs to its own chart domain. AT2: $\xi_{\mathbf{v}}$ is a bijection onto $\prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}_*^{\times r_\alpha}$ by Lemma 8.2.1. AT3: Lemma 8.3.1 gives $\mathcal{C}^\infty$

transitions. The fibre bundle structure follows because $\varrho_{\mathfrak{r}}$ acts locally as the projection onto $\prod_\alpha \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha)$ (the Grassmannian chart domain), with fibre $\mathbb{R}_*^{\times r_\alpha}$. $\square$

Remark 8.3.3 (Independence from the norm). The manifold structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is independent of the choice of $\|\cdot\|_D$: the chart maps $\xi_{\mathbf{v}}$ depend only on the splitting $V_{\alpha,\|\cdot\|_\alpha} = U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v})$, and the $\mathcal{C}^\infty$ regularity follows from the continuity of the tensor product map, which holds for any crossnorm. Different norms give equivalent smooth structures.

Corollary 8.3.4 (Hilbert manifold)**.** *If $V_{\alpha,\|\cdot\|_\alpha}$ is a Hilbert space for each $\alpha \in D$, then $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a $\mathcal{C}^\infty$-Hilbert manifold modelled on $\prod_\alpha W_\alpha^{r_\alpha} \times \mathbb{R}^{\times r_\alpha}$, where $W_\alpha^{r_\alpha} = (U_\alpha)^\perp \times \cdots$ (each $\mathcal{L}(U_\alpha, W_\alpha) \cong W_\alpha^{r_\alpha}$ via the chosen orthonormal basis of U_α).*

Example 8.3.5. For $1 < p < \infty$, the L^p tensor space $L^p(\prod_\alpha I_\alpha)$ satisfies the hypotheses of Theorem 8.3.2. Hence $\mathfrak{M}_{\mathfrak{r}}(()_a \bigotimes_\alpha L^p(I_\alpha))$ is a $\mathcal{C}^\infty$-Banach manifold. For $p = 2$ it is a Hilbert manifold.

Example 8.3.6. For $V_1 = H^{1,p}(I_1)$, $V_2 = H^{1,p}(I_2)$, and either the full Sobolev norm $\|\cdot\|_{H^{1,p}}$ or the partial norm $\|\cdot\|_{(0,1),p}$ on $V_1 \otimes_a V_2$, both norms make the tensor product map continuous (see [Hac19, Examples 4.41, 4.42]). Hence $\mathfrak{M}_{(r,r)}(H^{1,p}(I_1) \otimes_a H^{1,p}(I_2))$ is a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathcal{L}(U_1, W_1) \times \mathcal{L}(U_2, W_2) \times \mathrm{GL}(\mathbb{R}^r)$.

8.4 The tangent space

Definition 8.4.1 (Tangent subspace $Z^{(D)}(\mathbf{v})$)**.** For $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, define the linear subspace

$$Z^{(D)}(\mathbf{v}) := {}_a \bigotimes_{\alpha \in D} U_\alpha^{\min}(\mathbf{v}) \oplus \bigoplus_{\alpha \in D} W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \ \subset\ \mathbf{V}_{D,\|\cdot\|_D}. \tag{8.7}$$

$Z^{(D)}(\mathbf{v})$ is a direct sum of $d+1$ subspaces: the core subspace ${}_a \bigotimes_\alpha U_\alpha^{\min}(\mathbf{v})$ (dimension $\prod r_\alpha$) and d "tangent directions" $W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$.

Proposition 8.4.2 (Tangent space identification, [FHN19, Proposition 4.13])**.** *Under the continuity assumption, the tangent map of the inclusion $\mathtt{i} : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D,\|\cdot\|_D}$ at $\mathbf{v}$ is injective, and*

$$T_{\mathbf{v}}\mathtt{i}\Big(\mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))\Big) = Z^{(D)}(\mathbf{v}).$$

More explicitly, for $(\dot{\mathfrak{L}}, \dot{C}^{(D)}) \in \mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$:

$$T_{\mathbf{v}}\mathtt{i}(\dot{\mathfrak{L}}, \dot{C}^{(D)}) = \sum_{(i_\alpha)} \dot{C}^{(D)}_{(i_\alpha)} \bigotimes_\alpha u_{i_\alpha}^{(\alpha)} + \sum_{\alpha \in D} \sum_{i_\alpha = 1}^{r_\alpha} \dot{u}_{i_\alpha}^{(\alpha)} \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}, \tag{8.8}$$

where $\dot{u}_{i_\alpha}^{(\alpha)} = \dot{L}_\alpha(u_{i_\alpha}^{(\alpha)}) \in W_\alpha^{\min}(\mathbf{v})$ and $\mathbf{U}_{i_\alpha}^{(\alpha)} \in U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ is given by (3.11).

Proof. The formula (8.8) follows by differentiating $(\mathfrak{i} \circ \xi_{\mathbf{v}}^{-1})$ at the origin $(\mathfrak{o}, C^{(D)})$ using Proposition 3.4.2 and the Leibniz rule for tensor products. Injectivity of $T_{\mathbf{v}}\mathfrak{i}$ follows from the linear independence of $\{\bigotimes_\alpha u_{i_\alpha}^{(\alpha)}\}$ (a basis of ${}_a\bigotimes_\alpha U_\alpha^{\min}(\mathbf{v})$) and of $\{\mathbf{U}_{i_\alpha}^{(\alpha)}\}_{i_\alpha}$ for each α (a basis of $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$). See [FHN19] for the complete proof. $\square$

8.5 Immersion in the ambient Banach space

To show that $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$, it remains to show that $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$, i.e., that $Z^{(D)}(\mathbf{v})$ is a split subspace of the ambient Banach space. This requires condition (Inj).

Lemma 8.5.1 ([FHN19, Lemma 4.13])**.** *Assume* (Inj) *holds. Let $\beta \in D$ and $W_\beta \in \mathbb{G}(V_{\beta,\|\cdot\|_\beta})$ with $V_{\beta,\|\cdot\|_\beta} = U_\beta \oplus W_\beta$ for some finite-dimensional U_β. Then for every finite-dimensional subspace $U_{[\beta]} \subset \mathbf{V}_\alpha^{[\beta]}$:*

$$W_\beta \otimes_a U_{[\beta]} \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D}).$$

Proof. The proof uses the continuity of the projection $\mathrm{id}_\beta \otimes P_{U_{[\beta]}\oplus W_{[\beta]}}$ on $\mathbf{V}_{D,\|\cdot\|_D}$ (which follows from (Inj) and a norm estimate via [FH12, Lemma 3.18]), and then decomposes $\mathrm{id}_\beta = P_{U_\beta\oplus W_\beta} + P_{W_\beta\oplus U_\beta}$ to construct an explicit bounded projection onto $W_\beta \otimes_a U_{[\beta]}$. See [FHN19] for details. $\square$

Lemma 8.5.2 ([FHN19, Lemma 4.14])**.** *Let X be a Banach space and $U, V \in \mathbb{G}(X)$ with $U \cap V = \{0\}$. Then $U \oplus V \in \mathbb{G}(X)$ and $U \cap V \in \mathbb{G}(X)$.*

Proof. Since $U, V \in \mathbb{G}(X)$ there exist $U', V' \in \mathbb{G}(X)$ with $X = U \oplus U' = V \oplus V'$. Then $U \oplus V \oplus (U' \cap V') = (U \cap V') \oplus (V \cap U') \oplus (U' \cap V') = X$, proving $U \oplus V \in \mathbb{G}(X)$. The second claim follows from $X = (U \cap V) \oplus (U \cap V') \oplus (V \cap U') \oplus (U' \cap V')$. $\square$

Theorem 8.5.3 (Immersion theorem, [FHN19, Theorem 4.15])**.** *Assume* (Inj) *holds. Then for each $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$:*

$$Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D}). \tag{8.9}$$

Consequently, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$.

Proof. For each $\alpha \in D$: $W_\alpha^{\min}(\mathbf{v}) \in \mathbb{G}(V_{\alpha,\|\cdot\|_\alpha})$ and $U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v})$ is finite-dimensional. By Lemma 8.5.1, $W_\alpha^{\min}(\mathbf{v}) \otimes_a U_{D\setminus\{\alpha\}}^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. Also, ${}_a\bigotimes_\alpha U_\alpha^{\min}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ (finite-dimensional, hence split by Proposition 2.2.3). The $d+1$ summands in (8.7) have pairwise trivial intersection (by the linear independence properties of minimal subspaces and the direct sum structure), so Lemma 8.5.2 applied iteratively gives $Z^{(D)}(\mathbf{v}) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. $\square$

Corollary 8.5.4 (Best projection onto tangent space)**.** *If $\mathbf{V}_{D,\|\cdot\|_D}$ is additionally reflexive, then for any $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $\dot{\mathbf{u}} \in \mathbf{V}_{D,\|\cdot\|_D}$, there exists $\dot{\mathbf{v}}_{\text{best}} \in Z^{(D)}(\mathbf{v})$ such that*

$$\|\dot{\mathbf{u}} - \dot{\mathbf{v}}_{\text{best}}\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v})} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

Proof. Since $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive, every closed subspace is proximinal (proximinal = best approximations exist). $Z^{(D)}(\mathbf{v})$ is closed (it is a split subspace by Theorem 8.5.3). $\square$

8.6 The Tucker fibre bundle

Theorem 8.3.2(ii) asserts that $(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D), \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha), \varrho_{\mathfrak{r}})$ is a fibre bundle. In this section we develop this structure in detail: we identify the transition functions, the structure group, and the local trivialisations explicitly in terms of the chart maps, and we compare the bundle structure with the quotient description available in finite dimensions.

Local trivialisations

Fix $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and recall the chart domain $\mathcal{U}(\mathbf{v})$ from (8.3). The rank map $\varrho_{\mathfrak{r}}$ restricted to $\mathcal{U}(\mathbf{v})$ takes values in

$$\mathcal{B}(\mathbf{v}) := \prod_{\alpha \in D} \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha) \subset \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha),$$

which is the product of the Grassmannian chart domains at $(U_\alpha^{\min}(\mathbf{v}))_{\alpha \in D}$, an open subset of the base.

Proposition 8.6.1 (Local trivialisation)**.** *For each $\mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, the map*

$$\chi_{\mathbf{v}} : \varrho_{\mathfrak{r}}^{-1}\big(\mathcal{B}(\mathbf{v})\big) \xrightarrow{\sim} \mathcal{B}(\mathbf{v}) \times F_{\mathfrak{r}}, \qquad \mathbf{w} \longmapsto \Big((U_\alpha^{\min}(\mathbf{w}))_{\alpha \in D},\ E^{(D)}(\mathbf{w})\Big), \tag{8.10}$$

where $F_{\mathfrak{r}} := \mathbb{R}_*^{\times_{\alpha \in D} r_\alpha}$ and $E^{(D)}(\mathbf{w})$ is the core tensor of $\mathbf{w}$ in the chart at $\mathbf{v}$ (Lemma 8.2.1), is a $\mathcal{C}^\infty$ diffeomorphism satisfying $\mathrm{pr}_1 \circ \chi_{\mathbf{v}} = \varrho_{\mathfrak{r}}$.

Proof. By Lemma 8.2.1, every $\mathbf{w} \in \mathcal{U}(\mathbf{v})$ is uniquely represented by $((L_\alpha)_\alpha, E^{(D)})$, and $\varrho_{\mathfrak{r}}(\mathbf{w}) = (U_\alpha^{\min}(\mathbf{w}))_\alpha = (G(L_\alpha))_\alpha \in \mathcal{B}(\mathbf{v})$. The Grassmannian chart maps $\Psi_{U_\alpha^{\min}(\mathbf{v}) \oplus W_\alpha^{\min}(\mathbf{v})}$ give a diffeomorphism $\mathcal{B}(\mathbf{v}) \cong \prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v}))$. Under this identification, $\chi_{\mathbf{v}}$ becomes the chart map $\xi_{\mathbf{v}}$ restricted to the base component (the L_α part) and the fibre component (the $E^{(D)}$ part). The chart map is a $\mathcal{C}^\infty$ diffeomorphism by Lemma 8.3.1 and Lemma 8.2.1, and $\mathrm{pr}_1 \circ \chi_{\mathbf{v}} = \varrho_{\mathfrak{r}}$ is immediate from the definition. $\square$

Transition functions and structure group

Proposition 8.6.2 (Transition functions). *For* $\mathbf{v}, \mathbf{v}' \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ *with* $\mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}') \neq \emptyset$, *the transition function*

$$g_{\mathbf{v}'\mathbf{v}} : \mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}') \longrightarrow \mathrm{GL}(F_{\mathfrak{r}}) \tag{8.11}$$

defined by $\chi_{\mathbf{v}'} \circ \chi_{\mathbf{v}}^{-1}(U, e) = (U, g_{\mathbf{v}'\mathbf{v}}(U) \cdot e)$ *is a* $\mathcal{C}^\infty$ *map taking values in the group*

$$G_{\mathfrak{r}} := \prod_{\alpha \in D} \mathrm{GL}(r_\alpha)\mathbb{R} \subset \mathrm{GL}(F_{\mathfrak{r}}), \tag{8.12}$$

acting on $F_{\mathfrak{r}} = \mathbb{R}_*^{\times r_\alpha}$ *by simultaneous change of basis: for* $g = (g_\alpha)_\alpha \in G_{\mathfrak{r}}$ *and* $E^{(D)} \in F_{\mathfrak{r}}$,

$$(g \cdot E^{(D)})_{i_1 \dots i_d} = \sum_{j_1, \dots, j_d} (g_1)_{i_1 j_1} \cdots (g_d)_{i_d j_d} \, E^{(D)}_{j_1 \dots j_d}. \tag{8.13}$$

Proof. For $\mathbf{w} \in \varrho_{\mathfrak{r}}^{-1}(\mathcal{B}(\mathbf{v}) \cap \mathcal{B}(\mathbf{v}'))$, write $\chi_{\mathbf{v}}(\mathbf{w}) = (U, E^{(D)})$ and $\chi_{\mathbf{v}'}(\mathbf{w}) = (U, E'^{(D)})$ (both have the same base component since $\varrho_{\mathfrak{r}}(\mathbf{w}) = U$ is intrinsic). The two core tensors $E^{(D)}$ and $E'^{(D)}$ represent the same tensor $\mathbf{w}$ in the two chart bases:

$$\mathbf{w} = \sum_{(i_\alpha)} E^{(D)}_{(i_\alpha)} \bigotimes_\alpha \exp(L_\alpha)(u^{(\alpha)}_{i_\alpha}) = \sum_{(i_\alpha)} E'^{(D)}_{(i_\alpha)} \bigotimes_\alpha \exp(L'_\alpha)(u'^{(\alpha)}_{i_\alpha}).$$

Since both $\exp(L_\alpha)(u^{(\alpha)}_{i_\alpha})$ and $\exp(L'_\alpha)(u'^{(\alpha)}_{i_\alpha})$ are bases of $U^{\min}_\alpha(\mathbf{w})$, there exist matrices $g_\alpha \in \mathrm{GL}(r_\alpha)\mathbb{R}$ relating them: $\exp(L_\alpha)(u^{(\alpha)}_{i_\alpha}) = \sum_j (g_\alpha)_{ij} \exp(L'_\alpha)(u'^{(\alpha)}_{j_\alpha})$. Substituting into the two representations gives $E'^{(D)}_{(i_\alpha)} = \sum_{(j_\alpha)} \prod_\alpha (g_\alpha)_{i_\alpha j_\alpha} E^{(D)}_{(j_\alpha)}$, which is exactly (8.13).

The map $U \mapsto g_{\mathbf{v}'\mathbf{v}}(U)$ is $\mathcal{C}^\infty$ because it is the composition of $\mathcal{C}^\infty$ Grassmannian transition maps (Lemma 8.3.1, Step 1). □

Remark 8.6.3 (The bundle is principal over its structure group). Proposition 8.6.2 shows that the Tucker bundle has structure group $G_{\mathfrak{r}} = \prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$, which is a *finite-dimensional* Lie group regardless of the (possibly infinite-dimensional) component spaces V_α. This is the key reason why the bundle has good properties (e.g., the fibre is a smooth manifold, and the bundle admits a $\mathcal{C}^\infty$ atlas) even in infinite dimensions.

The fibre $F_{\mathfrak{r}} = \mathbb{R}_*^{\times r_\alpha}$ is the set of arrays of full Tucker rank, i.e., the complement of a closed algebraic variety in $\mathbb{R}^{\times r_\alpha}$. The group $G_{\mathfrak{r}}$ acts on $F_{\mathfrak{r}}$ freely (a full-rank array cannot be fixed by a non-trivial gauge transformation) and properly (since $G_{\mathfrak{r}}$ is finite-dimensional Lie and acts by linear isomorphisms). Therefore the bundle is a *principal* $G_{\mathfrak{r}}$*-bundle* in the classical sense.

The quotient description and its limitations

In finite dimensions, the structure group $G_{\mathfrak{r}}$ acts on the total space of frames:

$$\widetilde{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} := \Big\{(\mathbf{v}, (u^{(\alpha)}_{i_\alpha})_{\alpha,i_\alpha}) : \mathbf{v} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D),\ \{u^{(\alpha)}_{i_\alpha}\}_{i_\alpha=1}^{r_\alpha} \text{ basis of } U^{\min}_\alpha(\mathbf{v})\Big\}, \tag{8.14}$$

and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) = \widetilde{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)}/G_{\mathfrak{r}}$, giving the quotient description of [KL07, UV13b].

In infinite dimensions, this quotient description breaks down for the following reason: the frame bundle $\widetilde{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)}$ has a natural topology as a product $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \times \prod_\alpha \mathrm{GL}(\mathbb{R}^{r_\alpha}, U^{\min}_\alpha(\mathbf{v}))$, but the group $G_{\mathfrak{r}}$ acts on the *second factor only*, and the projection $\widetilde{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} \to \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a principal $G_{\mathfrak{r}}$-bundle over $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ — which is what we are trying to construct. The quotient construction is circular in infinite dimensions.

The explicit atlas approach of Section 8.2 resolves this by working directly with the Grassmannian structure of the base $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$, bypassing the group action entirely. The bundle structure is then recovered a posteriori from Propositions 8.6.1 and 8.6.2.

Special cases

Example 8.6.4 ($d = 2$: line bundle and matrix bundle)**.** For $d = 2$:

(i) If $r_1 = r_2 = 1$ (rank-one matrices), the fibre is $F_{(1,1)} = \mathbb{R}^* = \mathbb{R} \setminus \{0\}$ and the structure group is $G_{(1,1)} = \mathbb{R}^* \times \mathbb{R}^*$. The Tucker bundle is a *line bundle* over $\mathbb{G}_1(V_1) \times \mathbb{G}_1(V_2) \cong \mathbb{P}(V_1) \times \mathbb{P}(V_2)$ (the product of projective spaces).

(ii) If $r_1 = r_2 = r \geq 2$ (matrices of rank r), the fibre is $F_{(r,r)} = \mathrm{GL}(r)\mathbb{R} \subset \mathbb{R}^{r\times r}$ and the structure group is $G_{(r,r)} = \mathrm{GL}(r)\mathbb{R} \times \mathrm{GL}(r)\mathbb{R}$, acting by $(g_1, g_2) \cdot M = g_1 M g_2^{-1}$. This recovers the principal bundle description of rank-r matrices as $\mathrm{GL}(r)\mathbb{R} \times \mathrm{GL}(r)\mathbb{R}$-equivariant maps.

Example 8.6.5 (Hilbert case: unitary structure group)**.** If each V_α is a Hilbert space and the bases $\{u^{(\alpha)}_{i_\alpha}\}$ are chosen to be orthonormal, the structure group reduces from $\mathrm{GL}(r_\alpha)\mathbb{R}$ to the orthogonal group $O(r_\alpha)$. In the complex Hilbert case it reduces to $U(r_\alpha)$ (unitary group). This is the structure group used in the Hilbert-space treatments of [KL07] for MCTDH and is the reason why gauge invariance in quantum chemistry can be phrased in terms of unitary transformations.

8.7 The tensor space as a Banach manifold

Corollary 8.7.1 ([FHN19, Corollary 3.19])**.** *Under the continuity assumption on the tensor product map, the algebraic tensor space* $\mathbf{V}_D$ *itself is a* $\mathcal{C}^\infty$*-Banach manifold (not modelled on a single Banach space), with atlas given by the disjoint union of the Tucker* *manifold atlases over all* $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$*:*

$$\mathbf{V}_D = \bigsqcup_{\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D).$$

8.8 Historical notes and comparison

The $\mathcal{C}^\infty$-Banach manifold structure of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ in infinite-dimensional spaces was established in [FHN19]. Prior work had treated only finite-dimensional or Hilbert space settings:

Koch and Lubich [KL07] proved that the Tucker manifold is a quotient manifold in finite dimensions and used it for the Dirac–Frenkel variational principle for the MCTDH method. Their approach, however, uses the Hilbert structure (metric projection) and does not extend to Banach spaces.

Uschmajew and Vandereycken [UV13a] studied the HT manifold in finite dimensions as a quotient manifold $\mathrm{GL}(r_1) \times \cdots \times \mathrm{GL}(r_d) \backslash \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)/G$ for a gauge group G. The description in [FHN19] is fundamentally different: it provides an *explicit atlas* with local charts given by formula (8.5), valid in the infinite-dimensional Banach setting.

The key novelties of [FHN19] are:

1. The chart maps $\xi_{\mathbf{v}}$ are given explicitly in terms of the Grassmann chart maps $\Psi_{\mathbf{v}}^{(\alpha)}$, making the geometric structure completely transparent.
2. The immersion theorem (Theorem 8.5.3) holds in general Banach spaces under condition (Inj), without requiring a Hilbert or reflexive structure.
3. The fibre bundle structure identifies the typical fibre as $\mathbb{R}_*^{\times r_\alpha}$, a finite-dimensional smooth manifold (an open subset of a Euclidean space).

Chapter 9

The Tree-Based Manifold

This chapter establishes the geometric counterpart of Chapter 8 for general tree-based tensor formats. The central result is that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ — when it is a *proper* set of tree-based tensors (Definition 9.1.1) — carries a natural $\mathcal{C}^\infty$-Banach manifold structure that is compatible with the Tucker manifold structures at each level of the tree, and is an immersed submanifold of the ambient Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$.

The key idea, due to [FHN23], is that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ (Theorem 5.4.1) is an intersection of Tucker manifolds, and its chart system is obtained by taking the intersection of the Tucker chart systems at each level. The resulting charts are natural and compatible with those of each Tucker factor.

Throughout, $(V_\alpha, \|\cdot\|_\alpha)$ are normed spaces, $\|\cdot\|_D$ satisfies (Inj) at each level of T_D, and $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$.

9.1 Proper sets of tree-based tensors

Not every $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a genuine intersection of Tucker manifolds — in degenerate cases it may coincide with a single Tucker manifold. The following definition isolates the non-degenerate case.

Definition 9.1.1 (Proper tree-based set)**.** The set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is called a *proper set of tree-based tensors* with rank $\mathfrak{r}$ if

$$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \neq \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D)) \quad \text{for all } 1 \leq k \leq \mathrm{depth}(T_D).$$

Example 9.1.2 (Degenerate case)**.** For $D = \{1, \dots, 6\}$ and the tree T_D with $S(D) = \{\{1\}, \{2,3,4,5,6\}\}$ (depth 2):

$$\begin{aligned}
\mathcal{P}_1(T_D) &= \Big\{\{1\}, \{2,3,4,5,6\}\Big\}, \\
\mathcal{P}_2(T_D) &= \mathcal{L}(T_D).
\end{aligned}$$

Since $\dim U^{\min}_{\{1\}}(\mathbf{v}) = \dim U^{\min}_{\{2,3,4,5,6\}}(\mathbf{v})$ for all $\mathbf{v} \in \mathbf{V}_D$ (Theorem 4.2.3), the constraint at level 1 is automatically satisfied whenever the leaf-level constraints hold. Hence $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) = \mathfrak{M}_{\mathfrak{r}_2}(\mathbf{V}_D, \mathcal{L}(T_D))$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is *not* a proper set. The geometry reduces to the Tucker manifold.

In all examples arising from balanced binary trees (HT format), Tensor Train, and most practical trees with $\operatorname{depth}(T_D) \geq 2$, the set $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is proper.

9.2 The Laplacian-like map and the intersection of model spaces

For each level k of the tree, the Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ has model space

$$\mathbf{E}_k := \Big(\bigoplus_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U^{\min}_\alpha(\mathbf{v}), W^{\min}_\alpha(\mathbf{v})) \otimes \operatorname{span}\{\mathbf{id}_{[\alpha]}\} \Big) \times \mathbb{R}^{\times_{\alpha \in \mathcal{P}_k(T_D)} r_\alpha},$$

where $\mathbf{id}_{[\alpha]} = \bigotimes_{\beta \in \mathcal{P}_k(T_D) \setminus \{\alpha\}} \mathrm{id}_{\mathbf{V}_\beta}$ is the identity on all factors except α.

The *Laplacian-like map* Δ_k embeds the product of $\mathcal{L}(U^{\min}_\alpha(\mathbf{v}), W^{\min}_\alpha(\mathbf{v}))$ spaces into $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$:

$$\Delta_k : \prod_{\alpha \in \mathcal{P}_k(T_D)} \mathcal{L}(U^{\min}_\alpha(\mathbf{v}), W^{\min}_\alpha(\mathbf{v})) \longrightarrow \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}), \quad (L_\alpha)_\alpha \mapsto \sum_{\alpha \in \mathcal{P}_k(T_D)} L_\alpha \otimes \mathbf{id}_{[\alpha]}. \tag{9.1}$$

Proposition 9.2.1 ([FHN23, Proposition 4.5])**.** *For $(L_\alpha)_{\alpha \in \mathcal{P}_k(T_D)} \in \prod_\alpha \mathcal{L}(U^{\min}_\alpha(\mathbf{v}), W^{\min}_\alpha(\mathbf{v}))$:*

$$\exp\Big(\Delta_k((L_\alpha)_\alpha)\Big) = \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} \exp(L_\alpha). \tag{9.2}$$

Proof. Put $L = \sum_\alpha L_\alpha \otimes \mathbf{id}_{[\alpha]}$. For $\alpha \neq \beta$: $(L_\alpha \otimes \mathbf{id}_{[\alpha]}) \circ (L_\beta \otimes \mathbf{id}_{[\beta]}) = L_\alpha \otimes L_\beta \otimes \mathbf{id}_{[\alpha,\beta]}$, and by the nilpotency of L_α (Proposition 2.2.4), $\exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \exp(L_\alpha) \otimes \mathbf{id}_{[\alpha]}$. Since the summands commute, $\exp(L) = \bigodot_\alpha \exp(L_\alpha \otimes \mathbf{id}_{[\alpha]}) = \bigotimes_\alpha \exp(L_\alpha)$. □

The critical consequence is that the local representation $\mathbf{w} = \exp(L)(\mathbf{u})$ for $L \in \mathbf{E}_k$ is *independent of* k: for any k and any $L \in \mathbf{E}_k$ with the same underlying tensor $L = \Delta_k((L_\alpha)_\alpha)$, the formula $\mathbf{w} = (\bigotimes_\alpha \exp(L_\alpha))(\mathbf{u})$ gives the same $\mathbf{w}$.

9.3 Charts for the tree-based manifold

Intersection of chart domains

For $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, define the *tree-based chart domain*:

$$\mathcal{W}(\mathbf{v}) := \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathcal{U}^{(k)}(\mathbf{v}), \tag{9.3}$$

where $\mathcal{U}^{(k)}(\mathbf{v})$ is the chart domain of $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ at $\mathbf{v}$ (Section 8.2).

Model space

Define the *tree-based model space*:

$$\mathbf{E}(\mathbf{v}) := \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathbf{E}_k \subset \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D}), \tag{9.4}$$

and the *tree-based core set*:

$$\mathcal{O}(\mathbf{v}) := \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathfrak{M}_{\mathfrak{r}_k}\Big(\Big)\Big({}_a \bigotimes_{\alpha\in\mathcal{P}_k(T_D)} U_\alpha^{\min}(\mathbf{v})\Big).$$

Lemma 9.3.1 ([FHN23, Lemma 4.6])**.** *Under condition* (Inj) *at each level,* $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ *is an open subset of the Banach space* $\mathbf{E}(\mathbf{v}) \times {}_a \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$.

Proof. We show separately that $\mathcal{O}(\mathbf{v})$ is open in ${}_a \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ and that $\mathbf{E}(\mathbf{v})$ is a Banach space.

Step 1: $\mathcal{O}(\mathbf{v})$ is open.

Fix $k \in \{1, \dots, \mathrm{depth}(T_D)\}$. The space $A_k := {}_a \bigotimes_{\alpha\in\mathcal{P}_k(T_D)} U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional, since each $U_\alpha^{\min}(\mathbf{v})$ is finite-dimensional (Theorem 4.2.3). The set $\mathcal{O}_k := \mathfrak{M}_{\mathfrak{r}_k}(()A_k)$ consists of tensors in A_k with α-ranks equal to r_α for all $\alpha \in \mathcal{P}_k(T_D)$. Its complement is the zero set of the rank-r_α minors of the unfolding matrices; since these are polynomial conditions on the coordinates of A_k, the complement is a closed algebraic variety. Hence $\mathcal{O}_k$ is open in A_k.

Now, all spaces A_k are identified with $A_1 = {}_a \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ via the canonical isomorphisms induced by the tree structure (each $\mathbf{V}_\alpha[\alpha]$ for $\alpha \in \mathcal{P}_k(T_D)$ is a subspace of $\mathbf{V}_\alpha[\beta]$ for the appropriate $\beta \in \mathcal{P}_1(T_D)$). Under this identification, $\mathcal{O}_k$ becomes an open subset of A_1, and $\mathcal{O}(\mathbf{v}) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} \mathcal{O}_k$ is a finite intersection of open sets in the finite-dimensional space A_1, hence open.

Step 2: $\mathbf{E}(\mathbf{v})$ is a Banach space.

For each level k, the model space $\mathbf{E}_k$ embeds into $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$ via the Laplacian map Δ_k (equation (9.1)). The image is:

$$\Delta_k\Big(\prod_{\alpha\in\mathcal{P}_k(T_D)} \mathcal{L}\Big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\Big)\Big) = \Big\{\sum_{\alpha\in\mathcal{P}_k(T_D)} L_\alpha \otimes \mathbf{id}_{[\alpha]} : L_\alpha \in \mathcal{L}\Big(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})\Big)\Big\},$$

which is a closed subspace of $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$: it is the image of a finite-dimensional Banach space under a bounded linear map, hence finite-dimensional and therefore closed. Therefore $\mathbf{E}_k \subset \mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$ is closed.

The intersection $\mathbf{E}(\mathbf{v}) = \bigcap_{k=1}^{\operatorname{depth}(T_D)} \mathbf{E}_k$ is a finite intersection of closed subspaces of $\mathcal{L}(\mathbf{V}_{D,\|\cdot\|_D})$, hence closed, and therefore a Banach space in its own right.

Conclusion.

Since $\mathbf{E}(\mathbf{v})$ is a Banach space and $\mathcal{O}(\mathbf{v})$ is open in the finite-dimensional space A_1, the product $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ is open in $\mathbf{E}(\mathbf{v}) \times A_1 = \mathbf{E}(\mathbf{v}) \times {}_a\bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$. $\square$

Tree-based chart map

Since $L \in \mathbf{E}(\mathbf{v})$ satisfies $L \in \mathbf{E}_k$ for all k, the formula $\mathbf{w} = \exp(L)(\mathbf{u})$ is well-defined and independent of the level index k by Proposition 9.2.1. This allows us to define:

Definition 9.3.2 (Tree-based chart)**.** For $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, the *tree-based chart map*

$$\boldsymbol{\xi}_{\mathbf{v}} : \mathcal{W}(\mathbf{v}) \longrightarrow \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v}) \tag{9.5}$$

is defined by $\boldsymbol{\xi}_{\mathbf{v}}(\mathbf{w}) = (L, \mathbf{u})$ where $\mathbf{w} = \exp(L)(\mathbf{u})$, with $L \in \mathbf{E}(\mathbf{v})$ and $\mathbf{u} \in \mathcal{O}(\mathbf{v})$ uniquely determined by any Tucker chart $\xi_{\mathbf{v}}^{(k)}$.

9.4 The $\mathcal{C}^\infty$-Banach manifold structure

Theorem 9.4.1 ($\mathcal{C}^\infty$ tree-based manifold, [FHN23, Theorem 4.7])**.** *Let T_D be a dimension partition tree over D with* $\operatorname{depth}(T_D) \geq 2$, *and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ be such that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a proper set of tree-based tensors. Assume* (Inj) *holds at each level of T_D. Then:*

(i) *The collection $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ is a $\mathcal{C}^\infty$-atlas on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, making it a $\mathcal{C}^\infty$-Banach manifold modelled on $\mathbf{E}(\mathbf{v}) \times {}_a\bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$.*

(ii) *The atlas is compatible with the Tucker manifold atlas at each level: the inclusion*

$$\mathfrak{i}_k : \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \hookrightarrow \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$$

is a $\mathcal{C}^\infty$-immersion for each $1 \leq k \leq \operatorname{depth}(T_D)$.

Proof. We verify the atlas conditions AT1–AT3 for the collection $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ and then establish the compatibility with the Tucker atlases.

AT1: Coverage.

For each $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, we have $\mathbf{v} \in \mathcal{W}(\mathbf{v})$ since $\mathbf{v}$ belongs to every Tucker chart domain $\mathcal{U}^{(k)}(\mathbf{v})$ (each Tucker chart is centred at $\mathbf{v}$ by definition). Hence $\bigcup_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)} \mathcal{W}(\mathbf{v}) = \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

AT2: $\boldsymbol{\xi}_{\mathbf{v}}$ is a bijection.

For $\mathbf{w} \in \mathcal{W}(\mathbf{v})$, we have $\mathbf{w} \in \mathcal{U}^{(k)}(\mathbf{v})$ for all k. By Lemma 8.2.1 applied at level k (Theorem 8.3.2), there exists a unique pair $(L^{(k)}, \mathbf{u}^{(k)})$ with $L^{(k)} \in \mathbf{E}_k$ and $\mathbf{u}^{(k)} \in \mathcal{O}_k$ such that $\mathbf{w} = \exp(L^{(k)})(\mathbf{u}^{(k)})$.

By Proposition 9.2.1, the map $\exp(L)$ for $L \in \mathbf{E}_k$ acts as $\bigotimes_{\alpha\in\mathcal{P}_k(T_D)} \exp(L_\alpha)$. Since the representation is independent of the level k used (the same tensor $\mathbf{w}$ is represented), we get $L^{(k)} = L^{(k')}$ and $\mathbf{u}^{(k)} = \mathbf{u}^{(k')}$ for all k, k'. In particular, $L := L^{(1)} \in \bigcap_k \mathbf{E}_k = \mathbf{E}(\mathbf{v})$ and $\mathbf{u} := \mathbf{u}^{(1)} \in \bigcap_k \mathcal{O}_k = \mathcal{O}(\mathbf{v})$.

The map $\boldsymbol{\xi}_{\mathbf{v}}(\mathbf{w}) = (L, \mathbf{u})$ is therefore well-defined and injective (since $(L, \mathbf{u})$ uniquely determines $\mathbf{w} = \exp(L)(\mathbf{u})$). It is surjective onto $\mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$ by the converse: for any $(L, \mathbf{u}) \in \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v})$, the tensor $\mathbf{w} = \exp(L)(\mathbf{u})$ belongs to every $\mathcal{U}^{(k)}(\mathbf{v})$ (since $L \in \mathbf{E}_k$ and $\mathbf{u} \in \mathcal{O}_k$ for all k), so $\mathbf{w} \in \mathcal{W}(\mathbf{v})$.

AT3: Smooth transitions.

Let $\mathbf{v}, \mathbf{v}' \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ with $\mathcal{W}(\mathbf{v}) \cap \mathcal{W}(\mathbf{v}') \neq \emptyset$. We must show that $\boldsymbol{\xi}_{\mathbf{v}'} \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1}$ is $\mathcal{C}^\infty$.

For any $\mathbf{w} \in \mathcal{W}(\mathbf{v}) \cap \mathcal{W}(\mathbf{v}')$, pick any level k. Since $\mathbf{w} \in \mathcal{U}^{(k)}(\mathbf{v}) \cap \mathcal{U}^{(k)}(\mathbf{v}')$, the transition map

$$\xi^{(k)}_{\mathbf{v}'} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}$$

is $\mathcal{C}^\infty$ by Lemma 8.3.1 (applied to the Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$). The tree-based chart maps satisfy $\boldsymbol{\xi}_{\mathbf{v}} = \xi^{(k)}_{\mathbf{v}}|_{\mathcal{W}(\mathbf{v})}$ for every k (by the uniqueness proved in AT2). Therefore

$$\boldsymbol{\xi}_{\mathbf{v}'} \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1} = \xi^{(k)}_{\mathbf{v}'} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}\Big|_{\boldsymbol{\xi}_{\mathbf{v}}(\mathcal{W}(\mathbf{v})\cap\mathcal{W}(\mathbf{v}'))},$$

which is the restriction of a $\mathcal{C}^\infty$ map to an open subset, hence $\mathcal{C}^\infty$.

Part (ii): Compatibility with Tucker atlases.

For each k, the inclusion $\mathfrak{i}_k : \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D) \hookrightarrow \mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$ satisfies, in charts,

$$\xi^{(k)}_{\mathbf{v}} \circ \mathfrak{i}_k \circ \boldsymbol{\xi}_{\mathbf{v}}^{-1} = \xi^{(k)}_{\mathbf{v}} \circ (\xi^{(k)}_{\mathbf{v}})^{-1}\Big|_{\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v})} = \text{inclusion } \mathbf{E}(\mathbf{v}) \times \mathcal{O}(\mathbf{v}) \hookrightarrow \mathbf{E}_k \times \mathcal{O}_k,$$

which is a bounded linear injection (hence $\mathcal{C}^\infty$ and an immersion). Since this holds for every choice of charts and every k, $\mathfrak{i}_k$ is a $\mathcal{C}^\infty$-immersion.

Manifold structure.

The atlas $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}_{\mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D)}$ satisfies AT1–AT3, so $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$-Banach manifold. The model space is $\mathbf{E}(\mathbf{v})\times\mathcal{O}(\mathbf{v})$, which is open in $\mathbf{E}(\mathbf{v})\times_a \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} U_\alpha^{\min}(\mathbf{v})$ by Lemma 9.3.1, so $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is modelled on this Banach space. □

9.5 Tangent space of the tree-based manifold

Proposition 9.5.1 (Tangent space). *Under the assumptions of Theorem 9.4.1, for $\mathbf{v} \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$:*

$$T_{\mathbf{v}}\mathbf{i}\Big(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D))\Big) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} T_{\mathbf{v}}\mathbf{i}_k\Big(\mathbb{T}_{\mathbf{v}}(\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D,\mathcal{P}_k(T_D)))\Big) = \bigcap_{k=1}^{\mathrm{depth}(T_D)} Z^{(D)}([)\mathcal{P}_k(T_D)]\mathbf{v}, \tag{9.6}$$

where $Z^{(D)}([)\mathcal{P}]\mathbf{v}$ is the tangent subspace (8.7) for the partition $\mathcal{P}$.

Proof. The model space of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is $\mathbf{E}(\mathbf{v}) \times \mathcal{O}_1$, where $\mathbf{E}(\mathbf{v}) = \bigcap_k \mathbf{E}_k$. Since $T_{\mathbf{v}}\mathbf{i}_k$ is injective (Proposition 8.4.2), the tangent map $T_{\mathbf{v}}\mathbf{i}$ is also injective and its image is $\bigcap_k T_{\mathbf{v}}\mathbf{i}_k(\mathbb{T}_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k()))$. The last equality follows from the explicit formula (8.8) for each level. □

9.6 Immersion in the ambient Banach space

Theorem 9.6.1 (Immersion of the tree-based manifold, [FHN23, Corollary 4.9]). *Under the assumptions of Theorem 9.4.1:*

$$T_{\mathbf{v}}\mathbf{i}\Big(\mathbb{T}_{\mathbf{v}}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D))\Big) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D}) \quad \textit{for all } \mathbf{v}\in\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D,T_D). \tag{9.7}$$

Consequently, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an immersed submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$.

Proof. From Theorem 8.5.3 applied at each level, $Z^{(D)}([)\mathcal{P}_k(])\mathbf{v} \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ for $1 \le k \le \mathrm{depth}(T_D)$. By Proposition 9.5.1, the tangent space of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is $\bigcap_k Z^{(D)}([)\mathcal{P}_k(])\mathbf{v}$. Applying Lemma 8.5.2 iteratively (intersection of split subspaces is split), we obtain $\bigcap_k Z^{(D)}([)\mathcal{P}_k(])\mathbf{v} \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$. □

Remark 9.6.2 (Embedded vs. immersed). Theorem 9.6.1 shows that $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an *immersed* submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$. Moreover, Theorem 9.4.1(ii) shows it is an *embedded* submanifold of each Tucker manifold $\mathfrak{M}_{\mathfrak{r}_k}(\mathbf{V}_D, \mathcal{P}_k(T_D))$. Whether $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is embedded in $\mathbf{V}_{D,\|\cdot\|_D}$ (i.e., the manifold topology coincides with the subspace topology) is a more delicate question related to the global topology discussed in Section 9.8.

9.7 Comparison with alternative descriptions

The geometric description of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in [FHN23] differs from earlier approaches in several respects.

Uschmajew–Vandereycken [UV13a]. For the HT format in finite-dimensional spaces, they construct $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as a quotient manifold $P\backslash\Big(\mathrm{GL}(r_\alpha) \times \cdots\Big)$ for a gauge group P. This gives a finite-dimensional manifold and is well-suited for numerical algorithms (Riemannian gradient methods). However, it does not extend to infinite-dimensional spaces and does not make the compatibility with Tucker manifolds at each level explicit.

Holtz–Rohwedder–Schneider [HRS12]. For the TT format in finite dimensions, a similar quotient construction is used. The connection to the Tucker manifold at each level is not made explicit.

Falcó–Hackbusch–Nouy (2015 preprint [FHN15]). An earlier version of the atlas in [FHN23] was proposed using a different chart system. The description in [FHN23] is more natural because the chart maps are directly inherited from the Tucker manifold charts via the Laplacian map (Proposition 9.2.1).

Key novelties of [FHN23].

1. The atlas $\{(\mathcal{W}(\mathbf{v}), \boldsymbol{\xi}_{\mathbf{v}})\}$ is *natural*: it is the restriction of the Tucker atlases to the intersection $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, without any additional construction.

2. The *compatibility* with Tucker manifold structures at each level is built into the definition, making the geometric relationships transparent.

3. The description holds in *infinite-dimensional Banach spaces* under the mild condition (Inj).

4. The tangent space formula (9.6) is explicit and computable.

9.8 Topology and connected components

Proposition 9.8.1. *For the Tucker manifold, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is connected for any $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$. This follows because $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ is connected (each $\mathbb{G}_r(V)$ is connected as a Banach manifold) and the fibre $\mathbb{R}_*^{\times r_\alpha}$ is connected.*

For general tree-based formats, the connectedness of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an open problem in general. The following is known:

Proposition 9.8.2. (i) *For the TT format and binary HT formats in* finite dimensions, $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ *is connected (see [UV13a, HRS12]).*

(ii) *For general T_D and infinite-dimensional spaces, connectedness of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an open question.*

Remark 9.8.3 (Stratification). The decomposition $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}\in\mathcal{AD}(\mathbf{V}_D,T_D)} \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ (Theorem 6.6.1) is a stratification of $\mathbf{V}_D$ by tree-based rank. Each stratum $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is an immersed manifold, and $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) = \overline{\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)}^{\text{weak}}$ is the weak closure. The boundary strata $\partial\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) = \mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D) \setminus \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ consist of tensors with strictly smaller rank. This stratification is the tensor-format analogue of the rank stratification of matrices.

9.9 Historical notes

The geometric structure of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ as a Banach manifold was established in [FHN23]. The fundamental observation — that the tree-based manifold is an intersection of Tucker manifolds and that its atlas can be obtained by intersecting the Tucker atlases — was made possible by the reformulation of the Tucker manifold charts using the Laplacian-like map Δ_k (Proposition 9.2.1).

The compatibility condition between the tree-based atlas and the Tucker atlases at each level (Theorem 9.4.1(ii)) is the key property that distinguishes the description of [FHN23] from earlier work: it allows one to apply the Tucker manifold geometry (tangent spaces, projections, Dirac–Frenkel principle) at each level of the tree independently, and then combine the results.

The Dirac–Frenkel variational principle for tree-based formats (Chapter 10) exploits this compatibility in an essential way: the reduced ODE on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is obtained by projecting the vector field onto the tangent space $\bigcap_k Z^{(D)}([])\mathcal{P}_k([])\mathbf{v}$, and the existence of this projection (Corollary 8.5.4 applied at each level) follows from the fact that each $Z^{(D)}([])\mathcal{P}_k([])\mathbf{v}$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$.

Part IV

Variational Principles and Applications

Chapter 10

The Dirac–Frenkel Variational Principle

The geometric framework developed in Chapters 8 and 9 has a direct dynamical application: it provides the rigorous foundation for *model reduction* of differential equations on tensor Banach spaces. Given an abstract Cauchy problem on $\mathbf{V}_{D,\|\cdot\|_D}$, one looks for an approximate solution that evolves on the manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (or $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$). The natural way to constrain the dynamics to the manifold is the *Dirac–Frenkel variational principle*: at each time t, the velocity $\dot{\mathbf{v}}_r(t)$ is chosen as the best approximation of the vector field $\mathbf{F}(t, \mathbf{v}_r(t))$ in the tangent space $Z^{(D)}(\mathbf{v}_r(t))$.

This principle was originally proposed by Dirac and Frenkel in 1930 for the time-dependent Schrödinger equation in Hilbert spaces. The present chapter extends it to general Banach tensor spaces, establishing the existence of the reduced ODE and characterising its solutions. Sections 10.1–10.2 set up the abstract framework. Section 10.4 proves the Tucker version (Theorems 5.2 and 5.4 of [FHN19]). Section 10.6 extends to the tree-based setting ([FHN23]).

10.1 The abstract Cauchy problem on tensor Banach spaces

Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces for $\alpha \in D$, and let $\mathbf{V}_{D,\|\cdot\|_D} = {}_{\|\cdot\|_D}\bigotimes_{\alpha\in D} V_\alpha$ be a reflexive Banach tensor space satisfying condition (Inj). Consider the abstract Cauchy problem:

$$\dot{\mathbf{u}}(t) = \mathbf{F}(t, \mathbf{u}(t)), \quad t \geq 0, \tag{10.1}$$

$$\mathbf{u}(0) = \mathbf{u}_0, \tag{10.2}$$

where $\mathbf{u}_0 \neq \mathbf{0}$, and $\mathbf{F} : [0,\infty) \times \mathbf{V}_{D,\|\cdot\|_D} \to \mathbf{V}_{D,\|\cdot\|_D}$ satisfies the usual conditions for existence and uniqueness of solutions (e.g., Lipschitz continuity in the second argument).

The goal is to approximate $\mathbf{u}(t)$ by a *differentiable curve* $t \mapsto \mathbf{v}_r(t)$ lying in $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ for some fixed $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, with initial condition $\mathbf{v}_r(0) = \mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ chosen as an approximation of $\mathbf{u}_0$.

Remark 10.1.1. By Theorem 4.5.4, a best approximation $\mathbf{v}_0 \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ of $\mathbf{u}_0$ exists when $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive. The initial condition $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (with exact rank $\mathfrak{r}$) can always be arranged by taking $\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}'}(\mathbf{V}_D)$ for some $\mathfrak{r}' \leq \mathfrak{r}$ and adjusting the rank.

10.2 The principle in Hilbert spaces

When $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space with inner product $\langle\cdot,\cdot\rangle_D$, the constraint to $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ takes the classical form. Since $Z^{(D)}(\mathbf{v}_r(t)) \in \mathbb{G}(\mathbf{V}_{D,\|\cdot\|_D})$ (Theorem 8.5.3), the orthogonal projection $\mathcal{P}_{\mathbf{v}_r(t)}$ onto $Z^{(D)}(\mathbf{v}_r(t))$ is well-defined and the *Dirac–Frenkel equation* reads:

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\Big(\mathbf{F}(t,\mathbf{v}_r(t))\Big), \tag{10.3}$$

which is equivalent to the variational condition:

$$\langle \dot{\mathbf{v}}_r(t) - \mathbf{F}(t,\mathbf{v}_r(t)),\, \dot{\mathbf{v}}\rangle_D = 0 \quad \text{for all } \dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t)). \tag{10.4}$$

The MCTDH (Multi-Configuration Time-Dependent Hartree) method of quantum dynamics [Lub08] is precisely (10.3) with $\mathbf{V}_{D,\|\cdot\|_D} = L^2(\mathbb{R}^{3d})$ and $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ the Hartree manifold ($\mathfrak{r} = (1,\ldots,1)$).

10.3 Projections in Banach spaces

In a general Banach space, there is no canonical inner product and hence no canonical projection. Two notions replace the orthogonal projection.

Metric projection

Definition 10.3.1. For $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $t \in I$, the *metric projection* $\mathcal{P}_{\mathbf{v}_r(t)}$ maps $\dot{\mathbf{u}} \in \mathbf{V}_{D,\|\cdot\|_D}$ to the nearest point in $Z^{(D)}(\mathbf{v}_r(t))$:

$$\mathcal{P}_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) := \operatorname*{arg\,min}_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{u}} - \dot{\mathbf{v}}\|_D.$$

By Corollary 8.5.4, when $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive the metric projection exists. When $\mathbf{V}_{D,\|\cdot\|_D}$ is strictly convex it is unique (since $Z^{(D)}(\mathbf{v}_r(t))$ is convex). The metric projection is, in general, a *nonlinear* operator on Banach spaces.

Normalised duality mapping

The duality mapping $J : \mathbf{V}_{D,\|\cdot\|_D} \to 2^{\mathbf{V}^*_{D,\|\cdot\|_D}}$ is defined by

$$J(\mathbf{u}) \coloneqq \Big\{ f \in \mathbf{V}^*_{D,\|\cdot\|_D} : \langle \mathbf{u}, f \rangle = \|\mathbf{u}\|_D^2 = \|f\|_*^2 \Big\}.$$

It is single-valued when $\mathbf{V}_{D,\|\cdot\|_D}$ is smooth (Definition 2.1.5), and uniformly continuous on bounded sets when $\mathbf{V}_{D,\|\cdot\|_D}$ is uniformly smooth. In a Hilbert space, J coincides with the Riesz isomorphism.

Theorem 10.3.2 (Metric projection characterisation, [FHN19, Theorem 5.2])**.** *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be reflexive and strictly convex, and assume* (Inj)*. For $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and fixed $t \in I$, the following are equivalent:*

(a) $\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}(\mathbf{F}(t, \mathbf{v}_r(t)))$.

(b) $\langle \dot{\mathbf{v}}_r(t) - \dot{\mathbf{v}},\ J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle \geq 0$ *for all* $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.

(c) $\langle \dot{\mathbf{v}},\ J(\mathbf{F}(t, \mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)) \rangle = 0$ *for all* $\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))$.

Proof. This is a direct consequence of the classical characterisation of the metric projection onto a closed convex set in a reflexive, strictly convex Banach space, combined with [Alb96, Proposition 2.10]. Condition (c) recovers the Hilbert-space variational form: when J is the Riesz map, (c) becomes the orthogonality condition (10.4). □

Generalised projection

When $\mathbf{V}_{D,\|\cdot\|_D}$ is additionally smooth, an alternative projection is available. Define $\phi : \mathbf{V}_{D,\|\cdot\|_D} \times \mathbf{V}_{D,\|\cdot\|_D} \to \mathbb{R}$ by

$$\phi(\mathbf{u}, \mathbf{v}) = \|\mathbf{u}\|_D^2 - 2\langle \mathbf{u}, J(\mathbf{v}) \rangle + \|\mathbf{v}\|_D^2.$$

The *generalised projection* $\Pi_{\mathbf{v}_r(t)}$ minimises ϕ over $Z^{(D)}(\mathbf{v}_r(t))$:

$$\Pi_{\mathbf{v}_r(t)}(\dot{\mathbf{u}}) \coloneqq \operatorname*{arg\,min}_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \phi(\dot{\mathbf{v}}, \dot{\mathbf{u}}).$$

It coincides with the metric projection when $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert space.

10.4 The Dirac–Frenkel principle for the Tucker manifold

Theorem 10.4.1 (Dirac–Frenkel for Tucker, [FHN19, Theorems 5.2, 5.4])**.** *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be reflexive and strictly convex, and assume* (Inj)*. Let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{r} \neq \mathbf{0}$, and let*

$\mathbf{v}_0 \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$.

(Existence) *There exists a differentiable curve* $t \mapsto \mathbf{v}_r(t)$ *in* $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, *defined for* t *in a neighbourhood of* 0, *satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\Big(\mathbf{F}(t, \mathbf{v}_r(t))\Big), \qquad \mathbf{v}_r(0) = \mathbf{v}_0. \tag{10.5}$$

(Characterisation) *The curve* $t \mapsto \mathbf{v}_r(t)$ *satisfies* (10.5) *if and only if* $\dot{\mathbf{v}}_r(t) \in Z^{(D)}(\mathbf{v}_r(t))$ *and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in Z^{(D)}(\mathbf{v}_r(t))} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \tag{10.6}$$

Sketch. **Existence.** The reduced ODE (10.5) is an ODE on the Banach manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. By Theorem 8.5.3, $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ for each $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. The right-hand side of (10.5) depends smoothly on $\mathbf{v}_r(t)$ (the metric projection onto a smoothly-varying split subspace is smooth), so local existence and uniqueness follow from the ODE existence theorem on Banach manifolds [Lan99, Theorem 4.1.5].

Characterisation. Since $Z^{(D)}(\mathbf{v}_r(t))$ is closed and convex, Corollary 8.5.4 gives existence of the minimum in (10.6). The equivalence with the metric projection condition (10.5) follows from the uniqueness (strict convexity) and Theorem 10.3.2. □

10.5 Kinematic equations for the Tucker manifold

The Dirac–Frenkel ODE (10.5) is an abstract equation on the Banach manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$. In this section we derive its explicit form in local chart coordinates, obtaining a coupled system of ODEs for the factor matrices and the core tensor. This system is the Banach-space analogue of the MCTDH equations of quantum dynamics.

Setup and notation

Fix $\mathbf{v}_r(t) \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and choose at each time t:

- Bases $\{u^{(\alpha)}_{i_\alpha}(t)\}_{i_\alpha=1}^{r_\alpha}$ of $U^{\min}_\alpha(\mathbf{v}_r(t))$ for each $\alpha \in D$, normalised so that the Gram matrices $G^{(\alpha)}_{ij}(t) := \langle u^{(\alpha)}_i(t), u^{(\alpha)}_j(t)\rangle_\alpha$ satisfy $G^{(\alpha)}(t) = I_{r_\alpha}$ when V_α is a Hilbert space (orthonormal basis), or more generally $G^{(\alpha)}(t) = I_{r_\alpha}$ modulo a choice of normalisation adapted to the Banach space structure.
- The core tensor $C^{(D)}(t) \in \mathbb{R}^{\times r_\alpha}_*$ uniquely determined by $\mathbf{v}_r(t) = \sum_{(i_\alpha)} C^{(D)}_{(i_\alpha)}(t) \bigotimes_\alpha u^{(\alpha)}_{i_\alpha}(t)$.

We also define, for each $\alpha \in D$ and $i_\alpha = 1, \ldots, r_\alpha$:

$$\mathbf{U}^{(\alpha)}_{i_\alpha}(t) := \sum_{(i_\beta)_{\beta\neq\alpha}} C^{(D)}_{(i_\alpha,(i_\beta)_{\beta\neq\alpha})}(t) \bigotimes_{\beta\neq\alpha} u^{(\beta)}_{i_\beta}(t) \in U^{\min}_{D\setminus\{\alpha\}}(\mathbf{v}_r(t)), \tag{10.7}$$

and the *mean-field matrix*

$$F^{(\alpha)}_{i_\alpha j_\alpha}(t) := \Big\langle \mathbf{U}^{(\alpha)}_{i_\alpha}(t),\ \Big(\mathrm{id}_\alpha \otimes T^{[\alpha]}_\psi\Big)(\mathbf{F}(t,\mathbf{v}_r(t)))[\psi = u^{(\alpha)}_{j_\alpha}(t)]\Big\rangle_{[\alpha]}, \tag{10.8}$$

where $T^{[\alpha]}_\psi(\mathbf{w}) := (\psi \otimes \mathrm{id}_{[\alpha]})(\mathbf{w}) \in \mathbf{V}^{[\alpha]}_\alpha$ denotes the contraction of $\mathbf{w}$ with $\psi \in V^*_\alpha$ in the α-direction.

Derivation of the kinematic system

By the tangent space identification (Proposition 8.4.2), any velocity $\dot{\mathbf{v}}_r(t) \in Z^{(D)}(\mathbf{v}_r(t))$ can be written uniquely as

$$\dot{\mathbf{v}}_r(t) = \sum_{(i_\alpha)} \dot{C}^{(D)}_{(i_\alpha)}(t) \bigotimes_\alpha u^{(\alpha)}_{i_\alpha}(t) + \sum_{\alpha\in D}\sum_{i_\alpha=1}^{r_\alpha} \dot{u}^{(\alpha)}_{i_\alpha}(t) \otimes \mathbf{U}^{(\alpha)}_{i_\alpha}(t), \tag{10.9}$$

where $\dot{u}^{(\alpha)}_{i_\alpha}(t) \in W^{\min}_\alpha(\mathbf{v}_r(t))$ (the complement of $U^{\min}_\alpha(\mathbf{v}_r(t))$ in V_α).

The Dirac–Frenkel condition (Theorem 10.3.2(a)⇔(c)):

$$\Big\langle \xi,\ J\Big(\mathbf{F}(t,\mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)\Big)\Big\rangle = 0 \quad \text{for all } \xi \in Z^{(D)}(\mathbf{v}_r(t)), \tag{10.10}$$

must hold for all basis elements of $Z^{(D)}(\mathbf{v}_r(t))$. Testing against each type of basis element yields two families of equations.

Equations for the core tensor.

Testing (10.10) against $\xi = \bigotimes_\alpha u^{(\alpha)}_{j_\alpha}(t)$ (the core basis elements) gives, for each multi-index $(j_\alpha)_{\alpha\in D}$:

$$\dot{C}^{(D)}_{(j_\alpha)}(t) = \Big\langle \bigotimes_\alpha u^{(\alpha)}_{j_\alpha}(t),\ J\Big(\mathbf{F}(t,\mathbf{v}_r(t))\Big)\Big\rangle - \sum_\alpha\sum_{i_\alpha} \Big\langle \mathbf{U}^{(\alpha)}_{i_\alpha}(t) \otimes u^{(\alpha)}_{j_\alpha}(t),\ J(\dot{u}^{(\alpha)}_{i_\alpha}(t))\Big\rangle C^{(D)}_{\dots i_\alpha \dots}(t), \tag{10.11}$$

where $J : \mathbf{V}_{D,\|\cdot\|_D} \to (\mathbf{V}_{D,\|\cdot\|_D})^*$ is the duality mapping. In a Hilbert space (where J is the Riesz isomorphism and $\langle \cdot, J(\cdot)\rangle = \langle \cdot,\cdot\rangle$), this reduces to

$$\dot{C}^{(D)}_{(j_\alpha)}(t) = \Big\langle \mathbf{F}(t,\mathbf{v}_r(t)),\ \bigotimes_\alpha u^{(\alpha)}_{j_\alpha}(t)\Big\rangle_D - \sum_\alpha\sum_{i_\alpha} \langle \dot{u}^{(\alpha)}_{i_\alpha}, u^{(\alpha)}_{j_\alpha}\rangle_\alpha C^{(D)}_{\dots i_\alpha \dots}(t). \tag{10.12}$$

Equations for the basis vectors.

Testing (10.10) against $\xi = w \otimes \mathbf{U}^{(\alpha)}_{j_\alpha}(t)$ for each $\alpha \in D$, $j_\alpha \in \{1,\dots,r_\alpha\}$, and $w \in W^{\min}_\alpha(\mathbf{v}_r(t))$ gives:

$$\Big\langle w \otimes \mathbf{U}^{(\alpha)}_{j_\alpha}(t),\ J\Big(\mathbf{F}(t,\mathbf{v}_r(t)) - \dot{\mathbf{v}}_r(t)\Big)\Big\rangle = 0 \quad \forall\, w \in W^{\min}_\alpha(\mathbf{v}_r(t)). \tag{10.13}$$

This can be rewritten, using the tensor product structure and the mean-field matrix (10.8), as:

$$P_W^{(\alpha)}(t)\left[\sum_{j_\alpha} F_{j_\alpha\cdot}^{(\alpha)}(t)\,u_{j_\alpha}^{(\alpha)}(t) - \sum_{j_\alpha} M_{j_\alpha\cdot}^{(\alpha)}(t)\,\dot{u}_{j_\alpha}^{(\alpha)}(t)\right] = 0, \tag{10.14}$$

where $P_W^{(\alpha)}(t) = P_{W_\alpha^{\min}(\mathbf{v}_r(t))\oplus U_\alpha^{\min}(\mathbf{v}_r(t))}$ is the projection onto the complement, and $M_{ij}^{(\alpha)}(t) := \langle \mathbf{U}_i^{(\alpha)}(t), \mathbf{U}_j^{(\alpha)}(t)\rangle_{[\alpha]}$ is the *mass matrix* (the Gram matrix of the $\mathbf{U}_{i_\alpha}^{(\alpha)}$ vectors).

Remark 10.5.1 (Gauge freedom and the MCTDH constraint). The kinematic equations (10.11) and (10.14) are underdetermined: one can add to $\dot{u}_{i_\alpha}^{(\alpha)}$ any vector in $U_\alpha^{\min}(\mathbf{v}_r(t))$ without changing $\dot{\mathbf{v}}_r(t)$ (since such a change stays within the core subspace and is absorbed by $\dot{C}^{(D)}$). This is the *gauge freedom* of the Tucker format (Remark 6.4.4 in Chapter 6).

The standard choice to remove this degeneracy is the *tangential constraint*:

$$\langle \dot{u}_{i_\alpha}^{(\alpha)}(t), u_{j_\alpha}^{(\alpha)}(t)\rangle_\alpha = 0 \quad \text{for all } i_\alpha, j_\alpha \in \{1,\dots,r_\alpha\},\ \alpha\in D, \tag{10.15}$$

which forces $\dot{u}_{i_\alpha}^{(\alpha)}(t) \in W_\alpha^{\min}(\mathbf{v}_r(t))$ (orthogonal to the current subspace). In the Hilbert case, this is the MCTDH tangential constraint of [KL07].

The kinematic system in the Hilbert case

In a Hilbert tensor space $\mathbf{V}_{D,\|\cdot\|_D} = \bigotimes_j H_j$ with orthonormal bases $\{u_{i_\alpha}^{(\alpha)}(t)\}$, imposing the tangential constraint (10.15) gives the closed system:

$$\dot{C}_{(j_\alpha)}^{(D)}(t) = \left\langle \mathbf{F}(t,\mathbf{v}_r(t)), \bigotimes_\alpha u_{j_\alpha}^{(\alpha)}(t)\right\rangle_D, \tag{10.16a}$$

$$\dot{u}_{j_\alpha}^{(\alpha)}(t) = P_W^{(\alpha)}(t)\left[M^{(\alpha)}(t)^{-1}\right]_{j_\alpha\cdot}\sum_{i_\alpha} F_{i_\alpha\cdot}^{(\alpha)}(t)\,u_{i_\alpha}^{(\alpha)}(t), \tag{10.16b}$$

where

$$F_{i_\alpha j_\alpha}^{(\alpha)}(t) = \left\langle \mathbf{U}_{i_\alpha}^{(\alpha)}(t), (\mathrm{id}_\alpha \otimes \langle u_{j_\alpha}^{(\alpha)}(t),\cdot\rangle_\alpha)\mathbf{F}(t,\mathbf{v}_r(t))\right\rangle_{[\alpha]}. \tag{10.17}$$

Remark 10.5.2 (Connection to MCTDH). Equations (10.16) are precisely the *Multi-Configuration Time-Dependent Hartree* (MCTDH) equations of [KL07], generalised from the rank-one case $\mathfrak{r} = \mathbf{1}$ to arbitrary Tucker rank $\mathfrak{r}$. For $\mathfrak{r} = \mathbf{1}$ (Hartree manifold, Example 10.5.4):

- The core reduces to a scalar $C^{(D)}(t) = \lambda(t) \in \mathbb{R}^*$, and (10.16a) gives $\dot{\lambda} = \langle \mathbf{F}, \bigotimes_\alpha v_\alpha\rangle_D$.
- The mean-field matrix $F^{(\alpha)}$ reduces to a scalar $F^{(\alpha)} = \langle \mathbf{u}_0, (\mathrm{id}_\otimes\langle v_\alpha,\cdot\rangle)\mathbf{F}\rangle$ (the

standard mean-field force), and (10.16b) gives $\dot{v}_\alpha = \lambda^{-1} P_W^{(\alpha)} F^{(\alpha)}$, recovering the classical Hartree equations.

Remark 10.5.3 (Banach case: the duality mapping replaces the inner product). In a non-Hilbert Banach tensor space, the kinematic equations carry through with the duality mapping J in place of the inner product. The core equation (10.11) involves $\langle \cdot, J(\cdot) \rangle$ instead of $\langle \cdot, \cdot \rangle_D$. The basis equations involve the duality pairing between V_α and V_α^*, which replaces the inner product. The mass matrix $M^{(\alpha)}$ must be replaced by the *duality mass matrix* $M_{ij}^{(\alpha)}(t) := \langle \mathbf{U}_i^{(\alpha)}(t), J(\mathbf{U}_j^{(\alpha)}(t)) \rangle_{[\alpha]}$, which is still invertible (since the $\mathbf{U}_{i_\alpha}^{(\alpha)}$ are linearly independent by Proposition 8.4.2) but no longer symmetric. The resulting system retains the same structure as (10.16) but with all inner products replaced by duality pairings — this is the general Banach-space form of the MCTDH equations.

The Hartree manifold

Example 10.5.4 (Time-dependent Hartree method)**.** For $\mathfrak{r} = \mathbf{1} = (1, \ldots, 1)$, the Tucker manifold $\mathfrak{M}_\mathbf{1}(\mathbf{V}_D)$ consists of elementary tensors $\mathbf{v} = \lambda \bigotimes_\alpha v_\alpha$ ($v_\alpha \in V_\alpha$, $\lambda \in \mathbb{R}$), called the *Hartree manifold* [Lub08]. The Dirac–Frenkel ODE (10.5) on the Hartree manifold gives the *time-dependent Hartree equations*. In a Hilbert space with $\|v_\alpha\|_\alpha = 1$ (unit sphere constraint):

$$\dot{\lambda}(t) = \langle \bigotimes_\alpha v_\alpha(t), \mathbf{F}(t, \mathbf{v}_r(t)) \rangle_D,$$
$$\lambda(t)\, \dot{v}_\alpha(t) = P_{W_\alpha(t)} \Big[(\mathrm{id}_\alpha \otimes \langle \cdot, \bigotimes_{\beta \neq \alpha} v_\beta(t) \rangle) \mathbf{F}(t, \mathbf{v}_r(t)) \Big],$$

where $P_{W_\alpha(t)}$ is the orthogonal projection onto the complement of $\mathrm{span}\{v_\alpha(t)\}$ in V_α. This is the system solved numerically by MCTDH.

10.6 Extension to tree-based formats

The extension of Theorem 10.4.1 to $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ is immediate given the immersion result of Chapter 9.

Theorem 10.6.1 (Dirac–Frenkel for tree-based formats, [FHN23, Section 5])**.** *Let T_D be a dimension partition tree, $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ such that $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ is a proper set of tree-based tensors. Assume $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive and strictly convex, and that* (Inj) *holds at each level of T_D. Let $\mathbf{v}_0 \in \mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$.*

(Existence) *There exists a differentiable curve $t \mapsto \mathbf{v}_r(t)$ in $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ satisfying*

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)} \Big(\mathbf{F}(t, \mathbf{v}_r(t)) \Big), \qquad \mathbf{v}_r(0) = \mathbf{v}_0, \tag{10.18}$$

where $\mathcal{P}_{\mathbf{v}_r(t)}$ *is the metric projection onto* $T_{\mathbf{v}_r(t)}\mathbf{i}(\mathbb{T}_{\mathbf{v}_r(t)}(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))) = \bigcap_k Z^{(D)}([)\mathcal{P}_k(T_D)]\mathbf{v}_r(t)$.

(Characterisation) *The curve satisfies* (10.18) *if and only if* $\dot{\mathbf{v}}_r(t) \in \bigcap_k Z^{(D)}([)\mathcal{P}_k(T_D)]\mathbf{v}_r(t)$ *and*

$$\|\dot{\mathbf{v}}_r(t) - \mathbf{F}(t, \mathbf{v}_r(t))\|_D = \min_{\dot{\mathbf{v}} \in \bigcap_k Z^{(D)}([)\mathcal{P}_k([)\mathbf{v}_r(t)} \|\dot{\mathbf{v}} - \mathbf{F}(t, \mathbf{v}_r(t))\|_D. \tag{10.19}$$

Proof. The proof is identical to that of Theorem 10.4.1, with $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ replaced by $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and $Z^{(D)}(\mathbf{v}_r(t))$ replaced by $\bigcap_k Z^{(D)}([)\mathcal{P}_k([)\mathbf{v}_r(t)$. The key inputs are: (i) $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is a $\mathcal{C}^\infty$-Banach manifold (Theorem 9.4.1); (ii) its tangent space is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ (Theorem 9.6.1); and (iii) the metric projection onto a split subspace in a reflexive strictly convex space is well-defined and unique. □

Remark 10.6.2 (Algorithmic interpretation). Theorem 10.6.1 provides the mathematical foundation for a class of algorithms that integrate tensor-format ODEs by evolving the transfer tensors $\{C^{(\alpha)}\}_{\alpha \in T_D}$ and the leaf basis vectors $\{u_{i_j}^{(j)}\}_{j \in \mathcal{L}(T_D)}$ along the Dirac–Frenkel trajectories. For the TT format, this is the starting point of the *projector-splitting integrator* of Lubich, Oseledets, and Vandereycken [LOV15]. For general tree-based formats, the corresponding algorithm was introduced by Ceruti, Lubich, and Walach [CLW21].

10.7 Kinematic equations for tree-based formats

The derivation of the kinematic system extends naturally to tree-based formats via the intersection structure of the tangent space.

Parametrisation of the tangent space

For $\mathbf{v}_r(t) \in \mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$, the tangent space is $\bigcap_{k=1}^{\mathrm{depth}(T_D)} Z^{(D)}([)\mathcal{P}_k(T_D)]\mathbf{v}_r(t)$ (Proposition 9.5.1). A natural parametrisation is provided by the tree-based chart coordinates:

- For each leaf $j \in \mathcal{L}(T_D)$: a basis $\{u_{i_j}^{(j)}(t)\}_{i_j=1}^{r_j}$ of $U_j^{\min}(\mathbf{v}_r(t))$.
- For each internal vertex $\alpha \in T_D \setminus (\{D\} \cup \mathcal{L}(T_D))$: a basis $\{U_{i_\alpha}^{(\alpha)}(t)\}_{i_\alpha=1}^{r_\alpha}$ of $U_\alpha^{\min}(\mathbf{v}_r(t)) \subset \bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}_r(t))$.
- The transfer tensors $C^{(\alpha)}(t)$ at each internal vertex, encoding the coefficients of $U_\alpha^{\min}(\mathbf{v}_r(t))$ in $\bigotimes_{\beta \in S(\alpha)} U_\beta^{\min}(\mathbf{v}_r(t))$.

A tangent vector $\dot{\mathbf{v}}_r(t) \in \bigcap_k Z^{(D)}([)\mathcal{P}_k([)\mathbf{v}_r(t)$ decomposes at each level k according to Proposition 8.4.2:

$$\dot{\mathbf{v}}_r(t) = \sum_{\alpha \in \mathcal{P}_k(T_D)} \sum_{(i_\alpha)} \dot{C}_{(i_\alpha)}^{(\alpha)}(t) \bigotimes_{\alpha \in \mathcal{P}_k(T_D)} u_{i_\alpha}^{(\alpha)}(t) + \sum_{\alpha \in \mathcal{P}_k(T_D)} \sum_{i_\alpha} \dot{u}_{i_\alpha}^{(\alpha)}(t) \otimes \mathbf{U}_{i_\alpha}^{(\alpha)}(t), \tag{10.20}$$

where the decomposition is independent of the level k by Proposition 9.2.1.

The kinematic system

The Dirac–Frenkel condition (10.10) tested against basis elements of $\bigcap_k Z^{(D)}([]\mathcal{P}_k([])\mathbf{v}_r(t)$ yields a system indexed by the tree structure.

Equations for the root transfer tensor. Testing against $\xi = \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} u^{(\alpha)}_{j_\alpha}(t)$ (for each multi-index $(j_\alpha)_{\alpha\in\mathcal{P}_1(T_D)}$):

$$\dot{C}^{(D)}_{(j_\alpha)}(t) = \Big\langle \bigotimes_{\alpha\in\mathcal{P}_1(T_D)} u^{(\alpha)}_{j_\alpha}(t),\ J\Big(\mathbf{F}(t,\mathbf{v}_r(t))\Big)\Big\rangle + (\text{gauge correction terms}). \tag{10.21}$$

Equations for the internal transfer tensors. For each internal vertex $\alpha \in T_D \setminus (\{D\}\cup\mathcal{L}(T_D))$, testing against $\xi = w^{(\alpha)} \otimes \mathbf{U}^{(\alpha)}_{j_\alpha}(t)$ with $w^{(\alpha)} \in W^{\min}_\alpha(\mathbf{v}_r(t))$:

$$P^{(\alpha)}_W(t)\Big[\sum_{i_\alpha} F^{(\alpha)}_{i_\alpha j_\alpha}(t)\,U^{(\alpha)}_{i_\alpha}(t) - M^{(\alpha)}(t)\,\dot{U}^{(\alpha)}_{j_\alpha}(t)\Big] = 0, \tag{10.22}$$

where $F^{(\alpha)}_{i_\alpha j_\alpha}(t)$ is the mean-field matrix at the α-level (defined analogously to (10.8)), and $M^{(\alpha)}(t)$ is the mass matrix of the $\mathbf{U}^{(\alpha)}_{i_\alpha}$.

Equations for the leaf basis vectors. For each leaf $j \in \mathcal{L}(T_D)$ and $\dot{u}^{(j)}_{i_j}(t) \in W^{\min}_j(\mathbf{v}_r(t))$, the equation has the same form as (10.14).

Remark 10.7.1 (Recursive structure and the projector-splitting integrator). The kinematic system (10.21)–(10.22) has a recursive structure matching the tree T_D: the equation at each internal vertex α involves only the transfer tensors of α and its children $S(\alpha)$. This recursive structure is precisely what the *projector-splitting integrator* of [CLW21] exploits: it solves the kinematic system level by level, from the root down to the leaves (forward sweep) and back (backward sweep), at cost $O(\mathrm{depth}(T_D)\cdot dr^3)$ per time step.

For the TT format ($T_D = T^{\mathrm{TT}}_D$, $\mathrm{depth}(T_D) = d-1$), the system reduces to d coupled equations — one per bond — each involving only adjacent components. This is the TT version of the projector-splitting integrator of [LOV15]:

$$\dot{C}^{(k)}(t) = P^{(k)}_W(t)\Big[M^{(k)}(t)^{-1}F^{(k)}(t)\Big], \quad k = 1,\dots,d, \tag{10.23}$$

where $C^{(k)}(t)$ is the k-th TT component and $F^{(k)}(t)$ is the mean-field force from the left and right environments.

10.8 Historical notes

The Dirac–Frenkel variational principle was proposed independently by Dirac [Dir30] and Frenkel [Fre34] in 1930 and 1934 for the time-dependent Schrödinger equation in Hilbert space. The formulation used here — projection of the vector field onto the tangent space of a manifold — is that of McLachlan et al. and Lubich.

Koch and Lubich [KL07] proved existence and uniqueness of solutions to the Dirac–Frenkel ODE on Tucker manifolds in finite-dimensional Hilbert spaces, and used it to analyse the MCTDH method. The extension to infinite-dimensional Banach spaces (Theorem 10.4.1) was established in [FHN19], where the key difficulty — showing that the tangent space $Z^{(D)}(\mathbf{v}_r(t))$ is a split subspace of $\mathbf{V}_{D,\|\cdot\|_D}$, without the Hilbert space structure — was resolved by the immersion theorem (Theorem 8.5.3).

The extension to tree-based formats (Theorem 10.6.1) is the main application result of [FHN23]: having established the geometry of $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ in Chapter 9, the Dirac–Frenkel principle follows with no additional effort.

Chapter 11

Open Problems and Perspectives

The results established in this monograph open several directions for further research. This chapter collects the most natural open problems, organised by theme, together with a discussion of connections to other active areas of mathematics and its applications.

11.1 Global topology of tensor manifolds

Connectedness of tree-based manifolds

We showed in Proposition 9.8.1 that each Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is connected. For tree-based manifolds with $\mathrm{depth}(T_D) \geq 2$, this question is open in the infinite-dimensional setting.

Open Problem 11.1.1. Let T_D be a dimension partition tree with $\mathrm{depth}(T_D) \geq 2$ and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D, T_D)$ with $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ a proper set. Is $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ connected?

In finite dimensions, connectedness has been established for the HT format [UV13b] and the TT format [HRS12], but the methods used there (algebraic geometry of determinantal varieties) do not directly extend to the Banach setting.

Higher homotopy groups

Even in finite dimensions, the homotopy type of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is not fully understood. Since $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a fibre bundle over the product of Grassmannians $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ (Theorem 8.3.2), the long exact sequence of homotopy groups relates $\pi_n(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$ to $\pi_n(\prod_\alpha \mathbb{G}_{r_\alpha}(\mathbb{R}^{n_\alpha}))$ and $\pi_n(\mathbb{R}_*^{\times r_\alpha})$.

Open Problem 11.1.2. Compute the fundamental group $\pi_1(\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D))$ and $\pi_1(\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D))$ in the finite-dimensional case. Are these manifolds simply connected?

Rank stratification

The decomposition $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a stratification of the algebraic tensor space. The boundary $\overline{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)} \setminus \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ consists of tensors with strictly smaller rank, and the closure $\overline{\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)}^{\|\cdot\|}$ in the Banach topology is the set $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ of tensors of bounded rank (Theorem 4.5.3).

Open Problem 11.1.3. Describe the local geometry of the boundary strata: what is the tangent cone to $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ at a boundary point $\mathbf{v} \in \mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D) \setminus \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$? Is the stratification Whitney regular?

11.2 Riemannian and Finsler geometry on tensor manifolds

In the Hilbert case, the Tucker manifold inherits a Riemannian metric from the ambient Hilbert tensor space. Riemannian gradient methods on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (ALS, Riemannian gradient, retraction-based methods) have been developed by Uschmajew and Vandereycken [UV20]. In the Banach setting, a natural replacement is a Finsler metric (a continuously varying norm on each tangent space).

Open Problem 11.2.1. Equip $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ with the Finsler metric induced by $\|\cdot\|_D$ (via the isomorphism $T_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \cong Z^{(D)}(\mathbf{v})$). Study the geodesics, curvature, and exponential map of this Finsler manifold. In particular: does the Dirac–Frenkel flow (10.5) preserve the Finsler structure in any sense?

11.3 Algorithms and convergence analysis

Projector-splitting integrators on tree-based manifolds

For the TT format, Lubich, Oseledets, and Vandereycken [LOV15] introduced the *projector-splitting integrator*, which integrates the Dirac–Frenkel ODE by splitting the projection onto $Z^{(D)}(\mathbf{v}_r(t))$ into a sequence of rank-one updates. For general tree-based formats, the corresponding integrator was introduced in [CLW21].

The convergence analysis of these algorithms in infinite-dimensional Banach spaces (as opposed to Hilbert spaces) requires understanding the stability of the Dirac–Frenkel flow on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ and the approximation properties of the splitting.

Open Problem 11.3.1. Extend the convergence analysis of the projector-splitting integrator to the setting of Banach tensor spaces. In particular, establish error bounds that depend explicitly on the geometry of the tree-based manifold $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ (curvature, injectivity radius, condition number of the rank map $\varrho_{\mathfrak{r}}$).

Alternating least squares in Banach spaces

The *alternating least squares* (ALS) algorithm and its variants (DMRG, HOSVD) minimise a functional over $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ by alternating optimisation over one factor at a time. In Hilbert spaces, convergence of ALS is well understood [UV20]. In Banach spaces, convergence requires different tools.

Open Problem 11.3.2. Prove convergence of ALS for minimisation over $\mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)$ in a reflexive Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$. What role does the geometry of the Tucker manifold (specifically, the structure of the tangent spaces $Z^{(D)}(\mathbf{v})$) play in the convergence rate?

11.4 Beyond Banach spaces

Quasi-Banach and Fréchet spaces

The component spaces V_α in this monograph are normed (Banach or at least normed). Many function spaces of interest — L^p for $0 < p < 1$, or spaces of smooth functions — are quasi-Banach or Fréchet but not Banach. The theory of minimal subspaces and Tucker manifolds should extend to these settings, but the proofs require substantial modifications.

Open Problem 11.4.1. Develop the theory of minimal subspaces and tensor manifolds for quasi-Banach spaces V_α with $0 < p < 1$. In particular, does the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ retain its smooth structure when the V_α are L^p spaces for $p < 1$?

Infinite-dimensional index sets

All results in this monograph assume $D = \{1, \ldots, d\}$ is finite. For problems in infinite dimensions — e.g., infinite particle quantum systems or PDEs in infinitely many variables — one would want D to be infinite. The algebraic tensor product over an infinite index set requires projective limits or nuclear spaces.

Open Problem 11.4.2. Extend the tree-based tensor framework to infinite index sets $D = \mathbb{N}$ (or more general sets). What is the natural analogue of a dimension partition tree, and what conditions on the norms guarantee the existence of minimal subspaces and best approximations?

11.5 Connections with quantum information

Tensor networks and entanglement

In quantum information, a *tensor network state* on a lattice is a tensor $\mathbf{v} \in \bigotimes_{j\in D} \mathbb{C}^{d_j}$ whose transfer tensors $\{C^{(\alpha)}\}$ are specified by a graph (not necessarily a tree). The tree-based format corresponds to *tree tensor networks* (TTN), which include MPS/TT as the linear-chain case and MERA (multi-scale entanglement renormalisation ansatz) as a more general tree.

The *entanglement entropy* of a bipartition $D = \alpha \cup \alpha^c$ is $S(\alpha) = -\mathrm{tr}(\rho_\alpha \log \rho_\alpha)$, where ρ_α is the reduced density matrix of the state on α. The *area law* for ground states of local Hamiltonians states that $S(\alpha)$ grows at most as $|\partial\alpha|$ (the boundary of α), not as $|\alpha|$ (the volume). This motivates the use of tree-based formats: the rank constraints in $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ directly control the entanglement structure.

Open Problem 11.5.1. Formulate the Dirac–Frenkel variational principle for tree tensor networks in the language of C^*-algebras and operator systems. What is the correct generalisation of the Tucker manifold in the non-commutative setting?

11.6 Learning on tensor manifolds

Statistical learning theory

In machine learning, tensor network states arise naturally as *function approximation schemes* for high-dimensional data. The generalisation capacity of these models depends on the geometry of the hypothesis class $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$.

The *Rademacher complexity* of $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ controls the generalisation error via the Dudley entropy integral. For finite D and fixed ranks, this can be bounded using the Weyl tube formula once the curvature of the manifold $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ is understood.

Open Problem 11.6.1. Compute the Rademacher complexity and metric entropy of $\mathcal{FT}_{\leq\mathfrak{r}}(\mathbf{V}_D, T_D)$ in terms of the tree-based rank $\mathfrak{r}$, the depth $\mathrm{depth}(T_D)$, and the dimensions $\dim V_j$. How does the choice of tree T_D affect the approximation-complexity tradeoff?

Optimal tree selection

Given a function $f : \prod_j I_j \to \mathbb{R}$ (a target function for approximation), which tree T_D gives the best approximation of f by a tree-based tensor of a given rank? This question connects the geometry of tensor manifolds to the combinatorics of tree structures.

Open Problem 11.6.2. Given $f \in \mathbf{V}_{D,\|\cdot\|_D}$ and a bound r on the tree-based rank, find the dimension partition tree T_D^* that minimises the approximation error $\inf_{\mathbf{v} \in \mathcal{FT}_{\leq \mathbf{r}}(\mathbf{V}_D, T_D^*)} \|f - \mathbf{v}\|_D$. Is this optimisation problem tractable?

Appendix A

Banach Manifolds: A Summary

This appendix collects the main definitions and results from infinite-dimensional differential geometry that are used throughout the monograph. The material is self-contained but terse; proofs are given only when they do not appear in Chapter 2. Standard references are [Lan99] and [Upm85].

A.1 Banach manifolds

Atlases and manifolds

Let $\mathbb{M}$ be a set. A $\mathcal{C}^p$-*atlas* on $\mathbb{M}$ (for $p \in \mathbb{N}_0 \cup \{\infty, \omega\}$, where ω denotes the analytic case) is a family of *charts* $\{(M_i, u_i)\}_{i \in A}$ where each $M_i \subset \mathbb{M}$ and $u_i : M_i \to U_i$ is a bijection onto an open set U_i in a Banach space X_i, subject to:

(AT1) $\bigcup_{i \in A} M_i = \mathbb{M}$.

(AT2) For all $i, j \in A$, the set $u_i(M_i \cap M_j)$ is open in X_i.

(AT3) Whenever $M_i \cap M_j \neq \emptyset$, the *transition map* $u_j \circ u_i^{-1} : u_i(M_i \cap M_j) \to u_j(M_i \cap M_j)$ is a $\mathcal{C}^p$-diffeomorphism.

Two atlases are *compatible* if their union satisfies (AT1)–(AT3). A $\mathcal{C}^p$-*Banach manifold* is a set together with a maximal compatible $\mathcal{C}^p$-atlas. When all model spaces X_i are isomorphic to a single Banach space X, the manifold is *modelled on* X.

Tangent spaces and differentiable maps

Let $\mathbb{M}$ be a $\mathcal{C}^p$-Banach manifold with $p \geq 1$, and let $m \in \mathbb{M}$. A *tangent vector* at m is an equivalence class of triples (M_i, u_i, v) where (M_i, u_i) is a chart at m and $v \in X_i$, under $(M_i, u_i, v) \sim (M_j, u_j, w)$ iff $D(u_j \circ u_i^{-1})(u_i(m)) \cdot v = w$. The *tangent space* $T_m\mathbb{M}$ is the vector space of all tangent vectors at m; it is isomorphic to X_i via any chart at m.

A map $f : \mathbb{M} \to \mathbb{N}$ between $\mathcal{C}^p$-manifolds is *of class* $\mathcal{C}^p$ if for all charts (M_i, u_i) on $\mathbb{M}$ and (N_j, v_j) on $\mathbb{N}$, the map $v_j \circ f \circ u_i^{-1}$ is $\mathcal{C}^p$ wherever defined. The *tangent map* $Tf(m) : T_m\mathbb{M} \to T_{f(m)}\mathbb{N}$ is the bounded linear map induced by differentiating the local representative.

Immersions and submanifolds

A $\mathcal{C}^1$-map $f : \mathbb{M} \to \mathbb{N}$ is an *immersion* at m if $Tf(m)$ is injective with closed split image in $T_{f(m)}\mathbb{N}$. It is a *(global) immersion* if this holds at every point. An *immersed submanifold* is the image of an injective immersion.

Remark A.1.1. In finite dimensions, or in Hilbert spaces, the split condition on $\operatorname{Im} Tf(m)$ is automatic. In general Banach spaces it is a genuine constraint: a closed subspace need not be complemented. The immersion theorems of Chapters 8 and 9 are precisely theorems establishing this split condition for the tensor manifolds.

A.2 The Grassmann–Banach manifold

Split subspaces

Let X be a Banach space. A closed subspace $U \subset X$ is *split* (or *complemented*) if there exists a closed subspace W with $X = U \oplus W$ (direct topological sum). Every finite-dimensional subspace is split. In a Hilbert space, every closed subspace is split (via orthogonal complement). In a general Banach space this fails; a Banach space in which every closed subspace is complemented is isomorphic to a Hilbert space (Lindenstrauss–Tzafriri).

The Grassmannian

The *Grassmannian* of X is

$$\mathbb{G}(X) := \{U \subset X : U \text{ is a split subspace}\},$$

with the *finite Grassmannian* $\mathbb{G}_r(X) := \{U \in \mathbb{G}(X) : \dim U = r\}$ for $r \in \mathbb{N}_0$.

Theorem A.2.1 (Douady [Dou66]). *$\mathbb{G}(X)$ is an analytic Banach manifold. For each splitting $X = U \oplus W$, the chart domain $\mathbb{G}(W, X) := \{V \in \mathbb{G}(X) : X = V \oplus W\}$ is mapped analytically to $\mathcal{L}(U, W)$ by*

$$\Psi_{U \oplus W}(V) := P_{W \oplus U}\big|_V \circ \left(P_{U \oplus W}\big|_V\right)^{-1}.$$

The transition maps are analytic, and $T_U\mathbb{G}_r(X) \cong \mathcal{L}(U, W)$ for any complement W of U.

Each connected component $\mathbb{G}_r(X)$ is an analytic Banach manifold modelled on $\mathcal{L}(U, W)$, whose isomorphism class depends only on r and X (not on the choice of U or W).

The nilpotent algebra and chart inverses

For a splitting $X = U \oplus W$, define

$$\mathcal{L}_{(U,W)}(X) := \left\{ P_{W\oplus U} \circ S \circ P_{U\oplus W} : S \in \mathcal{L}(X) \right\} \cong \mathcal{L}(U, W).$$

Every element $L \in \mathcal{L}_{(U,W)}(X)$ satisfies $L^2 = 0$, so $\exp(L) = \mathrm{id}_X + L$ and $(\mathrm{id}_X + L)^{-1} = \mathrm{id}_X - L$. The inverse chart is $\Psi^{-1}_{U\oplus W}(L) = \{(\mathrm{id}_X + L)(u) : u \in U\}$, a degree-one polynomial map; see Proposition 2.2.4.

A.3 Fréchet derivatives and regularity

Let X, Y be Banach spaces and $U \subset X$ open. A map $f : U \to Y$ is *Fréchet differentiable* at x_0 if there exists $A \in \mathcal{L}(X, Y)$ such that

$$\lim_{\|h\|\to 0} \frac{\|f(x_0 + h) - f(x_0) - Ah\|}{\|h\|} = 0.$$

The map $A = Df(x_0)$ is the *Fréchet derivative* at x_0. Higher derivatives $D^k f$ are defined inductively; f is $\mathcal{C}^k$ if $D^k f : U \to \mathcal{L}(X, \mathcal{L}(X, \ldots, \mathcal{L}(X, Y) \ldots))$ is continuous.

Proposition A.3.1 ([HN12])**.** *Every continuous multilinear map $M : X_1 \times \cdots \times X_d \to Y$ is $\mathcal{C}^\infty$, with $D^k M = 0$ for $k > d$.*

Proposition A.3.2 ([HN12])**.** *Let X, Y be complex Banach spaces and $U \subset X$ open. A map $f : U \to Y$ is analytic if and only if it is $\mathcal{C}^1$-Fréchet differentiable.*

Corollary A.3.3. *Every continuous multilinear map between complex Banach spaces is analytic. Consequently, the Tucker and tree-based tensor manifolds are analytic Banach manifolds when the component spaces are complex, and $\mathcal{C}^\infty$ in the real case.*

A.4 Vector fields and ODEs on Banach manifolds

A *(smooth) vector field* on a $\mathcal{C}^\infty$-Banach manifold $\mathbb{M}$ is a $\mathcal{C}^\infty$-section of the tangent bundle: a map $F : \mathbb{M} \to T\mathbb{M}$ with $F(m) \in T_m\mathbb{M}$ for all m. In a chart (M_i, u_i), the vector field is represented by a $\mathcal{C}^\infty$-map $f_i : u_i(M_i) \to X_i$.

An *integral curve* of F through $m_0 \in \mathbb{M}$ is a $\mathcal{C}^1$-curve $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{M}$ with $\gamma(0) = m_0$ and $\dot{\gamma}(t) = F(\gamma(t))$.

Theorem A.4.1 (Existence and uniqueness, [Lan99, Theorem 4.1.5])**.** *Let $\mathbb{M}$ be a $\mathcal{C}^\infty$-Banach manifold and F a $\mathcal{C}^\infty$ vector field on $\mathbb{M}$. For every $m_0 \in \mathbb{M}$ there exists $\varepsilon > 0$ and a unique integral curve $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{M}$ of F through m_0. The flow map $(t, m_0) \mapsto \gamma(t)$ is $\mathcal{C}^\infty$.*

This theorem is applied in Chapter 10 to the Dirac–Frenkel projected vector field on the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ and on $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$.

A.5 The tangent bundle

Construction

Let $\mathbb{M}$ be a $\mathcal{C}^p$-Banach manifold with $p \geq 1$. The *tangent bundle* of $\mathbb{M}$ is the disjoint union

$$T\mathbb{M} := \bigsqcup_{m \in \mathbb{M}} T_m\mathbb{M} = \{(m, v) : m \in \mathbb{M},\ v \in T_m\mathbb{M}\}, \tag{A.1}$$

equipped with the natural projection $\pi_{T\mathbb{M}} : T\mathbb{M} \to \mathbb{M}$, $(m, v) \mapsto m$.

Each chart (M_i, u_i) on $\mathbb{M}$, with model space X_i, gives rise to a *chart on* $T\mathbb{M}$:

$$Tu_i : \pi_{T\mathbb{M}}^{-1}(M_i) \longrightarrow u_i(M_i) \times X_i, \qquad (m, v) \longmapsto \big(u_i(m),\, [v]_i\big),$$

where $[v]_i$ is the representative of the tangent vector $v \in T_m\mathbb{M}$ in the chart (M_i, u_i). The collection $\{(\pi_{T\mathbb{M}}^{-1}(M_i), Tu_i)\}_{i \in A}$ is a $\mathcal{C}^{p-1}$-atlas on $T\mathbb{M}$, making it a $\mathcal{C}^{p-1}$-Banach manifold modelled on $X_i \times X_i$.

Proposition A.5.1 ([Lan99, Chapter 3])**.** *Let $\mathbb{M}$ be a $\mathcal{C}^p$-Banach manifold modelled on a Banach space X. Then:*

(i) *$T\mathbb{M}$ is a $\mathcal{C}^{p-1}$-Banach manifold modelled on $X \times X$, and $\pi_{T\mathbb{M}} : T\mathbb{M} \to \mathbb{M}$ is a $\mathcal{C}^{p-1}$-submersion.*

(ii) *Each fibre $\pi_{T\mathbb{M}}^{-1}(m) = T_m\mathbb{M}$ is a Banach space isomorphic to X.*

(iii) *The zero section $\sigma_0 : \mathbb{M} \to T\mathbb{M}$, $m \mapsto (m, 0)$, is a $\mathcal{C}^{p-1}$-embedding.*

Tangent maps and the chain rule

Given a $\mathcal{C}^p$-map $f : \mathbb{M} \to \mathbb{N}$ between Banach manifolds, the *tangent map* (or *differential*) of f is the $\mathcal{C}^{p-1}$-map

$$Tf : T\mathbb{M} \longrightarrow T\mathbb{N}, \qquad (m, v) \longmapsto \big(f(m),\, Tf(m)(v)\big),$$

where $Tf(m) : T_m\mathbb{M} \to T_{f(m)}\mathbb{N}$ is the bounded linear map defined in any pair of charts by the Fréchet derivative of the local representative. The *chain rule* holds: if $g : \mathbb{N} \to \mathbb{P}$ is $\mathcal{C}^p$, then

$$T(g \circ f)(m) = Tg(f(m)) \circ Tf(m).$$

Local triviality

Although $T\mathbb{M}$ is globally non-trivial in general (the tangent bundle of the sphere S^2 has no global frame, by the hairy ball theorem), it is always *locally trivial*: for each chart domain M_i, the map Tu_i gives a diffeomorphism $\pi_{T\mathbb{M}}^{-1}(M_i) \cong M_i \times X_i$. This local product structure is precisely the data of a *vector bundle*, which we define in the next section.

Remark A.5.2. For the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, the tangent bundle $T\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ has fibres $T_{\mathbf{v}}\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \cong \mathbf{E}_D$ (the model Banach space, see §A.6). The inclusion $i : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D,\|\cdot\|_D}$ induces a tangent map $Ti(\mathbf{v}) : \mathbf{E}_D \to \mathbf{V}_{D,\|\cdot\|_D}$ whose image is the subspace $Z^{(D)}(\mathbf{v})$ studied in Chapters 8–10. The immersion theorem (Theorem 8.5.3) asserts that $Ti(\mathbf{v})$ is injective with closed split image, i.e., that i is a genuine immersion in the sense of Definition A.1.

A.6 Fibre bundles and the Tucker bundle structure

Vector bundles

A $\mathcal{C}^p$*-vector bundle* over $\mathbb{M}$ with fibre F (a Banach space) is a $\mathcal{C}^p$-Banach manifold E together with a $\mathcal{C}^p$-surjection $\pi : E \to \mathbb{M}$ such that:

(VB1) Each fibre $E_m := \pi^{-1}(m)$ is a Banach space isomorphic to F.

(VB2) There is an open cover $\{M_i\}$ of $\mathbb{M}$ and $\mathcal{C}^p$-diffeomorphisms (local trivialisations) $\Phi_i : \pi^{-1}(M_i) \xrightarrow{\sim} M_i \times F$ such that $\mathrm{pr}_1 \circ \Phi_i = \pi$.

(VB3) The *transition maps* $\Phi_j \circ \Phi_i^{-1} : (M_i \cap M_j) \times F \to (M_i \cap M_j) \times F$ have the form $(m, v) \mapsto (m, g_{ij}(m)(v))$, where $g_{ij} : M_i \cap M_j \to \mathrm{GL}(F)$ is $\mathcal{C}^p$.

The tangent bundle $T\mathbb{M}$ is a vector bundle with fibre X (the model space). The trivial bundle is $\mathbb{M} \times F$.

Principal bundles

A $\mathcal{C}^p$*-principal bundle* with structure group G (a Lie group) is a fibre bundle $\pi : E \to \mathbb{M}$ on which G acts freely and properly on the right, with $E/G \cong \mathbb{M}$ and local trivialisations equivariant under the G-action.

Example A.6.1 (Frame bundle)**.** The *frame bundle* $\mathrm{Fr}(\mathbb{M})$ of a Banach manifold $\mathbb{M}$ modelled on X has fibre $\mathrm{Fr}_m(\mathbb{M}) = \mathrm{GL}(X, T_m\mathbb{M})$ (bounded linear isomorphisms from X to $T_m\mathbb{M}$). It is a principal $\mathrm{GL}(X)$-bundle over $\mathbb{M}$.

The Tucker manifold as a fibre bundle

The central structural theorem of Chapter 8 can be stated in this language as follows.

Theorem A.6.2 (Tucker fibre bundle, Theorem 8.3.2)**.** *Let $(V_\alpha, \|\cdot\|_\alpha)$ be normed spaces for $\alpha \in D$ and let $\mathfrak{r} \in \mathcal{AD}(\mathbf{V}_D)$ with $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ proper. Then the rank map*

$$\varrho_\mathfrak{r} : \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \longrightarrow \prod_{\alpha \in D} \mathbb{G}_{r_\alpha}(V_\alpha), \qquad \mathbf{v} \longmapsto \Big(U_\alpha^{\min}(\mathbf{v})\Big)_{\alpha \in D},$$

makes $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ into a $\mathcal{C}^\infty$-fibre bundle over the product of finite Grassmannians $\prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$, with typical fibre

$$F_\mathfrak{r} := \mathbb{R}_*^{\times_{\alpha\in D} r_\alpha} := \Big\{C^{(D)} \in \mathbb{R}^{\times_\alpha r_\alpha} : C^{(D)} \textit{ has full Tucker rank } \mathfrak{r}\Big\}.$$

Remark A.6.3 (Special cases).

(i) For $d = 2$ and any $r \geq 1$, the fibre $F_\mathfrak{r} = \mathrm{GL}(r)\mathbb{R}$ is the general linear group, making $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ a principal $\mathrm{GL}(r)$-bundle when $r_1 = r_2 = r$.

(ii) For $r_\alpha = 1$ for all $\alpha \in D$ (rank-one tensors), the fibre is $F_\mathfrak{r} = \mathbb{R}^* = \mathbb{R} \setminus \{0\}$, a principal $\mathbb{R}^*$-bundle (a line bundle).

(iii) In both cases (i) and (ii) the structure group is a Lie group, making the bundle a principal bundle. For general rank tuples $\mathfrak{r}$, the fibre $F_\mathfrak{r}$ is an open subset of a Euclidean space, hence a smooth finite-dimensional manifold but not a group.

Trivialisation over a chart of the base

Fix $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ and let $(U_\alpha)_\alpha = \varrho_\mathfrak{r}(\mathbf{v}) = (U_\alpha^{\min}(\mathbf{v}))_\alpha$. The chart domain at $\mathbf{v}$ in the Tucker manifold atlas is

$$\mathcal{U}(\mathbf{v}) = \varrho_\mathfrak{r}^{-1}\Big(\prod_\alpha \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha)\Big),$$

where $\mathbb{G}(W, V) = \{U \in \mathbb{G}(V) : V = U \oplus W\}$ is the chart domain of the Grassmannian of V_α at $U_\alpha^{\min}(\mathbf{v})$. The local trivialisation is

$$\chi_\mathbf{v} : \mathcal{U}(\mathbf{v}) \xrightarrow{\sim} \prod_\alpha \mathbb{G}(W_\alpha^{\min}(\mathbf{v}), V_\alpha) \times F_\mathfrak{r}, \qquad \mathbf{w} \longmapsto \Big((U_\alpha^{\min}(\mathbf{w}))_\alpha,\ C^{(D)}(\mathbf{w})\Big), \tag{A.2}$$

where $C^{(D)}(\mathbf{w})$ is the core tensor of $\mathbf{w}$ in the basis induced by the chart. This is exactly the chart map $\xi_{\mathbf{v}}$ of Theorem 8.3.2, split into its base and fibre components.

A.7 Why the quotient construction fails in infinite dimensions

In finite dimensions, $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is commonly described as a *quotient manifold*. This section explains why that description does not extend to infinite dimensions, and why the explicit atlas approach of Chapter 8 is necessary.

The finite-dimensional quotient picture

Let $d = 2$ and $V_1 = \mathbb{R}^{n_1}$, $V_2 = \mathbb{R}^{n_2}$. A rank-r matrix $M \in \mathbb{R}^{n_1 \times n_2}$ can be written as $M = AB^\top$ with $A \in \mathbb{R}_*^{n_1 \times r}$ (full column rank) and $B \in \mathbb{R}_*^{n_2 \times r}$. The pair (A, B) is not unique: for any $G \in \mathrm{GL}(r)\mathbb{R}$, the pair $(AG, B(G^{-\top}))$ gives the same product. Thus, in finite dimensions,

$$\mathfrak{M}_r(\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}) \cong \left(\mathbb{R}_*^{n_1 \times r} \times \mathbb{R}_*^{n_2 \times r}\right) \Big/ \mathrm{GL}(r)\mathbb{R}, \tag{A.3}$$

where $\mathrm{GL}(r)\mathbb{R}$ acts by simultaneous right multiplication. Since $\mathbb{R}_*^{n_j \times r}$ is an open subset of $\mathbb{R}^{n_j \times r}$ and $\mathrm{GL}(r)\mathbb{R}$ acts freely and properly, the quotient (A.3) is a smooth manifold by the principal bundle theorem.

The same construction works for the Tucker format with $d \geq 3$: one takes the product $\prod_{\alpha \in D} \mathcal{L}(U_\alpha, V_\alpha)_*$ of spaces of injective linear maps, modulo simultaneous gauge transformations by $\prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$.

Failure in infinite dimensions

When V_α are infinite-dimensional Banach spaces, the quotient construction (A.3) encounters two fundamental obstructions.

Obstruction 1: The stabiliser group is infinite-dimensional. Replace $\mathbb{R}_*^{n_j \times r}$ by the space of injective bounded operators $\mathcal{L}_{\mathrm{inj}}(U_\alpha, V_\alpha)$ (where $U_\alpha \cong \mathbb{R}^{r_\alpha}$ is finite-dimensional). The structure group $\prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$ remains finite-dimensional, so this obstruction does not arise directly for the Tucker case. However, for the generalisation to tree-based formats, the relevant group includes automorphisms of the component spaces V_α at internal nodes of the tree, which are infinite-dimensional Banach spaces. Their automorphism group $\mathrm{GL}(V_\alpha)$ is an infinite-dimensional Lie group whose quotient topology may not be Hausdorff.

Obstruction 2: The quotient topology may not be well-behaved. The *principal bundle theorem* (which guarantees that E/G is a smooth manifold when G acts freely and properly on E) requires either that G is a finite-dimensional Lie group, or that the action is a *proper and slice-admitting* action in the sense of Palais. For infinite-dimensional G, these conditions are non-trivial and frequently fail. In particular:

- The orbit map $G \to G \cdot e$ need not be a homeomorphism onto its image (the orbit may not be a submanifold).

- The quotient E/G equipped with the quotient topology may not be locally Euclidean (not locally modelled on a Banach space).

Remark A.7.1 (Tree-based formats). For tree-based tensor formats $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ with $\mathrm{depth}(T_D) \geq 2$, the natural gauge group includes transformations at internal nodes of the tree that act on the *transfer tensors* $C^{(\alpha)}$. The stabiliser of a generic element includes change-of-basis matrices at each internal vertex, which, at nodes α with $\#(\alpha) \geq 2$, are automorphisms of the infinite-dimensional space $\bigotimes_{\beta \in S(\alpha)} V_\beta$. The quotient construction therefore fails more severely for tree-based formats than for the Tucker format.

This is the precise reason why Chapters 8–9 construct the manifold structure via explicit local charts rather than via the quotient approach.

The explicit atlas as a resolution

The approach of [FHN19] (Chapter 8) bypasses both obstructions by working with the Grassmannian structure directly. The key observations are:

1. The map $\varrho_{\mathfrak{r}} : \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \to \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$ is well-defined without reference to any group action: it sends $\mathbf{v}$ to its minimal subspaces, which are intrinsic.

2. The fibre $\varrho_{\mathfrak{r}}^{-1}(U_\alpha)_\alpha$ over a fixed tuple of subspaces is isomorphic to $F_{\mathfrak{r}} = \mathbb{R}_*^{\times r_\alpha}$, a finite-dimensional smooth manifold.

3. The local trivialisations (A.2) are constructed from the Grassmann charts $\Psi_{U_\alpha \oplus W_\alpha}$, which are defined without any group action and are automatically $\mathcal{C}^\infty$.

The result is a description of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ as a fibre bundle over a product of Grassmannians, with finite-dimensional fibre and infinite-dimensional base, without any quotient. The structure group reduces to $\prod_\alpha \mathrm{GL}(r_\alpha)\mathbb{R}$ (a finite-dimensional Lie group acting on the fibres), so the bundle is in fact a principal bundle in the finite-dimensional-structure-group sense, even though the total space and base are infinite-dimensional.

Remark A.7.2 (The Uschmajew–Vandereycken approach). Uschmajew and Vandereycken [UV13b] studied the HT manifold in finite dimensions as a quotient by a gauge group. Their approach gives the same manifold structure in finite dimensions (where both constructions are available), but does not extend to infinite dimensions. The comparison is discussed further in Chapter 8 (historical notes).

A.8 Connections, parallel transport, and curvature

This section briefly describes connections on vector bundles, included for completeness; the material is not used in the proofs of this monograph but is relevant to the open problems of Chapter 11.

Connections on vector bundles

Let $\pi : E \to \mathbb{M}$ be a $\mathcal{C}^\infty$ vector bundle. A *(linear) connection* on E is a bilinear map $\nabla : \mathfrak{X}(\mathbb{M}) \times \Gamma(E) \to \Gamma(E)$, $(X, s) \mapsto \nabla_X s$, where $\mathfrak{X}(\mathbb{M})$ denotes the $\mathcal{C}^\infty$ vector fields and $\Gamma(E)$ the sections of E, satisfying:

(C1) $\nabla_{fX} s = f \nabla_X s$ for all $f \in \mathcal{C}^\infty(\mathbb{M})$.

(C2) $\nabla_X (fs) = (Xf)s + f \nabla_X s$ (Leibniz rule).

In a local trivialisation, ∇ is represented by a $\mathcal{L}(F)$-valued 1-form (the *connection form*), and $\nabla_X s = X(s) + \omega(X)(s)$ where ω is the connection 1-form.

Parallel transport

Given a connection ∇ and a $\mathcal{C}^1$-curve $\gamma : [0, 1] \to \mathbb{M}$, *parallel transport* along γ is the family of isomorphisms $P_t : E_{\gamma(0)} \to E_{\gamma(t)}$ defined by solving the ODE $\nabla_{\dot\gamma(t)} s(t) = 0$, $s(0) = e_0$. Parallel transport preserves the fibre structure and is used to define the covariant derivative of sections along curves.

The Levi-Civita connection on Riemannian manifolds

If $\mathbb{M}$ is a (possibly infinite-dimensional) Riemannian manifold — that is, each $T_m\mathbb{M}$ carries a smoothly varying inner product — then there is a unique torsion-free metric connection, the *Levi-Civita connection*. Its curvature tensor $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$ measures the failure of parallel transport to commute around infinitesimal loops.

Remark A.8.1 (Relevance to tensor manifolds). The Tucker manifold $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$ and the tree-based manifold $\mathcal{FT}_\mathfrak{r}(\mathbf{V}_D, T_D)$ inherit a Riemannian structure from the ambient space $\mathbf{V}_{D,\|\cdot\|_D}$ whenever $\mathbf{V}_{D,\|\cdot\|_D}$ is a Hilbert tensor space (e.g., $\bigotimes_{j=1}^d L^2(I_j)$). The Levi-Civita connection and its curvature tensor control the stability and accuracy of geometric integration schemes on these manifolds — in particular the projector-splitting integrator of Lubich [KL07]. In the Banach (non-Hilbert) setting, no natural Riemannian structure is available, and the curvature is replaced by more subtle nonlinear invariants. This is one of the sources of difficulty in Open Problem 11.3.1.

Holonomy

The *holonomy group* at $m \in \mathbb{M}$ is the group of parallel transport maps around closed loops based at m. For the Grassmannian $\mathbb{G}_r(X)$ (with the natural $U(r)$-invariant connection in the Hilbert case), the holonomy group is $U(r)$. Understanding the holonomy of the Tucker and tree-based manifolds is related to Open Problem 11.1.2 (fundamental group and higher homotopy groups).

Appendix B

Functional Analysis: Selected Results

This appendix collects classical results from Banach space theory and convex analysis that are used at various points in the monograph but whose proofs would interrupt the main narrative. Standard references are [Hac19] (tensor analysis viewpoint) and [Lan99] (differential-geometric perspective).

B.1 Geometry of Banach spaces

Strict and uniform convexity

Definition B.1.1. A normed space $(X, \|\cdot\|)$ is:

(i) *Strictly convex* (rotund) if $\|x+y\|/2 < 1$ whenever $\|x\| = \|y\| = 1$ and $x \neq y$.

(ii) *Uniformly convex* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x+y\|/2 \leq 1-\delta$ whenever $\|x\|, \|y\| \leq 1$ and $\|x-y\| \geq \varepsilon$.

(iii) *Smooth* if $\lim_{t\to 0}(\|x+ty\| - \|x\|)/t$ exists for all x, y with $\|x\| = 1$.

Uniform convexity implies strict convexity and reflexivity (Clarkson's theorem). The L^p spaces satisfy: L^p is uniformly convex for $1 < p < \infty$; L^1 and L^∞ are neither strictly convex nor smooth in general. The duality relations are: X is smooth $\Leftrightarrow$ X^* is strictly convex, and X is strictly convex $\Leftrightarrow$ X^* is smooth.

Duality mapping

Definition B.1.2. The *normalised duality mapping* $J : X \to 2^{X^*}$ of a Banach space X is

$$J(x) := \Big\{ f \in X^* : \langle x, f\rangle = \|x\|^2 = \|f\|^2_{X^*} \Big\}.$$

The duality mapping is:

- Always non-empty (Hahn–Banach theorem).
- Single-valued on $X \setminus \{0\}$ if and only if X is smooth.
- The Riesz isomorphism $J : H \to H^*$, $J(x) = \langle x, \cdot\rangle$, when $X = H$ is a Hilbert space.
- Uniformly continuous on bounded sets when X is uniformly smooth.

Metric and generalised projections

Let $C \subset X$ be a closed convex subset of a Banach space X.

Definition B.1.3. (i) The *metric projection* $P_C : X \to 2^C$ assigns to each $x \in X$ the set of nearest points in C:

$$P_C(x) \coloneqq \arg\min_{c \in C} \|x - c\|.$$

(ii) The *Lyapunov functional* is $\phi(x, y) \coloneqq \|x\|^2 - 2\langle x, J(y)\rangle + \|y\|^2$.

(iii) The *generalised projection* $\Pi_C : X \to 2^C$ is

$$\Pi_C(x) \coloneqq \arg\min_{c \in C} \phi(c, x).$$

Key properties:

- $P_C(x)$ is non-empty for all x if and only if C is *proximinal.* Every closed convex set in a reflexive Banach space is proximinal (see [Hac19, p. 61]).
- $P_C(x)$ is a singleton for all x if X is strictly convex.
- $\Pi_C(x)$ is non-empty whenever X is reflexive and smooth; it is a singleton whenever X is additionally strictly convex.
- In a Hilbert space, $\phi(x, y) = \|x - y\|^2$, so $P_C = \Pi_C$ (orthogonal projection).

Proposition B.1.4 ([Alb96])**.** *Let X be reflexive and strictly convex, and $C \subset X$ closed convex. For $x \in X$ and $c_* \in C$, the following are equivalent:*

(a) $c_* = P_C(x)$ *(metric projection).*

(b) $\langle c_* - c, J(x - c_*)\rangle \geq 0$ *for all* $c \in C$.

(c) $\langle c, J(x - c_*)\rangle = 0$ *for all* $c \in C$ *(when C is a closed linear subspace).*

Condition (c) is the Banach-space analogue of orthogonality and is used in Chapter 10 to characterise the Dirac–Frenkel projection onto the tangent space $Z^{(D)}(\mathbf{v})$.

B.2 Weak topology and weak convergence

Theorem B.2.1 (Eberlein–Šmulian)**.** *A Banach space X is reflexive if and only if every bounded sequence has a weakly convergent subsequence.*

Theorem B.2.2 (Mazur)**.** *A convex subset of a Banach space is weakly closed if and only if it is norm-closed.*

Corollary B.2.3. *The norm $\|\cdot\|$ is weakly lower semicontinuous: if $x_n \rightharpoonup x$, then $\|x\| \leq \liminf_n \|x_n\|$.*

These results are used in Chapter 4 (and Chapter 7) to prove existence of best Tucker and tree-based approximations: a minimising sequence in the weakly closed set $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ has a weakly convergent subsequence (by reflexivity), whose limit lies in the set (by weak closedness) and achieves the infimum (by weak lower semicontinuity of the norm).

B.3 Crossnorms and the injective norm condition

Let $V_1, \ldots, V_d$ be normed spaces and $\mathbf{V}_D = \bigotimes_{j=1}^d V_j$ their algebraic tensor product.

Definition B.3.1. A norm $\|\cdot\|_D$ on $\mathbf{V}_D$ is:

(i) A *crossnorm* if $\| \bigotimes_j v_j\|_D = \prod_j \|v_j\|_j$ for all $v_j \in V_j$.

(ii) A *reasonable crossnorm* (or satisfies (Inj)) if it is a crossnorm and $\|\mathbf{v}\|_D \geq \|\mathbf{v}\|_\varepsilon$ for all $\mathbf{v} \in \mathbf{V}_D$, where $\|\mathbf{v}\|_\varepsilon$ is the injective (or spatial) norm

$$\|\mathbf{v}\|_\varepsilon := \sup\Big\{|(\bigotimes_j \varphi_j)(\mathbf{v})| : \varphi_j \in V_j^*,\ \|\varphi_j\|_{V_j^*} \leq 1\Big\}.$$

The injective norm $\|\cdot\|_\varepsilon$ is the smallest reasonable crossnorm; the projective norm $\|\cdot\|_\pi$ is the largest crossnorm. For $d = 2$: $\|A\|_\varepsilon = \|A\|_{\mathcal{L}(V_1^*,V_2)} = \|A\|_{\mathcal{L}(V_2^*,V_1)}$ (operator norm), and $\|A\|_\pi = \inf\{\sum_k \|u_k\|\|v_k\| : A = \sum_k u_k \otimes v_k\}$ (nuclear norm).

Condition (Inj) plays a central role in:

- The weak closedness of $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$ (Theorem 4.5.3).
- The existence of best Tucker and tree-based approximations (Theorems 4.5.4 and 7.5.3).
- The Dirac–Frenkel immersion theorems (Theorems 8.5.3 and 9.6.1).

Remark B.3.2 (Examples of reasonable crossnorms). (i) The L^p-norm on $L^p(\prod_j I_j)$ is a reasonable crossnorm for $1 \leq p \leq \infty$.

(ii) The Sobolev norm on $H^{N,p}(\prod_j I_j)$ with $1 < p < \infty$, $N \geq 0$, is a reasonable crossnorm.

(iii) The partial-derivative norm $\|f\|^2 = \|f\|_{L^2}^2 + \|\partial_{x_j} f\|_{L^2}^2$ on $H^{1,2}(I_1) \otimes H^{1,2}(I_2)$ is a crossnorm satisfying (Inj).

B.4 Fredholm theory and split operators

Definition B.4.1. A bounded operator $T \in \mathcal{L}(X, Y)$ between Banach spaces is:

(i) *Fredholm* if $\ker T$ and $Y/\overline{\operatorname{Im} T}$ are finite-dimensional.

(ii) *Split* (or *normally solvable*) if $\operatorname{Im} T$ is closed and complemented in Y.

Proposition B.4.2. *Let $T \in \mathcal{L}(X, Y)$. The following are equivalent:*

(a) *T is split.*

(b) $\operatorname{Im} T$ *is closed in Y and split as a subspace of Y.*

(c) *There exists $S \in \mathcal{L}(Y, X)$ such that $T \circ S \circ T = T$ (T has a bounded generalised inverse).*

This characterisation is used in the proof of the immersion theorems in Chapters 8 and 9: the tangent map of the inclusion $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D) \hookrightarrow \mathbf{V}_{D,\|\cdot\|_D}$ is shown to be split by exhibiting an explicit bounded right inverse.

B.5 The Cauchy–Lipschitz theorem in Banach spaces

Theorem B.5.1 (Picard–Lindelöf in Banach spaces)**.** *Let X be a Banach space, $U \subset X$ open, and $F : [0, T] \times U \to X$ continuous with $F(t, \cdot)$ locally Lipschitz uniformly in t. For every $x_0 \in U$ there exist $\varepsilon > 0$ and a unique $\mathcal{C}^1$-solution $x : [0, \varepsilon) \to U$ to $\dot{x}(t) = F(t, x(t))$, $x(0) = x_0$.*

Applied to the Dirac–Frenkel ODE on the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Chapter 10), with $F(t, \mathbf{v}_r) = \mathcal{P}_{\mathbf{v}_r}(\mathbf{F}(t, \mathbf{v}_r))$. The local Lipschitz condition follows from the $\mathcal{C}^\infty$-regularity of the manifold and the vector field.

B.6 Reflexivity and the theorem of James

Reflexivity is the single most important hypothesis in the existence theorems of this monograph (Theorems 4.5.4 and 7.5.3). This section collects the main characterisations and stability properties of reflexive Banach spaces.

Characterisations of reflexivity

Recall that a Banach space X is *reflexive* if the canonical embedding $J_X : X \to X^{**}$, $J_X(x)(f) = f(x)$, is surjective. Equivalently, the closed unit ball $B_X = \{x \in X : \|x\| \leq 1\}$ is weakly compact (Alaoglu–Kakutani theorem). The following theorem gives a deeper characterisation.

Theorem B.6.1 (James, 1957; [Jam57])**.** *A Banach space X is reflexive if and only if every continuous linear functional $f \in X^*$ attains its norm on the closed unit ball, i.e., there exists $x_f \in B_X$ such that $f(x_f) = \|f\|_{X^*}$.*

Remark B.6.2. James's theorem is non-trivial even in separable spaces. The "only if" direction is an immediate consequence of the Krein–Milman theorem (every continuous linear functional on a weakly compact convex set attains its maximum). The "if" direction — that norm-attainment of all functionals forces reflexivity — is the hard part, proved by James in [Jam57] with a subtle diagonal argument.

Corollary B.6.3. *The spaces $L^p(\Omega, \mu)$ are reflexive for $1 < p < \infty$, and not reflexive for $p = 1$ or $p = \infty$. The Sobolev spaces $W^{k,p}(\Omega)$ are reflexive for $1 < p < \infty$. These are the standard component spaces V_j in applications of this monograph.*

Reflexivity of tensor products

The reflexivity of the ambient tensor space $\mathbf{V}_{D,\|\cdot\|_D}$ is not automatic even when all component spaces are reflexive. The following proposition summarises the known results.

Proposition B.6.4. *Let $(V_j, \|\cdot\|_j)$ be Banach spaces for $j = 1, \ldots, d$ and let $\|\cdot\|_D$ be a reasonable crossnorm on $\mathbf{V}_D$.*

(i) *If all V_j are reflexive and $\|\cdot\|_D$ is the* projective norm *$\|\cdot\|_\pi$, then the completion $V_1 \otimes_\pi V_2$ is* not *reflexive in general (it fails already for $V_1 = V_2 = \ell^2$).*

(ii) *If all V_j are Hilbert spaces and $\|\cdot\|_D$ is the Hilbert tensor norm (the unique crossnorm making $\mathbf{V}_{D,\|\cdot\|_D}$ a Hilbert space), then $\mathbf{V}_{D,\|\cdot\|_D}$ is reflexive.*

(iii) *If all V_j are reflexive and the norm $\|\cdot\|_D$ is both a crossnorm and satisfies* (Inj)*, the reflexivity of $\mathbf{V}_{D,\|\cdot\|_D}$ must be verified case by case. In practice (e.g., for the mixed Sobolev norm $\|\cdot\|_{\mathrm{mix}}$ on $\bigotimes_j H^N(I_j)$), it can be established by duality arguments; see [FH12].*

Remark B.6.5 (Reflexivity in Chapters 4 and 7)*.* The proofs of best-approximation existence (Theorem 4.5.4 for Tucker formats and Theorem 7.5.3 for tree-based formats) use reflexivity of $\mathbf{V}_{D,\|\cdot\|_D}$ in a single step: a minimising sequence $(\mathbf{v}_n) \subset \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$

is bounded (since the sequence approaches the infimum), hence weakly precompact by reflexivity. A weakly convergent subsequence $\mathbf{v}_{n_k} \rightharpoonup \mathbf{v}_*$ lies in $\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ (which is weakly closed by Theorem 4.5.3), and $\|\mathbf{v} - \mathbf{v}_*\| \leq \liminf_k \|\mathbf{v} - \mathbf{v}_{n_k}\|$ (weak lower semicontinuity of the norm, Corollary B.2.3) shows $\mathbf{v}_*$ is a best approximation.

Stability properties

Proposition B.6.6. *The class of reflexive Banach spaces is stable under:*

(i) *Closed subspaces: every closed subspace of a reflexive space is reflexive.*

(ii) *Quotients: every quotient of a reflexive space by a closed subspace is reflexive.*

(iii) *Finite direct sums: $X \oplus Y$ is reflexive if and only if both X and Y are reflexive.*

(iv) *Isomorphisms: reflexivity is preserved under bounded linear isomorphisms (but not mere homeomorphisms of the underlying topological spaces).*

Property (i) is relevant to the monograph since the Tucker manifold $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (as a subset of $\mathbf{V}_{D,\|\cdot\|_D}$, not a subspace) is not itself a Banach space; but the tangent spaces $Z^{(D)}(\mathbf{v})$, being closed subspaces of $\mathbf{V}_{D,\|\cdot\|_D}$, are reflexive when $\mathbf{V}_{D,\|\cdot\|_D}$ is. This matters for the existence of the metric projection in Chapter 10.

B.7 The Hahn–Banach theorem and its tensor consequences

The Hahn–Banach theorem and its geometric forms are the foundational separation tools used throughout the monograph — principally in the proofs of weak closedness (Chapter 4) and in the duality arguments for crossnorms.

The analytic Hahn–Banach theorem

Theorem B.7.1 (Hahn–Banach extension theorem)**.** *Let X be a real vector space, $p : X \to \mathbb{R}$ a sublinear functional ($p(x + y) \leq p(x) + p(y)$, $p(\lambda x) = \lambda p(x)$ for $\lambda \geq 0$), $Y \subset X$ a linear subspace, and $f : Y \to \mathbb{R}$ linear with $f(y) \leq p(y)$ for all $y \in Y$. Then there exists a linear extension $F : X \to \mathbb{R}$ with $F|_Y = f$ and $F(x) \leq p(x)$ for all $x \in X$.*

In particular, for any normed space $(X, \|\cdot\|)$, subspace Y, and bounded linear functional $f \in Y^*$, there exists $F \in X^*$ with $F|_Y = f$ and $\|F\|_{X^*} = \|f\|_{Y^*}$.

Corollary B.7.2 (Separation of points by functionals)**.** *Let X be a normed space. Then:*

(i) *For every $x \in X$ with $x \neq 0$, there exists $f \in X^*$ with $\|f\|_{X^*} = 1$ and $f(x) = \|x\|$.*

(ii) *The dual X^* separates points of X: if $f(x) = f(y)$ for all $f \in X^*$, then $x = y$.*

(iii) *The canonical embedding $J_X : X \to X^{**}$ is an isometry.*

Geometric forms

Theorem B.7.3 (Hahn–Banach separation theorem)**.** *Let X be a locally convex topological vector space (in particular, any normed space).*

(i) Strict separation*: If $A, B \subset X$ are convex, A is open, and $A \cap B = \emptyset$, then there exists a continuous linear functional $f : X \to \mathbb{R}$ and $c \in \mathbb{R}$ with $f(a) < c \leq f(b)$ for all $a \in A$, $b \in B$.*

(ii) Closed convex sets*: A closed convex subset $C \subset X$ is the intersection of the closed half-spaces $\{x : f(x) \leq c\}$ containing it, over all $f \in X^*$.*

Consequences for tensor spaces

The Hahn–Banach theorem has two direct consequences for the tensor theory of this monograph.

Proposition B.7.4 (Dual characterisation of the injective norm)**.** *For $\mathbf{v} \in \mathbf{V}_D = \bigotimes_{j=1}^d V_j$,*

$$\|\mathbf{v}\|_\varepsilon = \sup\Big\{\Big|(\bigotimes_j \varphi_j)(\mathbf{v})\Big| : \varphi_j \in V_j^*,\ \|\varphi_j\|_{V_j^*} \leq 1\Big\} = \sup\Big\{|F(\mathbf{v})| : F \in (\mathbf{V}_D)^*,\ F = \bigotimes_j \varphi_j,\ \|F\|_{(\mathbf{V}_D)^*} \tag{B.1}$$

In particular, $\|\mathbf{v}\|_\varepsilon = 0$ implies $\mathbf{v} = 0$, so the injective norm is indeed a norm. This follows from the Hahn–Banach theorem: for any $\mathbf{v} \neq 0$ there exists a functional F in the algebraic dual of $\mathbf{V}_D$ that does not vanish on $\mathbf{v}$, and this functional can be chosen to be a tensor product of functionals.

Proposition B.7.5 (Weak closedness via duality)**.** *Let $C \subset \mathbf{V}_{D,\|\cdot\|_D}$ be convex. The following are equivalent:*

(a) *C is weakly closed.*

(b) *C is norm-closed (by Mazur's theorem, Theorem B.2.2).*

(c) *For every $\mathbf{v} \notin C$, there exists $F \in (\mathbf{V}_{D,\|\cdot\|_D})^*$ such that $F(\mathbf{v}) > \sup_{\mathbf{u} \in C} F(\mathbf{u})$.*

The equivalence of (a) *and* (b) *uses Mazur's theorem; the equivalence of* (b) *and* (c) *is the geometric Hahn–Banach theorem applied to the pair $\{\mathbf{v}\}$ and C.*

The set $\mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ is shown to be weakly (hence norm-) closed in Theorem 4.5.3 using the characterisation of (b): one shows that weak limits of sequences in $\mathfrak{M}_{\le \mathfrak{r}}(\mathbf{V}_{D,\|\cdot\|_D})$ remain in the set by exploiting the semicontinuity of the minimal subspace dimension established in Theorem 7.5.1.

B.8 Grothendieck's theorem and tensor norms in the Hilbert case

Grothendieck's 1953 Résumé introduced the systematic study of tensor norms and established several deep results about their coincidence in special cases. This section summarises the results most directly relevant to the monograph.

The Grothendieck inequality

The foundational result of Grothendieck's tensor norm theory is the following inequality, which controls the gap between the injective and projective norms.

Theorem B.8.1 (Grothendieck inequality; [Gro53]). *There exists a universal constant $K_G > 0$ (the* Grothendieck constant*) such that for any Hilbert spaces H_1, H_2 and any bounded bilinear form $B : H_1 \times H_2 \to \mathbb{R}$,*

$$\sup\Big\{\Big|\sum_{i,j} a_{ij}\langle x_i, y_j\rangle\Big| : x_i \in H_1,\ y_j \in H_2,\ \|x_i\|, \|y_j\| \le 1\Big\} \le K_G \sup\Big\{\Big|\sum_{i,j} a_{ij} s_i t_j\Big| : |s_i|, |t_j| \le 1\Big\}. \tag{B.2}$$

Equivalently, for bilinear forms on Hilbert spaces, $\|\cdot\|_\pi \le K_G \|\cdot\|_\varepsilon$, so the projective and injective norms are equivalent on $H_1 \otimes H_2$.

Remark B.8.2. The exact value of K_G is unknown; the best known bounds are $1.676 \le K_G \le \pi/(2\ln(1+\sqrt{2})) \approx 1.782$ (real case). The significance of the theorem is that it shows the injective and projective norms on a tensor product of Hilbert spaces are equivalent, with a universal constant independent of the spaces. This does *not* hold for general Banach spaces: $c_0 \otimes c_0$ has non-equivalent injective and projective norms.

Coincidence of norms on Hilbert tensor products

The following result is more directly relevant to the monograph: for two-factor Hilbert tensor products, the injective norm, the Hilbert tensor norm, and the projective norm can all be identified.

Theorem B.8.3. *Let H_1, H_2 be Hilbert spaces with inner products $\langle\cdot,\cdot\rangle_j$. On $H_1 \otimes H_2$:*

(i) *The Hilbert tensor norm $\|\cdot\|_{H_1\otimes H_2}$ is the unique crossnorm induced by the inner product $\langle u_1 \otimes u_2, v_1 \otimes v_2\rangle = \langle u_1, v_1\rangle_1 \langle u_2, v_2\rangle_2$.*

(ii) *The completion $H_1 \otimes_H H_2$ under this norm is a Hilbert space, isometrically isomorphic to the space of Hilbert–Schmidt operators from H_1 to H_2: $H_1 \otimes_H H_2 \cong \mathcal{HS}(H_1, H_2)$.*

(iii) *The Hilbert tensor norm satisfies $\|\cdot\|_\varepsilon \le \|\cdot\|_H \le \|\cdot\|_\pi$, with equality $\|\cdot\|_H = \|\cdot\|_\varepsilon$ if and only if one of the spaces is finite-dimensional.*

(iv) *For d factors, $\bigotimes_{j=1}^d H_j$ under the Hilbert tensor norm is isometrically isomorphic to $L^2(\Omega_1 \times \cdots \times \Omega_d)$ when $H_j = L^2(\Omega_j)$, via the identification $f_1 \otimes \cdots \otimes f_d \leftrightarrow (x_1, \ldots, x_d) \mapsto \prod_j f_j(x_j)$.*

Part (iv) is the key fact underlying the identification of the algebraic tensor space $\bigotimes_j L^2(I_j)$ with a dense subspace of $L^2(I_1 \times \cdots \times I_d)$, which motivates the entire framework of tensor formats as low-complexity approximations of multivariate functions.

The Hilbert–Schmidt identification and rank

Proposition B.8.4 (Rank-one operators and elementary tensors)**.** *Under the identification $H_1 \otimes_H H_2 \cong \mathcal{HS}(H_1, H_2)$:*

(i) *An elementary tensor $u \otimes v$ corresponds to the rank-one operator $x \mapsto \langle x, u\rangle_1 v$.*

(ii) *An element $T \in \mathcal{HS}(H_1, H_2)$ has Tucker rank (r_1, r_2) if and only if it has operator rank r with $r = \min(r_1, r_2)$. The minimal subspaces satisfy $U_1^{\min}(T) = \ker(T)^\perp \subset H_1$ and $U_2^{\min}(T) = \overline{\operatorname{Im}(T)} \subset H_2$.*

(iii) *The Schmidt decomposition of T is the singular value decomposition: $T = \sum_{k=1}^r \sigma_k u_k \otimes v_k$ with $\{u_k\}$ orthonormal in H_1, $\{v_k\}$ orthonormal in H_2, and $\sigma_k > 0$.*

This proposition shows that the abstract minimal subspace theory of Chapter 4 reduces, in the Hilbert case $d = 2$, to the classical Schmidt decomposition (or singular value decomposition for matrices). The minimal subspaces are the column space and row space of the corresponding operator, and the admissible rank condition $r_1 = r_2 = r$ is the condition that the operator has rank exactly r.

Grothendieck's theorem for general tensor norms

We close with a statement of Grothendieck's classification theorem for tensor norms, which places the injective and projective norms as the two extremes of a rich landscape.

Theorem B.8.5 ([Gro53])**.** *Let α be a* tensor norm *(a rule assigning to each pair of Banach spaces X, Y a norm α on $X \otimes Y$, natural with respect to bounded linear maps). Then:*

(i) $\|\cdot\|_\varepsilon \leq \alpha \leq \|\cdot\|_\pi$ *for all Banach spaces* X, Y.

(ii) *There are exactly fourteen distinct tensor norms in the sense of Grothendieck for two-factor products* (*the "Grothendieck 14"*).

(iii) *In the Hilbert space case, any natural tensor norm* α *satisfies* $\|\cdot\|_\varepsilon \leq \alpha \leq K_G\|\cdot\|_\varepsilon$ (*by the Grothendieck inequality*), *so all reasonable crossnorms on* $H_1 \otimes H_2$ *are equivalent.*

Remark B.8.6 (Relevance to the monograph)*.* The condition (Inj) ($\|\cdot\|_D \geq \|\cdot\|_\varepsilon$) is precisely Grothendieck's condition that the tensor norm is "reasonable": it lies above the minimal reasonable crossnorm. Part (i) of Theorem B.8.5 guarantees that any reasonable crossnorm satisfies (Inj) automatically. Part (iii) implies that in the Hilbert case, the condition (Inj) imposes no effective constraint: all reasonable crossnorms are equivalent. This is why the Hilbert case of the theory (e.g., $\bigotimes_j L^2(I_j)$ with the natural norm) is somewhat simpler: the specific choice of norm $\|\cdot\|_D$ does not affect the qualitative conclusions, only the constants. In the Banach case (e.g., $\bigotimes_j L^p(I_j)$ with $p \neq 2$), the choice of norm matters more, and (Inj) is a genuine hypothesis.

Appendix C

Computational Aspects and Algorithms

The theory developed in this monograph — minimal subspaces, Tucker and tree-based manifolds, the Dirac–Frenkel variational principle — has direct computational consequences. This appendix describes the main algorithms used in practice, explains their connection to the mathematical structures of the main text, and identifies precisely where the Banach-space generalisations of Chapters 4–10 modify or constrain the algorithmic picture.

Each algorithm is stated with pseudocode, its cost is analysed, and its relationship to the abstract theory is made explicit. The principal references are [Hac19] (systematic treatment of tensor numerics), [BSU16] (survey of tensor network methods), and [UV20] (Riemannian optimisation on tensor manifolds).

C.1 Storage complexity of tensor formats

The curse of dimensionality, quantified

Let $V_j = \mathbb{R}^n$ for $j = 1, \ldots, d$. The full tensor space $\mathbf{V}_D = \bigotimes_{j=1}^d \mathbb{R}^n$ has dimension n^d: intractable for moderate n and d. Tensor formats replace this by structured low-dimensional parametrisations. The table below summarises storage cost in terms of maximal rank r and mode size n:

Format	**Parameters**	**Storage**	**Tree**
Full	n^d entries	$O(n^d)$	—
Tucker	core r^d + d factor matrices $n \times r$	$O(r^d + dnr)$	T_D^{Tucker}
HT (balanced binary)	transfer tensors at $O(d)$ nodes, each $r \times r^2$	$O(dr^3 + dnr)$	T_D^{HT}
TT (Tensor Train)	d core matrices, each $r \times n \times r$	$O(dr^2 n)$	T_D^{TT}

Remark C.1.1 (Exponential vs. polynomial scaling). The Tucker format reduces factor storage to $O(dnr)$ (linear in d) but retains the core at $O(r^d)$. The HT and TT formats remove this remaining exponential cost by imposing rank constraints at each level of the tree, reducing core storage to $O(dr^3)$ and $O(dr^2 n)$ respectively. This is precisely the content of the intersection representation (Theorem 5.4.1): $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ imposes rank constraints at *every* level of T_D, not only at the leaves.

Parameter count for tree-based formats

For a general dimension partition tree T_D and rank tuple $\mathfrak{r}$, the total parameter count is:

$$P(\mathfrak{r}, T_D) = \sum_{j \in D} r_{\{j\}} n_j + \sum_{\alpha \in T_D \setminus (\mathcal{L}(T_D) \cup \{D\})} r_\alpha \prod_{\beta \in S(\alpha)} r_\beta + \prod_{\beta \in S(D)} r_\beta, \tag{C.1}$$

where the three terms count, respectively, the leaf factor matrices, the internal transfer tensors, and the root tensor.

C.2 The Higher-Order SVD

Setup and goal

The *Higher-Order SVD* (HOSVD) of De Lathauwer, De Moor, and Vandewalle [DLDMV00] computes a Tucker approximation. Given $\mathbf{v} \in \mathbf{V}_D = \bigotimes_{j=1}^d \mathbb{R}^{n_j}$ and target ranks $\mathfrak{r} = (r_j)_{j \in D}$, it produces:

(i) Factor matrices $U_j \in \mathbb{R}^{n_j \times r_j}$ with orthonormal columns, approximating $U_j^{\min}(\mathbf{v})$.

(ii) A core tensor $C^{(D)} \in \mathbb{R}^{r_1 \times \cdots \times r_d}$ such that $\mathbf{v} \approx \sum_{(i_j)} C^{(D)}_{(i_j)} \bigotimes_j u^{(j)}_{i_j}$.

Algorithm

Algorithm D.1 (HOSVD).
Input: tensor $\mathbf{v} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$, target ranks $(r_1, \ldots, r_d)$.
Output: core tensor $C^{(D)}$, factor matrices $U_1, \ldots, U_d$.

1. **Mode unfoldings.** For each $j = 1, \dots, d$:

   (a) Compute the j-mode unfolding $M_j = \mathcal{M}_{\{j\}}(\mathbf{v}) \in \mathbb{R}^{n_j \times \prod_{k \neq j} n_k}$.

   (b) Compute the truncated SVD: $M_j \approx U_j \Sigma_j V_j^\top$, retaining the r_j largest singular values. Set $U_j \in \mathbb{R}^{n_j \times r_j}$ (left singular vectors).

2. **Core compression.** Compute

$$C^{(D)}_{i_1,\dots,i_d} = \sum_{k_1,\dots,k_d} (U_1)_{k_1 i_1} \cdots (U_d)_{k_d i_d} \, \mathbf{v}_{k_1,\dots,k_d},$$

   i.e., project $\mathbf{v}$ onto the subspaces $\mathrm{span}(U_j)$ in each mode.

3. **Output.** Return $(C^{(D)}, U_1, \dots, U_d)$.

Cost and approximation quality

The dominant cost is d SVDs of unfolding matrices of size $n_j \times \prod_{k \neq j} n_k$, giving arithmetic cost $O(dn^{d+1})$, which still scales exponentially in d. For tensors already in a low-rank format, recompression costs $O(dr^{d+1})$ or $O(dr^3)$ depending on the tree structure.

Theorem C.2.1 (HOSVD quasi-optimality; [DLDMV00, Theorem 4.2])**.** *Let* $\mathbf{v} \in \mathbf{V}_D$ *and let* $\mathbf{w}$ *be the output of* Algorithm D.1 *with ranks* $(r_j)_j$*. Then*

$$\|\mathbf{v} - \mathbf{w}\|_F \leq \sqrt{d} \min_{\mathbf{u} \in \mathfrak{M}_{\leq \mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{v} - \mathbf{u}\|_F. \tag{C.2}$$

The HOSVD is not a best-approximation algorithm, but provides a quasi-optimal approximation with factor $\sqrt{d}$*.*

Remark C.2.2 (Connection to minimal subspaces)*.* The factor matrix U_j is the best rank-r_j approximation to the column space of the j-mode unfolding M_j, whose column space is exactly $U_j^{\min}(\mathbf{v})$ when $r_j = r_j(\mathbf{v})$ (the exact rank). Algorithm D.1 is therefore a computational procedure for approximating the minimal subspaces of Chapter 4.

In the Banach setting (non-Hilbert V_j), the SVD is replaced by a best rank-r_j approximation in the operator norm of $\mathcal{L}(V_j^*, \mathbf{V}_\alpha^{[j]})$. Its existence is guaranteed by Theorem 4.5.4, but the quasi-optimality factor $\sqrt{d}$ may worsen depending on the Banach–Mazur distance of $\mathbf{V}_{D,\|\cdot\|_D}$ from a Hilbert space.

C.3 Alternating Least Squares

The ALS problem and microiterations

Given $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$ and rank tuple $\mathfrak{r}$, the best Tucker approximation problem is

$$\min_{\mathbf{w}\in\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{u}-\mathbf{w}\|_D^2. \tag{C.3}$$

Alternating Least Squares (ALS) solves this by fixing all factor matrices but one and optimising over the remaining one, cycling over all modes.

Algorithm D.2 (ALS for Tucker format).
Input: $\mathbf{u} \in \mathbf{V}_{D,\|\cdot\|_D}$, ranks $\mathfrak{r}$, initial factor matrices $\{U_j^{(0)}\}_{j=1}^d$.
Repeat until convergence:

1. For $j = 1, \dots, d$:

   (a) **Fix** all U_k for $k \neq j$ at their current values.

   (b) **Form the reduced right-hand side:** compute the array

   $$\left(R_j\right)_{i_j,(i_k)_{k\neq j}} := \sum_{(k_l)} \mathbf{u}_{k_1,\dots,k_d} \prod_{l\neq j} (U_l)_{k_l,i_l},$$

   which is the tensor $\mathbf{u}$ projected onto all modes except j. This gives $R_j \in \mathbb{R}^{n_j \times \prod_{k\neq j} r_k}$.

   (c) **Solve the mode-j subproblem:**

   $$U_j^{\text{new}} = \arg\min_{U\in\mathbb{R}^{n_j\times r_j}} \left\|R_j - U\,H_j\right\|_F^2,$$

   where $H_j \in \mathbb{R}^{r_j \times \prod_{k\neq j} r_k}$ is the Gram factor assembled from the other modes.

   (d) **Orthonormalise:** $U_j \leftarrow U_j^{\text{new}}$ after a QR step.

2. **Update the core:** $C^{(D)}_{i_1,\dots,i_d} \leftarrow \sum_{k_1,\dots,k_d} \mathbf{u}_{k_1,\dots,k_d} \prod_j (U_j)_{k_j,i_j}$.

Convergence

Theorem C.3.1 (ALS convergence in Hilbert spaces; [UV20, Theorem 4.1])**.** *Let* $\mathbf{V}_{D,\|\cdot\|_D}$ *be a Hilbert tensor space and let* $(\mathbf{w}^{(k)})$ *be the sequence generated by* Algorithm D.2. *Then:*

(i) *The residuals* $\|\mathbf{u}-\mathbf{w}^{(k)}\|$ *are monotonically non-increasing.*

(ii) *Every accumulation point* $\mathbf{w}^*$ *satisfies the first-order optimality conditions in each mode direction:* $\langle \mathbf{u}-\mathbf{w}^*, v\otimes \mathbf{w}^*_{[j]}\rangle = 0$ *for all* $v \in V_j$, $j = 1, \dots, d$.

(iii) *Global convergence to a best approximation is not guaranteed; ALS may converge to a local minimum or saddle point.*

Remark C.3.2 (ALS in Banach spaces). In a reflexive Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$, the mode-$j$ subproblem in step 1(c) becomes: find the best rank-r_j operator approximation in the norm $\|\cdot\|_D$ projected onto mode j. This is a convex problem whose unique solution exists by Theorem 4.5.4. Monotone decrease of part (i) holds in full generality. The characterisation of accumulation points in part (ii) must be rephrased using the duality mapping J (Appendix B) in place of the inner product:

$$\langle v \otimes \mathbf{w}^*_{[j]},\, J(\mathbf{u} - \mathbf{w}^*)\rangle = 0 \quad \forall\, v \in V_j,\ j = 1, \dots, d.$$

The convergence analysis in this Banach setting is Open Problem 11.3.2.

C.4 The projector-splitting integrator

The Dirac–Frenkel ODE

Consider the Dirac–Frenkel reduced model on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Chapter 10):

$$\dot{\mathbf{v}}_r(t) = \mathcal{P}_{\mathbf{v}_r(t)}\Big(\mathbf{F}(t, \mathbf{v}_r(t))\Big), \quad \mathbf{v}_r(0) = \mathbf{v}_0. \tag{C.4}$$

Direct integration requires evaluating $\mathcal{P}_{\mathbf{v}_r(t)}$ at each step. The *projector-splitting integrator* of [LOV15] avoids this by decomposing the projection into a sequence of simpler updates.

The TT case

For the TT format ($T_D = T_D^{\mathrm{TT}}$, bond dimensions $r_1, \dots, r_{d-1}$), the tangent space projection splits as

$$\mathcal{P}_{\mathbf{v}_r} = \sum_{k=1}^{d} P_k - \sum_{k=1}^{d-1} P_{k,k+1}, \tag{C.5}$$

where P_k projects onto the subspace with the k-th component free and $P_{k,k+1}$ is the correction at the $(k, k{+}1)$ interface.

Algorithm D.3 (Projector-splitting, one step h).
Input: $\mathbf{v}_r^n$ in TT format, vector field $\mathbf{F}$, step h.
Output: $\mathbf{v}_r^{n+1}$ in TT format.

1. **Forward sweep** ($k = 1, \dots, d$): integrate the k-th component ODE

$$\dot{C}^{(k)}(t) = \Big\langle \mathbf{F}(t, \mathbf{v}_r(t)),\, P_k(\,\cdot\,) \Big\rangle$$

from t^n to $t^n + h$; left-orthogonalise via QR.

2. **Correction sweep** ($k = d-1, \dots, 1$): apply the correction steps $-P_{k,k+1}$ symmetrically.

3. **Return** the updated TT tensor $\mathbf{v}_r^{n+1}$.

Theorem C.4.1 (Error bound in Hilbert spaces; [LOV15, Theorem 4.1])**.** *Let $\mathbf{V}_{D,\|\cdot\|_D}$ be a Hilbert tensor space and $\mathbf{F}$ Lipschitz with constant L. Then there exists $C > 0$ (depending on L, the curvature of $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$, and the initial data) such that*

$$\|\mathbf{u}(t^n) - \mathbf{v}_r^n\| \le Ch + O\Big(\delta_{\mathfrak{r}}(\mathbf{u})\Big), \tag{C.6}$$

where $\delta_{\mathfrak{r}}(\mathbf{u}) = \inf_{\mathbf{w}\in\mathfrak{M}_{\le\mathfrak{r}}(\mathbf{V}_D)} \|\mathbf{u} - \mathbf{w}\|$ is the best-approximation error.

The curvature term in C is the quantity whose Banach-space analogue appears in Open Problem 11.3.1. For general tree-based formats, the splitting integrator of [CLW21] applies a level-by-level sweep along T_D at cost $O(\mathrm{depth}(T_D) \cdot dr^3)$ per time step.

C.5 DMRG and the Tucker manifold

DMRG as ALS on the TT manifold

The *Density Matrix Renormalisation Group* (DMRG), introduced by White [Whi92], is the most widely used algorithm for ground-state computation in quantum many-body systems. In its modern formulation [HRS12], DMRG is equivalent to ALS on the TT manifold applied to the eigenvalue problem $H\mathbf{u} = \lambda\mathbf{u}$: each microiteration updates one TT component while fixing all others, producing an effective eigenvalue problem of size $r^2 n$.

Remark C.5.1 (Gauge conditions and the Tucker bundle)*.* The *left-orthogonal gauge* in DMRG requires each component tensor $C^{(k)} \in \mathbb{R}^{r_{k-1}\times n_k\times r_k}$ to satisfy

$$\sum_{i_k=1}^{n_k} \left(C^{(k)}\right)^{\top}_{:,i_k,:} \left(C^{(k)}\right)_{:,i_k,:} = I_{r_k}.$$

This is the computational analogue of fixing the complement W_k at each tree node to be the orthogonal complement of $\mathrm{span}(C^{(k)})$, giving a canonical trivialisation of the Tucker bundle (Theorem A.6.2 in Appendix A). The gauge group $\prod_k \mathrm{GL}(r_k)\mathbb{R}$ of Remark 6.4.4 is quotiented out by the QR orthogonalisation step.

Two-site DMRG and rank adaptation

One-site DMRG cannot change the bond dimensions r_k. *Two-site DMRG* optimises two adjacent components simultaneously, forming an effective tensor of rank $r_k \cdot r_{k+1}$ which is then truncated by SVD to obtain updated bond dimensions. Geometrically, this corresponds to moving between strata of the stratification $\mathbf{V}_D = \bigsqcup_{\mathfrak{r}} \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ (Theorem 4.4.3).

C.6 Retraction-based optimisation on tensor manifolds

Riemannian gradient methods

In the Hilbert case, the Tucker manifold is a Riemannian submanifold of $\mathbf{V}_{D,\|\cdot\|_D}$. Riemannian gradient descent [UV20] for minimising $f : \mathbf{V}_{D,\|\cdot\|_D} \to \mathbb{R}$ over $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ reads:

$$\mathbf{v}_{k+1} = \mathrm{Ret}_{\mathbf{v}_k}\Big(-\alpha_k\, \mathcal{P}_{\mathbf{v}_k}(\nabla f(\mathbf{v}_k))\Big), \tag{C.7}$$

where $\mathcal{P}_{\mathbf{v}}(\nabla f(\mathbf{v}))$ is the Riemannian gradient and $\mathrm{Ret}_{\mathbf{v}} : Z^{(D)}(\mathbf{v}) \to \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is a retraction.

Definition C.6.1 (Retraction)**.** A *retraction* on a Banach manifold $\mathbb{M}$ is a $\mathcal{C}^1$-map $\mathrm{Ret} : T\mathbb{M} \to \mathbb{M}$ such that, for each $m \in \mathbb{M}$:

(i) $\mathrm{Ret}_m(0_m) = m$,

(ii) $D(\mathrm{Ret}_m)(0_m) = \mathrm{id}_{T_m\mathbb{M}}$.

Example C.6.2 (Tucker retraction via HOSVD truncation)**.** A practical retraction on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$ is:

$$\mathrm{Ret}_{\mathbf{v}}(\dot{\mathbf{v}}) = \mathrm{HOSVD}_{\mathfrak{r}}(\mathbf{v} + \dot{\mathbf{v}}),$$

i.e., apply Algorithm D.1 to $\mathbf{v} + \dot{\mathbf{v}}$ and truncate to rank $\mathfrak{r}$. This satisfies (i) and (ii) with quasi-optimality constant $\sqrt{d}$ (Theorem C.2.1), and remains valid in Banach spaces as long as the best-approximation step is solvable — guaranteed by Theorem 4.5.4.

The Banach case: duality-mapping gradient

In a non-Hilbert Banach tensor space $\mathbf{V}_{D,\|\cdot\|_D}$, the Riemannian gradient is replaced by the *duality-mapping gradient*: the unique element $\nabla_B f(\mathbf{v}) \in Z^{(D)}(\mathbf{v})$ satisfying

$$\Big\langle \nabla_B f(\mathbf{v}),\, J\Big(\mathbf{v} - \mathbf{u}\Big)\Big\rangle = Df(\mathbf{v})[\mathbf{v} - \mathbf{u}] \qquad \forall\, \mathbf{u} \in \mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D), \tag{C.8}$$

where $J : \mathbf{V}_{D,\|\cdot\|_D} \to (\mathbf{V}_{D,\|\cdot\|_D})^*$ is the normalised duality mapping (Appendix B, §B.1). The iteration (C.7) becomes a generalised gradient step using the generalised projection $\Pi_{\mathbf{v}}$ of Appendix B in place of $\mathcal{P}_{\mathbf{v}}$. Convergence analysis in this setting is related to Open Problem 11.2.1.

C.7 Historical notes

Tucker decomposition and HOSVD. The Tucker decomposition [Tuc66] was introduced as a psychometric tool and has been used in data analysis since the 1970s. The HOSVD as an algorithm for best Tucker approximations was introduced by De Lathauwer, De Moor, and Vandewalle [DLDMV00]; the quasi-optimality bound (C.2) is from the same paper. A comprehensive survey of tensor decompositions is given in Kolda and Bader [KB09].

Tensor Train and DMRG. The TT decomposition was introduced independently by Vidal [Vid03] (matrix product states, MPS) and by Oseledets and Tyrtyshnikov [Ose11]. The DMRG algorithm of White [Whi92] predates both and was recognised in [HRS12] as ALS on the TT manifold. The TT-SVD of [Ose11] provides quasi-optimal TT approximations analogous to the HOSVD for Tucker.

Projector-splitting integrators. The projector-splitting integrator for TT formats was introduced by Lubich, Oseledets, and Vandereycken [LOV15], building on the matrix case of Koch and Lubich [KL07] (MCTDH). The extension to general tree-based formats is in [CLW21]. Convergence in the Banach setting is Open Problem 11.3.1.

Riemannian optimisation. Riemannian gradient methods on Tucker and TT manifolds were systematically developed by Uschmajew and Vandereycken [UV20], using the Riemannian structure in a way that parallels Chapters 8–9 but restricted to the Hilbert case. The Banach generalisation is related to Open Problem 11.2.1.

Appendix D

Index of Notation

All macros are defined in `notation.tex`. Entries are grouped thematically and listed in the order they first appear in the text. For each entry, the rightmost column indicates the chapter or appendix where the symbol is first introduced; a blank entry means the symbol is standard notation used throughout without introduction.

Sets, fields, and standard spaces

$\mathbb{K}$	field ($\mathbb{R}$ or $\mathbb{C}$)	
$\mathbb{R}$, $\mathbb{C}$	real and complex numbers	
$\mathbb{N}$, $\mathbb{N}_0$, $\mathbb{Z}$	positive integers, non-negative integers, integers	
$D = \{1, \dots, d\}$	dimension index set	Ch. 3
$d = \#(D)$	number of dimensions	Ch. 3
$\alpha^c = D \setminus \alpha$	complement of α in D	Ch. 3
2^D	power set of D	Ch. 5
$\#(S)$	cardinality of set S	
$\overline{S}$	closure of S	
$B_X = \{x \in X : \|x\| \leq 1\}$	closed unit ball of X	App. B
$L^p(\Omega)$, $W^{k,p}(\Omega)$	Lebesgue and Sobolev spaces	Ch. 2
$\mathcal{HS}(H_1, H_2)$	Hilbert–Schmidt operators $H_1 \to H_2$	App. B
K_G	Grothendieck constant	App. B

Norm conditions and crossnorms

The following conditions on a norm $\|\cdot\|_D$ on $\mathbf{V}_D$ are used throughout; they are formally introduced in Chapter 4 and App. B.

(Cross)	$\Vert \bigotimes_j v_j \Vert_D = \prod_j \Vert v_j \Vert_j$ (crossnorm condition)	Ch. 3
(Inj)	$\Vert \cdot \Vert_D \geq \Vert \cdot \Vert_\varepsilon$ (reasonable crossnorm / injective condition)	Ch. 4
$\Vert \mathbf{v} \Vert_\varepsilon$	injective (spatial / Grothendieck) norm	Ch. 4
$\Vert \mathbf{v} \Vert_\pi$	projective (nuclear) norm	Ch. 3
$\Vert \cdot \Vert_H$	Hilbert tensor norm on $\bigotimes_j H_j$	App. B
(Inj)$[T_D]$	T_D-local version of (Inj), one condition per level of tree T_D	Ch. 7

Banach spaces and linear maps

$(X, \Vert \cdot \Vert)$	normed (Banach) space	Ch. 2
$X^* = \mathcal{L}(X, \mathbb{K})$	topological dual of X	Ch. 2
X'	algebraic dual of X	Ch. 2
$\langle x, f \rangle$	duality pairing, $f \in X^*$, $x \in X$	
$\mathcal{L}(X, Y)$	bounded linear maps $X \to Y$	Ch. 2
$\mathcal{L}(X)$	$\mathcal{L}(X, X)$	
$\Vert A \Vert_{Y \leftarrow X}$	operator norm of $A \in \mathcal{L}(X, Y)$	Ch. 2
$\mathrm{GL}(X)$	bounded linear automorphisms of X (general linear group)	Ch. 2
$\mathrm{GL}(r)\mathbb{R}$	invertible $r \times r$ real matrices	App. A
id_X	identity map on X	
$P_{U \oplus W}$	projection onto U along W	Ch. 2
$J : X \to 2^{X^*}$	normalised duality mapping	Ch. 10
$J_X : X \to X^{**}$	canonical embedding into bidual	App. B
$\phi(x, y)$	Lyapunov functional $\Vert x \Vert^2 - 2\langle x, J(y) \rangle + \Vert y \Vert^2$	Ch. 10
$\mathcal{L}_{(U,W)}(X)$	nilpotent algebra associated to splitting $X = U \oplus W$	Ch. 2

Grassmannians and subspaces

$\mathbb{G}(X)$	Grassmannian of all split subspaces of X	Ch. 2
$\mathbb{G}_r(X)$	r-dimensional split subspaces of X	Ch. 2
$\mathbb{G}(W, X)$	split subspaces of X with complement W (chart domain of $\mathbb{G}(X)$)	Ch. 2
$\Psi_{U\oplus W}$	Grassmann chart map at U with complement W	Ch. 2
$T_U\mathbb{G}_r(X) \cong \mathcal{L}(U, W)$	tangent space to Grassmannian	Ch. 2
$\mathrm{Fr}(\mathbb{M})$	frame bundle of manifold $\mathbb{M}$	App. A

Component spaces and tensor products

V_j, V_α	component spaces ($j \in D$, $\alpha \subset D$)	Ch. 3
$\\|\cdot\\|_j$, $\\|\cdot\\|_\alpha$	norms on V_j, V_α	
$\mathbf{V}_D = \bigotimes_{j\in D} V_j$	algebraic tensor product	Ch. 3
$\mathbf{V}_\alpha = \bigotimes_{j\in\alpha} V_j$	partial algebraic tensor product	Ch. 3
$v \otimes_a w$	algebraic tensor product of vectors	Ch. 3
$\mathbf{v}, \mathbf{u}, \mathbf{w}$	elements of $\mathbf{V}_D$ (tensors)	
$\mathbf{V}_{D,\\|\cdot\\|_D}$	completion of $\mathbf{V}_D$ under $\\|\cdot\\|_D$	Ch. 3
$\\|\cdot\\|_D$	norm on $\mathbf{V}_D$ (or $\mathbf{V}_{D,\\|\cdot\\|_D}$)	
$H_1 \otimes_H H_2$	Hilbert tensor product (completion under $\\|\cdot\\|_H$)	App. B
$\mathcal{M}_\alpha(\mathbf{v})$	α-matricisation (unfolding) of $\mathbf{v}$	Ch. 4

Minimal subspaces and Tucker format

$U_\alpha^{\min}(\mathbf{v})$	minimal subspace of $\mathbf{v}$ w.r.t. direction α	Ch. 4
$W_\alpha^{\min}(\mathbf{v})$	complement of $U_\alpha^{\min}(\mathbf{v})$ used in local charts	Ch. 8
$r_\alpha(\mathbf{v}) = \dim U_\alpha^{\min}(\mathbf{v})$	α-rank of $\mathbf{v}$	Ch. 4
$\mathfrak{r} = (r_\alpha)_{\alpha\in D}$	Tucker rank tuple	Ch. 4
$\mathcal{AD}(\mathbf{V}_D)$	admissible rank tuples for $\mathbf{V}_D$	Ch. 4
$\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Tucker tensors with fixed rank $\mathfrak{r}$	Ch. 4
$\mathfrak{M}_{\leq\mathfrak{r}}(\mathbf{V}_D)$	Tucker tensors with bounded rank $\leq \mathfrak{r}$	Ch. 4
$\varrho_\mathfrak{r}$	rank map $\mathfrak{M}_\mathfrak{r}(\mathbf{V}_D) \to \prod_\alpha \mathbb{G}_{r_\alpha}(V_\alpha)$	Ch. 8
$Z^{(D)}(\mathbf{v})$	tangent subspace of $\mathbf{V}_{D,\|\cdot\|_D}$ at $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Ch. 8
$\mathbf{E}_D$	model Banach space of Tucker manifold, $\prod_\alpha \mathcal{L}(U_\alpha^{\min}(\mathbf{v}), W_\alpha^{\min}(\mathbf{v})) \times \mathbb{R}^{\times r_\alpha}$	Ch. 8
$\xi_\mathbf{v}$	local chart map at $\mathbf{v} \in \mathfrak{M}_\mathfrak{r}(\mathbf{V}_D)$	Ch. 8
$\mathcal{U}(\mathbf{v})$	chart domain at $\mathbf{v}$	Ch. 8
$\chi_\mathbf{v}$	fibre-bundle trivialisation map at $\mathbf{v}$	App. A
$F_\mathfrak{r} = \mathbb{R}_*^{\times_\alpha r_\alpha}$	typical fibre of the Tucker bundle (full-rank core tensors)	App. A

Trees and tree-based formats

T_D	dimension partition tree on D	Ch. 5
$S(\alpha)$	sons of internal vertex α in T_D	Ch. 5
$\mathcal{L}(T_D)$	leaves of T_D (singletons $\{j\}$, $j \in D$)	Ch. 5
$\mathrm{level}(\alpha)$	level of vertex α	Ch. 5
$\mathrm{depth}(T_D)$	depth of the tree T_D	Ch. 5
$\mathcal{P}_k(T_D)$	level-k partition of D induced by T_D	Ch. 5
T_D^{Tucker}	Tucker tree (star, depth 1)	Ch. 5
T_D^{TT}	Tensor Train tree (linear chain, depth $d-1$)	Ch. 5
T_D^{HT}	Hierarchical Tucker tree (complete binary, depth $\lceil \log_2 d \rceil$)	Ch. 5
$\mathfrak{r} = (\mathfrak{r}_\alpha)_{\alpha \in T_D \setminus \mathcal{L}(T_D)}$	tree-based rank tuple	Ch. 6
$\mathcal{AD}(\mathbf{V}_D, T_D)$	admissible tree-based rank tuples	Ch. 6
$\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$ $=$ $\mathcal{FT}_{\mathfrak{r}}(\mathbf{V}_D, T_D)$	tree-based tensors with fixed rank $\mathfrak{r}$	Ch. 6
$\mathcal{FT}_{\leq \mathfrak{r}}(\mathbf{V}_D, T_D)$	tree-based tensors with bounded rank $\leq \mathfrak{r}$	Ch. 7
$C^{(\alpha)}$	transfer tensor at vertex $\alpha \in T_D$	Ch. 6
Δ_k	Laplacian-like map at level k of T_D, used in construction of tree-based atlas	Ch. 9

Differential geometry and tangent bundle

$\mathbb{M}$, $\mathbb{N}$	Banach manifolds	Ch. 2
$T_m\mathbb{M}$	tangent space at $m \in \mathbb{M}$	Ch. 2
$T\mathbb{M}$	tangent bundle of $\mathbb{M}$, $\bigsqcup_m T_m\mathbb{M}$	App. A
$\pi_{T\mathbb{M}} : T\mathbb{M} \to \mathbb{M}$	bundle projection	App. A
$\sigma_0 : \mathbb{M} \to T\mathbb{M}$	zero section, $m \mapsto (m, 0)$	App. A
$Tf(m)$	tangent map (derivative) of f at m	Ch. 2
$Df(m)$	Fréchet derivative of f at m	Ch. 2
$\mathcal{C}^p$	p times continuously Fréchet differentiable	
$\mathcal{C}^\omega$	analytic (real or complex)	
$\mathfrak{X}(\mathbb{M})$	smooth vector fields on $\mathbb{M}$	App. A
$\Gamma(E)$	smooth sections of vector bundle E	App. A
∇	linear connection on a vector bundle	App. A
$R(X, Y)Z$	curvature tensor of connection ∇	App. A
$\mathrm{Hol}_m(\nabla)$	holonomy group of ∇ at m	App. A

Dirac–Frenkel variational principle

$\mathbf{F}(t, \mathbf{v})$	vector field (right-hand side of ODE)	Ch. 10
$\mathbf{v}_r(t)$	solution curve on $\mathfrak{M}_{\mathfrak{r}}(\mathbf{V}_D)$	Ch. 10
$\mathcal{P}_{\mathbf{v}}$	metric projection onto $Z^{(D)}(\mathbf{v})$	Ch. 10
$\Pi_{\mathbf{v}}$	generalised projection onto $Z^{(D)}(\mathbf{v})$ (Banach analogue)	Ch. 10
$P_C(x)$, $\Pi_C(x)$	metric and generalised projection onto convex set C	App. B
$\arg\min$	argument of the minimum	Ch. 10

Assumptions and conditions (summary)

The following labelled conditions appear repeatedly across chapters. A condition tagged with $[T_D]$ indicates the tree-based version, which specialises to the Tucker version when $T_D = T_D^{\text{Tucker}}$.

(Cross)	tensor product map is continuous ($\|\cdot\|_D$ is a crossnorm)	Ch. 3
(Inj)	$\|\cdot\|_D \geq \|\cdot\|_\varepsilon$ (reasonable crossnorm; implies (Cross))	Ch. 4
(Inj)$[T_D]$	(Inj) holds at every level of T_D: $\|\cdot\|_\alpha \geq \|\cdot\|_{\varepsilon,\alpha}$ for $\alpha \in T_D$	Ch. 7

Common abbreviations

TT	Tensor Train
HT	Hierarchical Tucker
TBT	Tree-Based Tensor
HOSVD	Higher-Order Singular Value Decomposition
MCTDH	Multiconfiguration Time-Dependent Hartree
MPS	Matrix Product State
DMRG	Density Matrix Renormalisation Group
ALS	Alternating Least Squares
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
SVD	Singular Value Decomposition

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